Commutative Algebra

— ***EYNTKA*** **Series** —

Nathanael Chwojko-Srawley

August 18, 2026

Contents

Contents 3

Introduction 7

1 Noetherian Rings, Hilbert's Basis Theorem, and Localization 9

1.1 Noetherian Rings and Modules 9
1.2 Hilbert Basis Theorem 12
1.3 Gröbner Bases 16
1.4 Localization 19
1.4.1 Localization of Rings 20
1.4.2 Localization of Modules 26
1.4.3 Local Properties 28
1.4.4 Local Rings 32
1.4.5 Nakayama's Lemma 34

2 Primary Decomposition and Associated Primes 41

2.1 Primary Ideals 41
2.2 Primary Decomposition 43
2.3 Associated Primes 43
2.4 The Lasker-Noether Theorem 44
2.5 Support and the Isolated Components 46
2.6 Localization and Primary Decomposition 48

3 Completion **51**
3.1 Constructing Completions . . . 52
3.2 Filtrations . . . 59
3.3 More on Graded Algebras . . . 60
3.4 Artin-Rees Lemma . . . 62
3.5 Associated Graded Rings: Noetherian Closed under Completion . . . 67
3.6 The Cohen Structure Theorem . . . 72
3.7 Weierstrass Preparation and Division . . . 72

4 Ring Extension and Integral Extension **75**
4.1 Ring and Integral Extensions . . . 75
4.2 Extension and Contraction of Ideals . . . 82
4.3 Integral Extensions and Prime Ideals . . . 85

5 Valuation Ring and Local Rings **89**
5.1 Valuations . . . 89
5.2 Discrete Valuation Ring . . . 92

6 Artinian Rings **97**
6.1 Artinian Rings and Modules . . . 97

7 Dimension Theory **105**
7.1 Noether Normalization . . . 105
7.2 Krull Dimension . . . 106
7.3 The Hilbert-Samuel Function . . . 106
7.4 The Dimension Theorem . . . 108
7.5 Systems of Parameters and Krull's Principal Ideal Theorem . . . 109
7.6 Dimension and Transcendence Degree . . . 109
7.7 Regular Local Rings . . . 110
7.8 Regular Local Rings Are Factorial . . . 114
7.9 Normal Domains in Codimension One . . . 118

8 Flatness and Faithfully-Flat Descent **121**
8.1 Flat Modules . . . 121
8.2 Faithfully Flat Modules and Algebras . . . 122
8.3 Faithfully-Flat Descent . . . 123
8.4 The Local Criterion for Flatness . . . 124

8.5 Fitting Ideals 124
8.6 Generic Flatness 125

What Next? 127

Index 128

Bibliography 131

Introduction

Commutative algebra is the language in which algebraic geometry and algebraic number theory are written. Both begin from one move: to a commutative ring one attaches a geometric object whose points are the ring's prime ideals, turning a question about equations into a question about rings, ideals, and modules. This book develops the ring theory that translation requires, and develops it as *infrastructure*: the machinery the geometry and arithmetic volumes cite rather than reprove.

Four threads run through the book, each a way a ring behaves like a space. *Finiteness*, the Noetherian and Artinian chain conditions and Hilbert's basis theorem, keeps everything else in hand. *Locality*, through localization, Nakayama's lemma, and completion, is the discipline of studying a ring one prime at a time and passing to its formal-local model. *Size and singularity* is measured by dimension theory here, and by the depth, Cohen-Macaulay, and regular-local theory of the companion *Homological Algebra* [5]. And *maps between rings* (integral and flat extensions, going-up and going-down, faithfully flat descent) record how properties travel along a homomorphism. The threads meet in the dictionary the geometry books spend: prime ideals are points, Krull dimension is dimension, regularity is smoothness, and flatness is the continuous variation of a family.

The book is a graduate text; it assumes the rings, modules, and field theory of *Core Algebra* [8] and reproves none of it. What it builds is the shared foundation of the algebraic- and arithmetic-geometry towers (the varieties of *Classical Algebraic Geometry* [9], the number fields of *Algebraic Number Theory* [3], and the schemes beyond them) which return to it for localization, completion, dimension, and descent.

1

Noetherian Rings, Hilbert's Basis Theorem, and Localization

As with groups an modules, all R-algebras are quotient of a polynomial ring over many variables. If the polynomial ring is over finitely many variables. The fact that all algebras are quotient of polynomial rings will let the study of geometry and algebra.

Next, we recall some important definition and results that shall be central to the following section:

1.1 Noetherian Rings and Modules

Definition 1.1.1: Noetherian Ring

Let R be a commutative ring. Then if R satisfies the ascending chain condition on Ideals, then R is Noetherian. In particular, if $\mathcal{I}$ is a collection of ideals ordered by inclusion $(I_1 \subseteq I_2 \subseteq \cdots I_k \cdots)$, then there exists an m such that for all $n \geq m$, $I_m = I_{m+1} = \cdots$.

Proposition 1.1.2: Extension of Noetherian Rings

Let R be a Noetherian ring. Then R/I is Noetherian for any ideal I. Conversely, if I and R/I is Noetherian, then R is Noetherian.

***Proof*:**
Ideals of R/I are exactly the ideals of R containing I, so an ascending chain in R/I pulls back to an ascending chain in R, which stabilizes; thus R/I is Noetherian. Conversely, suppose I and R/I are Noetherian and let $J_1 \subseteq J_2 \subseteq \cdots$ be ideals of R. The chains $J_n \cap I$ (in I) and $(J_n + I)/I$ (in R/I) both stabilize, say for $n \geq m$. For such n, if $x \in J_{n+1}$ write $x = j + a$ with $j \in J_n$, $a \in I$ (possible since $J_n + I = J_{n+1} + I$); then $a = x - j \in J_{n+1} \cap I = J_n \cap I \subseteq J_n$, so $x \in J_n$. Hence $J_n = J_{n+1}$ and R is Noetherian.

As a consequence the image of any Noetherian ring is Noetherian.

Proposition 1.1.3: Equivalence of Noetherian Ring

Let R be a commutative ring. Then the following are equivalent

1. R is Noetherian
2. every nonempty collection of ideals of R has a maximal element under inclusion
3. every ideal of R is finitely generated

***Proof*:**
(1) $\Rightarrow$ (2): a nonempty collection of ideals with no maximal element would yield a strictly ascending chain, contradicting the ascending chain condition. (2) $\Rightarrow$ (3): given an ideal J, the collection of finitely generated subideals of J has a maximal element J_0; if $J_0 \neq J$, then for $x \in J \setminus J_0$ the ideal $J_0 + (x)$ is finitely generated and strictly larger, a contradiction, so $J = J_0$ is finitely generated. (3) $\Rightarrow$ (1): for an ascending chain $J_1 \subseteq J_2 \subseteq \cdots$, the union $J = \bigcup_n J_n$ is an ideal, hence finitely generated; its finitely many generators lie in some J_m, so $J = J_m$ and the chain stabilizes.

Example 1.1.4: Exercise
Show that if R is a Noetherian ring and M is a finitely generated R-module, then M is Noetherian (Hint: remember the universal property of free-modules)

The maximal-element form of the condition, part (2) above, is the one that does the work in practice. Here is a first instance, and the finiteness statement the factorization theory of Dedekind domains rests on: an ideal of a Noetherian ring need not factor into primes, but it always *contains* a product of them.

Proposition 1.1.5: Every Proper Ideal Contains a Product of Primes

Let R be a Noetherian ring and let $I \subsetneq R$ be a proper ideal. Then there are finitely many prime ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$, each containing I, with

$$\mathfrak{p}_1 \mathfrak{p}_2 \cdots \mathfrak{p}_r \subseteq I$$

***Proof*:**
Let Σ be the collection of proper ideals of R admitting no such family of primes, and suppose $\Sigma \neq \emptyset$. Since R is Noetherian, Σ has a maximal element I by proposition 1.1.3(2).

Step 1: the maximal counterexample is not prime. Were I prime, $r = 1$ and $\mathfrak{p}_1 = I$ would satisfy the conclusion. So there are $a, b \in R \setminus I$ with $ab \in I$. Put $J = I + (a)$ and $J' = I + (b)$; both strictly contain I. Since $ab \in I$ and I is an ideal,

$$JJ' = (I + (a))(I + (b)) \subseteq I \cdot I + I(b) + (a)I + (ab) \subseteq I$$

Step 2: both are proper. If $J = R$ then $J' = JJ' \subseteq I$, contradicting $J' \supsetneq I$; symmetrically $J' \neq R$. So J and J' are proper ideals strictly containing I.

Step 3: descend. By maximality of I in Σ, neither J nor J' lies in Σ, so there are primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_s \supseteq J$ with $\mathfrak{p}_1 \cdots \mathfrak{p}_s \subseteq J$ and primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_t \supseteq J'$ with $\mathfrak{q}_1 \cdots \mathfrak{q}_t \subseteq J'$. Then

$$\mathfrak{p}_1 \cdots \mathfrak{p}_s \mathfrak{q}_1 \cdots \mathfrak{q}_t \subseteq JJ' \subseteq I$$

and each of these primes contains J or J', hence contains I. So $I \notin \Sigma$, a contradiction, and $\Sigma = \emptyset$ as we sought to show.

Over a domain the primes so produced are automatically nonzero, since they contain I. That is the form the theory of Dedekind domains actually consumes:

Corollary 1.1.6: Nonzero Ideals Contain Products of Nonzero Primes

Let R be a Noetherian domain and let $I \subsetneq R$ be a nonzero proper ideal. Then I contains a product of finitely many *nonzero* prime ideals, each containing I.

***Proof*:**
Take $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ as in proposition 1.1.5. Each $\mathfrak{p}_i$ contains I, and $I \neq 0$, so each $\mathfrak{p}_i$ is nonzero.

We will soon encounter rings that satisfy the descending chain condition (commonly abbreviated to D.C.C). Such rings are called *Artinian rings*. For the following result, let us upgrade **??** for modules:

Definition 1.1.7: Chain of Modules and Composition Series

Let M be a module. Then a *chain* of submodule of M is a sequence (M_i) of submodules st

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$$

where the length of the chain is the number of $\supsetneq$. A maximal chain is called a *composition series* of M (equivalently, each M_{i-1}/M_i is *simple*. IF the chain is of infinite length, we shall say that M does not have a composition series.

Proposition 1.1.8: Finite Length, then Length is Invariant

Let M have a composition series of length n. Then every composition series of M has length n, and every chain in M can be extended to a composition series.

***Proof*:**
Let $\ell(M)$ denote the least length of composition series of a module M To get the desired result, let us prove a few results:

1. $N \subseteq M \Rightarrow \ell(N) \leq \ell(M)$, with equality iff $N = M$. Let (M_i) be a composition series of M of minimum length, and take $N_i = N \cap M_i$. Then $N_{i-1}/N_i \subseteq M_{i-1}/M_i$, which is simple, so each N_{i-1}/N_i is 0 or all of M_{i-1}/M_i; dropping repetitions gives a composition series of N of length $\leq \ell(M)$, and $\ell(N) = \ell(M)$ forces $N_i = M_i$ for all i, i.e. $N = M$

2. each $M_i \subsetneq M$ have length $\leq \ell(M)$. To see this, we may make a new chain with M_0 as the starting point:
$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_k = 0$$
which by part 1 give us:
$$\ell(M) = \ell(M_0) \geq \ell(M_1) \geq \cdots \geq \ell(M_k) = 0$$

3. Take any composition series of M. Then if it has length k, it must be that $k \leq \ell(M)$ by part (2). As $\ell(M)$ is the least length of a composition series, also $k \geq \ell(M)$, so $k = \ell(M)$. Hence, all composition series have the same length.

Finally, consider any chain for M. If its length is $\ell(M)$, by (2) it must be a composition series, if its length is less than $\ell(M)<$ it cannot be a composition series, and hence cannot be maximal. There does exists a lengthening by a new term, which as the number of possible terms that can be added is of finite length we may continue until we have a series of length $\ell(M)$, completing the proof.

1.2 Hilbert Basis Theorem

Recall from **??** that if F is a field, then $F[x]$ is a Euclidean domain. By [8, Polynomial Rings in Several Variables over a UFD are UFDs], $F[x_1, x_2, \dots]$ is a Unique Factorization Domain. However, of all the condition's on rings we have covered they are not necessarily stronger than Unique Factorization Domains (with the counter-example being **??** where $\mathbb{Z}[x, y]$ is a UFD but not a PID). If we loosen the condition from the ideals be generated by a single element to being generated by *finitely many* elements (i.e. a Noetherian ring), then we in fact get that extending by an indeterminate x does not affect the finiteness condition on ideals.

Theorem 1.2.1: Hilbert's Basis Theorem (HBT)

Let R be a Noetherian ring. Then $R[x]$ is a Noetherian ring

If you know some linear algebra, the term "basis" is not like the term basis in linear algerba. It is because of an archaic use of the word basis when Hilbert formulated the theorem near the end of the 19th century, where it had a closer to "generation" than basis.

***Proof*:**
Recall that if $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, then a_n is the leading coefficient of $p(x)$. This terminology will often be used in this proof.

Let $I \subseteq R[x]$ be an ideal. The trick is to somehow make I linked to an ideal of R (which is finitely generated since R is Noetherian). Let L be the set of all leading coefficients of the ideal I. Then L is an ideal. First, $0 \in I$, so $0 \in L$. Next, take $f, g \in I$ be two polynomials in I so that $f = ax^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ and $g = bx^m + b_{n-1}x^{m-1} + \cdots + b_1 x + b_0$ with leading coefficients a and b and degrees n and m respectively, with $n \geq m$, so that $a, b \in L$. Multiplying the lower-degree polynomial by a power of x to equalize degrees, $rx^{n-m}g \in I$ has degree n and leading coefficient rb, so $f - rx^{n-m}g \in I$ has degree-n coefficient $a - rb$; hence $a - rb \in L$ whenever it is nonzero. Thus L is closed under subtraction and absorbs multiplication by R, so L is an ideal. Now, since R is a Noetherian ring and L is an ideal of R, L is finitely generated, say $L = \langle a_1, a_2, ..., a_n \rangle$. Then for each a_i, there is a polynomial f_i that has a_i as it's coefficient. Let's say e_i is the degree of f_i, so that the set $e_1, e_2, ..., e_n$ represents the degree's of the polynomials representing the generators of L. Let $N = \max\{e_1, e_2, ..., e_n\}$.

Next, we will construct a couple of smaller ideals. Let $d \in \{0, 1, 2, ..., N-1\}$ (i.e. d is strictly smaller than the largest degree N) and define $L_d \subseteq L$ to be the set of all leading coefficients of polynomials of I of degree d, together with 0. Then L_d is also an ideal of R as can be checked by a similar method as last proof. Since R is Noetherian, there exists a set of generators for L_d, say $L_d = \{b_{d,1}, b_{d,2}, ..., b_{d,k_d}\}$. For each generator of L_d, there is a polynomial $f_{d,i} \in I$ of degree d that has the generator $b_{d,i}$ as its leading coefficient.

Now, we will show that

$$I = \langle f_1, f_2, ..., f_n, f_{1,1}, f_{1,2}, ...f_{1,k_1}, f_{2,1}, ..., f_{N-1,k_{N-1}} \rangle$$

that is, I is generated by the polynomial f_i and the polynomials $f_{d,i}$. Since there are finitely many of them, I would be finitely generating completing the proof. To show this, let I' represent the right hand side. Then $I' \subseteq I$, since every generator of I' is in I, and I' is the smallest set that contains those generators. So, let's prove $I \subseteq I'$, and take a $f \in I$.

If $\deg(f) \geq N$, then since $f \in I$, it's leading coefficient (say a) is in L: $a \in L$. Then since L is finitely generated, it is an R-linear combination of the generators of L: $r_1 a_1 + r_2 a_2 + \cdots + r_n a_n$. From this, we can define a new polynomial $g_1 = r_1 x^{d-e_1} f_1 + r_2 x^{d-e_2} f_2 + \cdots + r_n x^{d-e_n} f_n$ with the same leading coefficient a of f that is in I'. Then

$$f - g_1$$

is a polynomial in I of degree less than $\deg(f)$. If $\deg(f - g_1)$ is still greater than or equal to N, then repeat the same process, making the polynomial $(f - g_1) - g_2$. Continue on this way till we get a polynomial of degree less than N:

$$f - g_1 - g_2 - \cdots - g_\ell = f - (g_1 + g_2 + \cdots + g_\ell)$$

Let's label this above polynomial f'. Note that the sum of the g_i's can be re-written as an R-linear combination of elements of I' since they all belong to I':

$$f - (\alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_n f_n)$$

Now, $\deg(f') < N$. Since $f' \in I'$, it's leading coefficient (say a) is in L_d: $a \in L_d$. Then since L_d is finitely generated, it is an R-linear combination of the generators of L_d: $a = r_1 b_{d,1} + r_2 b_{d,2} + \cdots + r_{k_d} b_{d,k_d}$. From this, we can define a new polynomial $h_1 = r_1 f_{d,1} + r_2 f_{d,2} + \cdots + r_{k_d} f_{d,k_d}$ with the same leading coefficient as f' that is in I'. Then

$$f' - h_1$$

Is of degree less than f'. Just as before, we can continue this procedure, which will terminate once the degree reaches 0:

$$f' - h_1 - h_2 - \cdots - h_p = f' - (h_1 + h_2 + \cdots + h_p)$$

Note that $\sum_i h_i$ can be decomposed into a sum of polynomials with leading coefficients from L_d:

$$f' - (\beta_{1,1} f_{1,1} + \beta_{1,2} f_{1,2} + \cdots + \beta k, k_d f_{k,k_d})$$

Since our algorithm terminated after we have subtacted the constant (that is, the leading term of L_0), we are left with:

$$f - (g_1 + g_2 + \cdots + g_\ell + h_1 + h_2 + \cdots + h_p) = 0$$

which implies $f = g_1 + g_2 + \cdots + g_\ell + h_1 + h_2 + \cdots + h_p$. Re-writing slightly, we get that

$$f = \alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_n f_n + \beta_{1,1} f_{1,1} + \beta_{1,2} f_{1,2} + \cdots + \beta k, k_d f_{k,k_d}$$

which shows that f is a combination of elements of I' and so $f \in I'$. Since f was arbitrary, we cna conclude that $I \subseteq I'$.

Along with the quick assertion earlier that $I' \subseteq I$, that shows us that $I = I'$, showing that I was finitely generated. Since I was an arbitrary ideal, we are left with saying that every ideal of $R[x]$ is finitely generated, showing that $R[x]$ is also Noetherian, as we sought to show.

The result is not necessarily true if R is not Noetherian

In the exercises there will be the outlining of an alternative proof which will give a different insight in the exercises.

Corollary 1.2.2: HBT on multi-variable Polynomials

Let R be a Noetherian ring. Then $R[x_1, x_2, ..., x_n]$ is a Noetherian ring

***Proof*:**
Since R is Noetherian, by the Hilbert Basis Theorem, $R[x_1]$ is Noetherian. Since $R[x_1]$ is Noetherian, $(R[x_1])[x_2] = R[x_1, x_2]$ is Noetherian. Continuing inductively, we see that $R[x_1, x_2, ..., x_n]$ is Noetherian, as we sought to show

Corollary 1.2.3: f.g. K-algebra then Finitely Presented

Let R be a finitely generated k-algebra. Then R is *finitely presented*, meaning

$$R \cong \frac{k[x_1, ..., x_n]}{(f_1, ..., f_m)} = \frac{k[x_1, ..., x_n]}{I}$$

where I is finitely generated.

***Proof*:**
As R is finitely generated as a k algebra:

$$R \cong \frac{k[x_1, ..., x_n]}{I}$$

for some finitely many variable $x_1, ..., x_n$ and an ideal $I \subseteq R$. By corollary 1.2.2 R is Noetherian, hence I is *finitely generated*: $I = (f_1, ..., f_m)$. But then R is finitely presented, completing the proof.

You can now return to **??** to fully appreciate how there exist Atomic Domains which are not Noetherian Domains.

This theorem is very powerful. Since $R[x]$ is an ideal of itself, it is also finitely generated and the polynomials which generate it finitely can be characterized by their leading coefficients. In section 1.3, we will expand this theory further find a computational way of getting these polynomials.

Since a field is Noetherian (only ideal is (1) and (0), both clearly finite), Hilbert's Basis Theorem and induction immediately give:

Corollary 1.2.4: HBT on Polynomial Rings of Several Variables

Every ideal in the polynomial ring $F[x_1, x_2, ..., x_n]$ with coefficients from a field F is finitely generated

***Proof*:**
A field is Noetherian, so by corollary 1.2.2 the ring $F[x_1, \ldots, x_n]$ is Noetherian; every ideal is therefore finitely generated by proposition 1.1.3.

Recall that finitely generated R-algebras are quotient of polynomial rings R. By HBT, we have the following

Corollary 1.2.5: Finitely Generated Algebras Are Noetherian

Let R be Noetherian and let A be a finitely generated R-algebra. Then A is Noetherian.

***Proof*:**
Write $A \cong R[x_1, \ldots, x_n]/I$ as a quotient of a polynomial ring. By corollary 1.2.2 the polynomial ring is Noetherian, and by proposition 1.1.2 so is its quotient A.

Polynomial rings being over a Noetherian ring being Noetherian is central to to the study of algebras thanks to the universal property of Algebras

(A word on how Artinian rings are the opposite with D.C.C. We won't study them till chapter 6, since they sort of seem like a "oh wow, Noetherian rings can do so much, what about the opposite?" and then we realise that Artinian ring implies Noetherian ring, and it has so much structure that we can easily classify them all with no further fought, so we'll just leave that for later. All in all, the Noetherian condition turns out to be "more interesting")

1.2.6 Without Assuming Noetherian Let $R = k[x_1, ..., x_n, ...]$ and $A = R/(x_1, ..., x_n, ...)$. Then A is finitely generated, but the ideal is not finitely generated. In algebraic geometry, there is often the need to work with rings that look like quotients of polynomial rings (ex. polynomial rings over local rings). Thus, it common to work with *finitely presented* rings.

Exercise 1.2.1

1. Alternative HBT: (the one uses Diksons lemma)
2. Consider the chain $k[[x]]] \subseteq k[[x^{1/2}]] \subseteq k[[x^{1/6}]] \subseteq \cdots \subseteq k[[x^{1/n!}]]$. Taking the union of all of these forms an object called the *Puisseaux series*. Is the Puisseaux seris a ring, and if so is the Puisseaux series Noetherian?
3. Let S be a finitely generated R-algebra, where R is Noetherian. Prove that S is Noetherian.
4. We know that ideals of $k[x]$ are generated by 1 element if k is a field. By Hilbert's basis theorem, we know that every ideal of $k[x, y]$ is finitely generated. Is it true that ideals of $k[x, y]$ is generated by at most 2 elements if k is a field?
5. Which of the following are Noetherian Rings:
   1. $\mathbb{Z}[x, 1/x]$ as a subring of $\mathbb{Q}$
   2. $R := \mathbb{R} + x\mathbb{C}[x]$
   3. $R := k + xL[x]$ where $[L : k] < \infty$ is a finite field extension
   4. $R := k + xk[x, y]$ as a subring of $k[x, y]$, k a field.
6. Let M be an R-module over a Noetherian ring R. Show that there exists a $x \in M$ such that $\mathrm{Ann}_R(x)$ is a prime ideal (large hint: consider the set $\mathcal{S} = \{\mathrm{Ann}_A(x) \mid x \in M\}$ of ideals, and use the fact that R is Noetherian to conclude $\mathcal{S}$ has a maximal element. Is this element prime?)

1.3 Gröbner Bases

Hilbert's basis theorem guarantees that every ideal of $k[x_1, \ldots, x_n]$ is finitely generated, but its proof is not constructive: it exhibits no canonical generating set and no way to decide whether a given polynomial lies in an ideal. *Gröbner bases* repair both defects. They provide, for each ideal and each choice of monomial order, a distinguished finite generating set against which a multivariate division algorithm gives unique remainders, turning ideal membership into a finite computation. This is the computational basis of commutative algebra and algebraic geometry; the standard reference is Cox, Little, and O'Shea, *Ideals, Varieties, and Algorithms*.

Fix the polynomial ring $S = k[x_1, \dots, x_n]$ over a field k, and write a monomial as $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ for $\alpha \in \mathbb{N}^n$.

Definition 1.3.1: Monomial Order

A *monomial order* on S is a total order $\preceq$ on the monomials of S (equivalently on $\mathbb{N}^n$) such that

1. $\preceq$ is a well-order: every nonempty set of monomials has a least element; and
2. it is compatible with multiplication: if $x^\alpha \preceq x^\beta$ then $x^\gamma x^\alpha \preceq x^\gamma x^\beta$ for all $\gamma \in \mathbb{N}^n$.

The prototype is the *lexicographic order*: $x^\alpha \succ x^\beta$ when the leftmost nonzero entry of $\alpha - \beta$ is positive. The *graded* and *graded reverse* lexicographic orders first compare total degree $|\alpha| = \sum_i \alpha_i$ and break ties lexicographically; the latter is usually fastest in practice. Fixing a monomial order, every nonzero $f \in S$ has a largest monomial.

Definition 1.3.2: Leading Term

Let $\preceq$ be a monomial order and $0 \neq f \in S$. The *multidegree* $\operatorname{multideg}(f)$ is the $\preceq$-largest α with x^α occurring in f; the corresponding term is the *leading term* $\mathrm{LT}(f)$ and its monomial the *leading monomial* $\mathrm{LM}(f)$. For an ideal I, its *leading term ideal* is $\mathrm{LT}(I) = (\mathrm{LT}(f) : 0 \neq f \in I)$, a monomial ideal.

Compatibility makes leading terms multiplicative, $\mathrm{LT}(fg) = \mathrm{LT}(f)\,\mathrm{LT}(g)$, and the well-order axiom is exactly what makes the following algorithm terminate.

Theorem 1.3.3: Multivariate Division Algorithm

Fix a monomial order and an ordered tuple $(g_1, \dots, g_s)$ of nonzero polynomials. Then every $f \in S$ can be written

$$f = q_1 g_1 + \cdots + q_s g_s + r$$

where no monomial of the remainder r is divisible by any $\mathrm{LM}(g_i)$. It is produced by repeatedly subtracting the appropriate multiple of some g_i whenever $\mathrm{LM}(g_i)$ divides the current leading monomial, and otherwise moving the leading term into r.

***Proof*:**
Each step strictly lowers the $\preceq$-largest monomial not yet moved into r, so by the well-order property the process terminates; the stated condition on r holds by construction.

The remainder generally depends on the *order* of the g_i, and $r = 0$ is sufficient but not necessary for $f \in (g_1, \dots, g_s)$. Both defects vanish for a special generating set.

Definition 1.3.4: Gröbner Basis

Let $I \subseteq S$ be a nonzero ideal and $\preceq$ a monomial order. A finite set $G = \{g_1, \ldots, g_s\} \subseteq I$ is a *Gröbner basis* of I if the leading terms of G generate the leading term ideal:

$$\mathrm{LT}(I) = (\mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_s))$$

Theorem 1.3.5: Existence and Membership

Every nonzero ideal $I \subseteq S$ has a Gröbner basis for any monomial order, and any Gröbner basis of I generates I. Dividing $f \in S$ by a Gröbner basis of I gives a remainder that is independent of the order of the divisors and is zero if and only if $f \in I$.

***Proof*:**
The monomial ideal $\mathrm{LT}(I)$ is finitely generated by finitely many $\mathrm{LT}(f_i)$ with $f_i \in I$ (*Dickson's lemma*, equivalently theorem 1.2.1); these f_i are a Gröbner basis. If G is a Gröbner basis and $f \in I$, division gives $f = \sum q_i g_i + r$ with $r \in I$ and no monomial of r divisible by any $\mathrm{LM}(g_i)$; were $r \neq 0$, then $\mathrm{LT}(r) \in \mathrm{LT}(I) = (\mathrm{LM}(g_i))$ would be divisible by some $\mathrm{LM}(g_i)$, a contradiction. Hence $r = 0$ and $f \in (G)$, so G generates I and membership is detected by the remainder. Two remainders of the same f differ by an element of I with no monomial divisible by a leading term, hence by 0; the remainder is unique.

Thus a Gröbner basis solves the *ideal membership problem*: $f \in I$ iff its Gröbner-basis remainder is 0. To compute one, we measure the failure of a generating set to be Gröbner through cancellations of leading terms.

Definition 1.3.6: S-polynomial

For nonzero $f, g \in S$ with $L = \mathrm{lcm}(\mathrm{LM}(f), \mathrm{LM}(g))$, the *S-polynomial* is

$$S(f, g) = \frac{L}{\mathrm{LT}(f)}\, f - \frac{L}{\mathrm{LT}(g)}\, g,$$

built so the leading terms of the two products cancel.

Theorem 1.3.7: Buchberger's Criterion

A finite generating set $G = \{g_1, \ldots, g_s\}$ of an ideal I is a Gröbner basis if and only if, for every pair $i \neq j$, the remainder of $S(g_i, g_j)$ on division by G is zero.

***Proof*:**
Necessity is immediate, since $S(g_i, g_j) \in I$ has zero remainder by theorem 1.3.5. For sufficiency, the content is that every cancellation of leading terms among the g_i is an S-polynomial syzygy; when all S-polynomial remainders vanish this forces $\mathrm{LT}(I) = (\mathrm{LM}(g_i))$. The details are in Cox, Little, and O'Shea.

The criterion is a decision procedure, and completing a set to satisfy it is an algorithm.

Theorem 1.3.8: Buchberger's Algorithm

Given generators $\{f_1, \ldots, f_m\}$ of I, start with $G = \{f_1, \ldots, f_m\}$; repeatedly compute the remainder r of each $S(g_i, g_j)$ on division by G, adjoining r to G whenever $r \neq 0$; stop when all S-polynomial remainders are zero. This terminates and returns a Gröbner basis of I.

***Proof*:**
Each adjoined remainder r has leading monomial outside the current $(\mathrm{LM}(g))$, so the leading term ideals strictly increase; by the ascending chain condition in the Noetherian ring S (theorem 1.2.1) this stabilizes, so finitely many elements are added and the loop halts. On termination all S-polynomial remainders vanish, so G is a Gröbner basis by theorem 1.3.7.

Discarding redundant generators and scaling leading coefficients to 1 yields the unique *reduced* Gröbner basis of I for a given order, a genuine canonical form: two ideals coincide iff their reduced Gröbner bases (for the same order) are equal.

Gröbner bases make ideal membership *decidable*, but not cheap. In the worst case the degrees occurring in a reduced Gröbner basis, and hence the running time of Buchberger's algorithm, are doubly exponential in the number of variables, and ideal membership for $k[x_1, \ldots, x_n]$ is EXPSPACE-complete (Mayr and Meyer). The abstract finiteness handed to us by Hilbert's basis theorem thus conceals real computational hardness: knowing that an ideal is finitely generated is a long way from being able to compute with it.

1.4 Localization

Localization shall be the generalization of **??** to commutative rings and modules in general (including ones with zero-divisors). With the introduction of the tensor product, localization shall be shown to have the important property of producing *flat module*, which shall have important geometric consequences in algebraic geometry. From this, we shall define the notion of local properties and the famous Nakayama lemma which shall allow many proofs to be done "locally". Concluding this section, we shall wrap up the correspondence between primes of a ring R and primes in its integral extension.

The term "localization" comes from algebraic geometry, where this process of localizing mirrors the process of "localizing" for manifolds. If you have done some differential geometry, recall that the tangent space at a point p has *cotangent* space $\mathfrak{m}_p/\mathfrak{m}_p^2$, where $\mathfrak{m}_p$ is the maximal ideal of germs vanishing at p (the tangent space is its dual). Localizing shall correspond to this process, where "inverting" elements around a point is the same as a function on a manifold being invertible on a small enough local neighborhood where $f(p) \neq 0$. The reader may question this correspondence, for in differential geometry we are working with smooth functions over a [smooth] manifold, while in the algebraic case we simply have a [commutative] ring. However, the correspondence is more equivalent than it may at first seem. It shall be particularly clear by the end of [9, The Dictionary], where it is shown that there is a way of thinking of all commutative rings as a set of "functions" on a special geometric space (the geometric space is called an [affine] *scheme*), hence there is a direct parallel between smooth functions on the geometric object of a manifold and commutative rings where each element is interpreted as a function on a scheme.

1.4.1 Localization of Rings

This is a generalization of the constructing of ring of fraction from **??** by allowing the set that's closed under multiplication D to have zero divisors. This will imply that R need not be embedded into $D^{-1}R$, but the universal property is still satisfied. Furthermore, we will have to solve the problem of elements that "should" be zero being equal to nonzero elements. If $a \in R$ and $d \in D$ where $as = 0$, then:

$$a = \frac{a}{1} = \frac{as}{s} = \frac{0}{s} = 0$$

this issue will be addressed in the proof:

Theorem 1.4.1: Universal Property of Localization

Let R be a commutative ring and D a multiplicatively closed subset containing 1. Then there is a ring $D^{-1}R$ and a homomorphism $\iota\colon R \to D^{-1}R$ with $\iota(D) \subseteq (D^{-1}R)^\times$ such that for every ring S and homomorphism $\psi\colon R \to S$ with $\psi(D) \subseteq S^\times$, there is a *unique* homomorphism $\Psi\colon D^{-1}R \to S$ making the following diagram commute:

$$\begin{tikzcd}
D^{-1}R \arrow[dr, "\Psi"] & \\
R \arrow[u, "\iota"] \arrow[r, "\psi"'] & S
\end{tikzcd}$$

The pair $(D^{-1}R, \iota)$ is universal among all such (S, ψ).

***Proof*:**
A lot of this proof is similar to **??** (Ring of Fractions) and so it will be quickly presented pointing out where the generalization of D having zero divisors makes a difference.

Let $R \times D$ have the equivalence relation

$$(a, b) \sim (c, d) \;\Leftrightarrow\; x(ad - bc) = 0 \text{ for some } x \in D$$

Which is essentially the same as in **??**. The difference between the proof in the case where D does not have zero divisors' is with the fact that the terms are scaled by some $x \in D$. This is to make sure elements that are zero-divisors will belong to the $[0]$ (as we'll show in the proceeding corollary). Usually, the equivalence class $[(r, d)]$ is denoted $\frac{r}{s}$ or r/s so that

$$\frac{a}{b} = \frac{c}{d} \text{ if and only if } x(ad - bc) = 0 \qquad \text{for some } x \in D$$

Define

$$\frac{r}{d} + \frac{s}{e} := \frac{re + sd}{de} \qquad \frac{r}{d}\frac{s}{e} := \frac{rs}{de}$$

Then these two operations form a well-defined commutative ring $(R \times D)/\sim$; this ring is denoted $D^{-1}R$. Note that $\frac{d}{1}$ is a unit in $D^{-1}R$ for every $d \in D$, with inverse $\frac{1}{d}$.

Define $\iota : R \to D^{-1}R$ by $\iota(r) = \frac{r}{1}$. This map defines a natural ring homomorphism. Using this map, if the map $\varphi : R \to S$ maps D to units, we can define the map $\Psi : D^{-1}R \to S$ by defining:

$$\Psi(\frac{r}{d}) = \psi(r)\psi(d)^{-1}$$

This map is well-defined: Notice if $\frac{r}{d} = \frac{s}{e}$ then by definition of the equivalence relation, for some $x \in D$, $x(re - sd) = 0$ Thus $\varphi(x(re - sd)) = \varphi(x)(\varphi(re) - \varphi(sd)) = 0$. Since x is a unit in S and S is a commutative ring, s is not a zero-divisor, and so $\varphi(re) - \varphi(sd) = 0$ Moving values around, we get

$$\varphi(r)\varphi(d)^{-1} = \varphi(s)\varphi(e)^{-1}$$

Thus, Ψ is a well-defined ring homomorphism. Furthermore, by construction, $\Psi \circ \iota = \varphi$

Finally, Ψ is unique. If there was another Ψ' such that $\Psi' \circ \iota = \varphi$, then:

$$\varphi(r) = \Psi'(\iota(r)) = \Psi'\left(\frac{r}{1}\right)$$

Thus Ψ' maps $\frac{r}{1}$ in the same way as Ψ. For the $\frac{1}{d}$ terms, notice that

$$\frac{1}{d} = \left(\frac{d}{1}\right)^{-1}$$

and so $\varphi(d) = \Psi'(\frac{d}{1})$, and so, by properties of ring homomorphisms

$$\Psi'\left(\frac{d}{1}\right)^{-1} = \Psi'\left(\frac{1}{d}\right)$$

showing that $\Psi' = \Psi$, fulfilling the last condition needed to prove for the universal property

As before, $D^{-1}R$ is still called the ring of fractions. However, there is a more common name with a geometric intuition:

Definition 1.4.2: Localization of R at D

Let R be a commutative ring and D be a multiplicatively closed subset of R. Then the ring $D^{-1}R$ produced in the previous theorem is the *ring of fractions of R with respect to D*, or the *localization of R at D*. This ring is also sometimes denoted $R[D^{-1}]$.
If $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$ for some $f \in R$, we usually denote $D^{-1}R = R_f$.

If D is not a multiplicatively closed set, we define $R[D^{-1}] := R[\overline{D}^{-1}]$ where $\overline{D^{-1}}$ is the multiplicative closure of D.

The fact that D can contain zero divisors means that ι can now have non-trivial kernel:

Corollary 1.4.3: Kernel of ι

Let ι be the natural inclusion of R into $D^{-1}R$ for some appropriate D. Then

1. $\ker(\iota) = \{r \in R \mid xr = 0 \text{ for some } x \in D\}$. In particular, ι is an injection if and only if D doesn't contain 0 or zero-divisors
2. $D^{-1}R = 0$ if and only if D contains 0 or a nilpotent element (since D is multiplicatively closed)

***Proof*:**

1. If $\iota(r) = \frac{r}{1} = 0$, then we have that $(r,1) \sim (0,1)$ and so $x(1r - 0) = xr = 0$ for some $x \in D$. Thus, ι is an injection if and only if D does not contain 0 or zero-divisors
2. $D^{-1}R = 0$ if and only if $r/1 = 0/1$ for all r i.e. $(r,1) \sim (0,1)$ if and only if $(1,1) \sim (0,1)$ if and only if $x \cdot 1 = 0$. If $x = 0$, we're done, and if not then as D is multiplicatively closed then $x^n \in D$ for each $n \in \mathbb{N}$ and so if $x^n = 0$ then x is nilpotent and $x^n \cdot 1 = 0$, completing the proof.

This let's us get a sense of how adding zero-divisors to D changes the resulting localization of R at D.

Example 1.4.4: Localization

All examples in **??** are valid examples, including the trivial one of any field $\mathbb{F}$, $\mathbb{Z}$ to $\mathbb{Q}$ with $D = \mathbb{Z} - \{0\}$, and $\mathbb{F}[x]$ to $\mathbb{F}(x)$ with $D = F[x] - \{0\}$. We shall give some more interesting examples here (all rings are commutative):

1. If R is an integral domain, then we can always take $D = R - \{0\}$. The ring $D^{-1}R$ is then always a field, and is called the *field of fractions of* R, and is sometimes denoted Q. Then for any smaller multiplicatively closed subset E, $E^{-1}R$ is a subring of Q with elements $r/e \in Q$ with $r \in R$ and $e \in E$
2. Let R be a ring and $f \in R$. Define $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$. Then we'll denote $D^{-1}R = R_f$. Then R_f is a ring in which the element f has become a unit. In a sense, it is now possible to solve the question
$$fx - 1 = 0$$
In fact, this intuition can be made more rigorous showing that
$$R_f \cong \frac{R[x]}{fx-1}$$
This is similar to how we worked with field extensions, and in fact is an important link to connect the intuition of localization to a geometric intuition. Keep this example in the back of your mind for now.
3. Consider $R = k[x,y]/(xy)$, take $\overline{x} \in R$, and consider $R_{\overline{x}}$. Then the the map $\iota : R \to R_{\overline{x}}$ is no longer injective as:
$$\overline{y} = \frac{\overline{xy}}{\overline{x}} = \frac{\overline{0}}{\overline{x}} = \overline{0}$$
4. This example is called *localizing at a prime*. Let R be a ring and P a prime ideal of R. Then the set $D = R - P$ is multiplicatively closed. To see that, let $x, y \in D$. If it were the case that $xy \in P$, then either $x \in P$ or $y \in P$. But then one of them wouldn't be in D, a contradiction. Thus, D is indeed multiplicatively closed. The localization $D^{-1}R$ is called the *localization of* R *at* P and is commonly denoted as R_P. In this ring, all elements not in P become units. As a simple example, take $R = \mathbb{Z}$ and $P = (p)$ for some prime p. Then by the nice divisibility properties of integers, the set $\mathbb{Z}_{(p)}$ can with a little work be shown to be:
$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} \;\middle|\; p \nmid b \right\}$$

that is, it is all elements of $\mathbb{Q}$ for which the denominator is not divisible by p. Can you prove that $\mathbb{Z}_{(p)}$ has only one maximal ideal (and hence, is a local ring)? Is this true of all localization at primes?

It is tempting to say that if $R_{(p)}$ is the localization of a ring R at the prime p and $a/b \in R_{(p)}$ if $b \notin (p)$, however this is not necessarily the case as we are dealing with an equivalence class! The reader may then ask if we can take a better representative (like the representative a/b in $\mathbb{Z}$ such that $\gcd(a,b) = 1$), however R is not necessarily a UFD and so this may not be well-defined.

5. The following examples is a way to "localise" functions. Let k be a field and V be any set. Define R to be functions from V to k containing the constant functions ($F \subseteq \mathrm{Hom}^{+\text{ constants}}_{\mathbf{Set}}(V,k)$) with $+$ and $\cdot$ defined by component-wise addition and multiplication on the image: $R = (F, +, \cdot)$. For example, take all continuous functions from $[0,1]$ to $\mathbb{R}$. For any $a \in V$, let $M_a \subseteq R$ be the set of all functions that vanish at a: $f(a) = 0$. Then M_a is the kernel of the evaluation map from R to k:

$$\mathrm{ev}_a : R \to k \qquad f \mapsto f(a) \qquad \ker(\mathrm{ev}_a) = M_a$$

   Since R contains constant functions, ev_a is surjective and so $k \cong R/M_a$. By **??**, M_a is a maximal idea, and hence a prime ideal. Thus, we can define the localization of R at M_a to be:

$$R_{M_a} = \left\{ \frac{f}{g} \;\middle|\; f, g \in R,\ g(a) \neq 0 \right\}$$

   where ev_a is extended as:

$$(f/g)(a) = f(a)/g(a)$$

   This then makes R_{M_a} a way of defining rational functions.

6. Let $R = \mathbb{C}[x]$, and take $p = (x-a)$. Since $\mathbb{C}[x]$ is a Principal Ideal Domain and $(x-a)$ is irreducible, $(x-a)$ is prime, and so we can define

$$\mathbb{C}[x]_{(x-a)} = \left\{ p(x)/q(x) \;\middle|\; x - a \not| q(x) \right\}$$

   which can be thought of all rational functions which do not vanish at a.

Localization has many nice geometric interpretations. As we will see in **??**, all algebraic varieties (i.e. irreducible algebraic sets) are related to the prime ideals. Hence, understanding how ideals of R correspond to ideals of $D^{-1}R$ will bring us to understanding a geometric connection.

For an ideal $I \subseteq R$, its *extension* ${}^{\iota}I$ is the ideal of $D^{-1}R$ generated by $\iota(I)$; since every such element can be written as a/d with $a \in I$ and $d \in D$, the notation $D^{-1}I$ is also used. For an ideal $J \subseteq D^{-1}R$, its *contraction* ${}^{\pi}J = \iota^{-1}(J)$ is an ideal of R, being the preimage of an ideal under the ring homomorphism ι.

Proposition 1.4.5: Ideals of Localization

Let R be a ring and $D^{-1}R$ be the localization of R at D for appropriate D. Then:

1. For every ideal $J \subseteq D^{-1}R$,
$$J = J^{ce}$$
As a consequence, every ideal of $D^{-1}R$ is an extension of an ideal in R, and every ideal of D^{-1} contracts to a distinct ideal in R

2. If $I \subseteq R$ is an ideal of R, then:
$$^{\pi}(^{\iota}I) = \{r \in R \mid dr \in I \text{ for some } d \in D\}$$
As a consequence, $I^e = D^{-1}R$ if and only if $I \cap D \neq \emptyset$

3. There is a bijection between the set of prime ideals with $P \cap D = \emptyset$ and the set of prime ideals of $D^{-1}R$:
$$\left\{\begin{matrix}\text{prime ideals of } R \text{ with} \\ P \cap D = \emptyset\end{matrix}\right\} \underset{(-)^c}{\overset{(-)^e}{\rightleftarrows}} \left\{\text{prime ideals of } D^{-1}R\right\}$$

4. If R is Noetherian, then $D^{-1}R$ is Noetherian. Similarly if R is Artinian

***Proof*:**

1. Notice that by construction, $^{\iota}(^{\pi}J) \subseteq J$. For the other direction, notice that for any $a/d \in J$ it is always the case that $a/1 = d(a/d) \in J$, that is there exists a a/d such that $d(a/d) = a/1$. Since $\iota(\pi(a/d)) = a/1$. and so by definition, of $^{\iota}(^{\pi}J)$, $(1/d)(a/1) = a/d \in {^{\iota}(^{\pi}J)}$ (since it closed by multiplication of elements from $D^{-1}R$.

2. Let $I' = \{r \in R \mid dr \in I \text{ for some } d \in D\}$ be the set in the question. We'll show $I' = {^{\pi}(\lim I)}$. For the $\subseteq$ direction, take $r \in I'$. Then by definition there exists a $d \in D$ such that $dr = a \in I$. Then $r/1 = a/d \in \lim I$, so contracting we get $r \in {^{\pi}(\lim I)}$. For the $\supseteq$ direction, take $r \in {^{\pi}(\lim I)}$. Then $r/1 = a/d \in \lim I$ for some $a \in I$ and $d \in D$. Then by the definition of the equivalence class, we have $x(dr - a) = 0$ for some $x \in D$, and hence $xdr = xa$ where $xa \in I$. Since $xd \in D$, $r \in I'$. The rest of the assertions of (2) follow quickly.

3. To map prime ideals from $D^{-1}R$ to R, recall that preimages of prime ideals are prime. Thus, by construction, the contraction must map prime ideals of $D^{-1}R$ to prime ideals that are disjoint from D.

   For the converse, let P be a prime ideal of R that is disjoint from D, and consider $Q = {^{\iota}P}$. We'll show Q is prime. Suppose $(a/d_1)(b/d_2) = (ab/d_1d_2) \in Q$. Then $ab/d_1d_2 = c/d$ for some $c \in P$ and $d \in D$. Then by definition of the equivalent relation, $x(dab - d_1d_2c) = 0$ for some $x \in D$. Since $xdab - xd_1d_2c = 0 \in P$ and P is closed under subtraction and left-multiplication of elements of R, it must be that $xdab = (xd)(ab) \in P$. Since P is disjoint from D, it must be that $ab \in P$. Since P is prime, either a or b is an element of P. Thus, either a/d_1 or b/d_2 is an element of Q, showing that Q is a prime ideal. Thus, extending prime ideals that are disjoint from D map to prime ideals in $D^{-1}R$.

Finally, by (2), since P is disjoint from D we have $P = {}^{\pi}({}^{\iota}P)$, showing that extending and contracting are inverses of each other, finishing the proof for (3).

4. Given some ascending chain of ideals in $D^{-1}R$, they contract to ascending chain of ideals in R. Since R is Noetherian, they stabilize. Thus, the ideals that get contracted must also stabilize. Similarly for Artinian Rings.

The ideal in part (2) of proposition 1.4.5 is given a name:

Definition 1.4.6: Saturated Ideal

Let R be a commutative ring, $I \subseteq R$ is an ideal and $D^{-1}R$ is the localization of R at D for appropriate D. Then *saturation* of the ideal I with respect to D is ${}^{\pi}(\lim I)$. If $I = {}^{\pi}(\lim I)$, then I is said to be *saturated* or *D-saturated* with respect to D

Geometrically, the prime-ideal bijection of proposition 1.4.5(3) identifies $\operatorname{Spec}(D^{-1}R)$ with the subspace of $\operatorname{Spec} R$ consisting of the primes disjoint from D. For $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$ this is the distinguished open set $D(f) = \{\mathfrak{p} \in \operatorname{Spec} R \mid f \notin \mathfrak{p}\}$, so $\operatorname{Spec} R_f \cong D(f)$; dually $\operatorname{Spec} R/(f) \cong V(f)$, the closed set of primes containing f. Thus $\operatorname{Spec} R = D(f) \sqcup V(f)$ as sets, the two pieces being homeomorphic to $\operatorname{Spec} R_f$ and $\operatorname{Spec} R/(f)$ as subspaces (though $\operatorname{Spec} R$ is their topological coproduct only when $D(f)$ is also closed).

Proposition 1.4.7: Localization Property Preservation

Let R be a ring, $D \subseteq R$ a multiplicative set with $0 \notin D$, and $D^{-1}R$ the localization. Then the following properties of R are preserved in $D^{-1}R$:

1. no zero divisors
2. Noetherian
3. Integrally closed

For (3), R is understood to be an integrally closed domain, so that (1) makes $D^{-1}R$ a domain with the same fraction field.

***Proof*:**

1. Suppose $(a/s)(b/t) = 0$ in $D^{-1}R$. Then $uab = 0$ for some $u \in D$ by corollary 1.4.3, and $u \neq 0$ because $0 \notin D$, so $ab = 0$ in the domain R and hence $a = 0$ or $b = 0$. Thus $a/s = 0$ or $b/t = 0$.

2. Let $J \subseteq D^{-1}R$ be an ideal and put $I = \iota^{-1}(J)$, where $\iota\colon R \to D^{-1}R$. By proposition 1.4.5 every ideal of the localization is the extension of its contraction, so $J = D^{-1}I$. Since R is Noetherian, $I = (a_1, \ldots, a_n)$ for finitely many a_i, and then J is generated by $a_1/1, \ldots, a_n/1$: an element of J is a/s with $a \in I$, and writing $a = \sum r_i a_i$ gives $a/s = \sum (r_i/s)(a_i/1)$. Every ideal of $D^{-1}R$ is therefore finitely generated.

3. Write $K = \operatorname{Frac}(R)$, which by (1) is also the fraction field of $D^{-1}R$. Let $z \in K$ be integral

over $D^{-1}R$, say
$$z^n + \frac{a_{n-1}}{s_{n-1}} z^{n-1} + \cdots + \frac{a_1}{s_1} z + \frac{a_0}{s_0} = 0$$
with $a_i \in R$ and $s_i \in D$. Put $s = s_0 s_1 \cdots s_{n-1} \in D$ and multiply the relation by s^n, collecting powers of sz:
$$(sz)^n + \sum_{i=1}^{n} \frac{a_{n-i}\, s^i}{s_{n-i}} (sz)^{n-i} = 0$$
Each coefficient lies in R, since $s^i / s_{n-i} = s^{i-1} \prod_{j \neq n-i} s_j$ and $i \geq 1$. So sz is integral over R, hence $sz \in R$ because R is integrally closed, and therefore $z = (sz)/s \in D^{-1}R$, as we sought to show.

Exercise 1.4.1

1. Let R be a ring, f be an element of R ,and R_f be the localization of R at $\{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$. Then
   1. $$R_f \cong \frac{R[x]}{fx - 1}$$
   2. $R_f \cong R_{f^n}$ for $n \geq 1$
   3. If f is nilpotent, then $R_f = 0$
2. Let R be a normal domain and S a multiplicatively closed set. Then $S^{-1}R$ is a normal domain

1.4.2 Localization of Modules

Since every ring is a module over itself, we can see how localization works for this special case of modules. We extend the concept of localization now to more general modules. The construction is essentially the same thing: if M is an R-module, and $D \subseteq R$ is multiplicatively closed, define the same equivalence class on $M \times D$ and the operation's on $(M \times D)/\sim$ as
$$\frac{m}{d} + \frac{m'}{d'} = \frac{d'm + dm'}{dd'} \quad \text{and} \quad r \cdot \frac{m}{d} = \frac{r \cdot m}{d}$$
Then this is indeed a module with the proof proceeeding essentially the same as theorem 1.4.1, and so is left as a sanity check. Just like $R[D^{-1}]$ is the localization of R at D, we will denote $M[D^{-1}]$ to be the localization of M at D. Thus, an R-module M is extended to an $D^{-1}R$-module $M[D^{-1}]$.

Proposition 1.4.8: Zeros of $M[D^{-1}]$

Let R be a ring, $D \subseteq R$ be a multiplicatively closed subset, and M be a module over R. Then m goes to 0 in $M[U^{-1}]$ if and only if m is annihilated by an element $d \in D$. If M is finitely generated, then $M[U^{-1}] = 0$ if and only if M is annihilated by an element in D.

If R lies in a field K and M is an R-module that is contained in a k-vector space, then $\iota : M \to D^{-1}M$ is injective iff $M \xrightarrow{\cdot s} M$ is injective for all $s \in D$.

***Proof*:**
Exercise

Localization modules works well with homomorphisms: let $\varphi : M \to N$ be an R-module homomorphism. Then by the universal property, we see that this is extended to an $D^{-1}R$-module homomorphism $D^{-1}\varphi : M[D^{-1}] \to N[D^{-1}]$ by mapping:

$$D^{-1}\varphi\left(\frac{m}{d}\right) = \frac{\varphi(m)}{d}$$

Furthermore, $D^{-1}(\varphi \circ \psi) = (D^{-1}\varphi) \circ (D^{-1}\psi)$, i.e. $D^{-1}(-)$ is a functor from Mod_R to $\text{Mod}_{D^{-1}R}$. We may ask for certain properties of this functor, most basic perhaps would be if $D^{-1}(-)$ is an exact functor, that it preserves kernels and cokernels. This is the case:

Proposition 1.4.9: Localization Is Exact

Let M_1, M_2, M_3 be R-modules such that $0 \to M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \to 0$ is a short exact sequence. Then

$$0 \to M_1[D^{-1}] \xrightarrow{D^{-1}f} M_2[D^{-1}] \xrightarrow{D^{-1}g} M_3[D^{-1}] \to 0$$

is a short exact sequence, that is $D^{-1}(\underline{\quad})$ is an exact functor.

***Proof*:**
Since $g \circ f = 0$, $D^{-1}g \circ D^{-1}f = D^{-1}(0) = 0$, and so $\text{im}(D^{-1}f) \subseteq \ker(D^{-1}g)$. For the reverse inclusion, let $m/d \in \ker(D^{-1}g)$ so that $g\left(\frac{m}{d}\right) = \frac{g(m)}{d} = 0$ in $M_3[D^{-1}]$. Thus, there exists an s such that $sg(m) = 0$ in M_3. Since g is an R-module homomorphism, $sg(m) = g(sm)$, and so $sm \in \ker(g) = \text{im}(f)$, implying for some $m' \in M_1$, $f(m') = sm$. Putting it all together, we get that in $M_2[D^{-1}]$:

$$\frac{m}{d} = \frac{f(m')}{ds} = (D^{-1}f)\left(\frac{m'}{ds}\right) \in \text{im}(D^{-1}f)$$

showing that $\ker(D^{-1}g) \subseteq \text{im}(D^{-1}f)$, which along with the fact that $\text{im}(D^{-1}f) \subseteq \ker(D^{-1}g)$ show's that

$$\ker(D^{-1}g) = \text{im}(D^{-1}f)$$

For exactness at the ends: if $(D^{-1}f)(m/d) = f(m)/d = 0$ then $sf(m) = f(sm) = 0$ for some $s \in D$, and injectivity of f gives $sm = 0$, so $m/d = 0$; thus $D^{-1}f$ is injective. And any m_3/d equals $(D^{-1}g)(m_2/d)$ where $g(m_2) = m_3$ (surjectivity of g), so $D^{-1}g$ is surjective. Hence the localized sequence is short exact, completing the proof.

Notice that as we are defining a $D^{-1}R$-module, we are extending scalars. If we use **??**, then we can use the following observation to arrive at this result more quickly:

Proposition 1.4.10: Characterization of Localization

Let M be an R-module and $M[D^{-1}]$ a localiation. Then:

$$M[D^{-1}] \cong_R D^{-1}R \otimes_R M$$

***Proof*:**
Define the map:

$$(r/d) \times m \mapsto \frac{r \cdot m}{d}$$

then it is easy to that this map is R-bilinear, and hence by the universal property of the tensor-product, there exists an R-module homomorphism:

$$f : D^{-1}R \otimes M \to M[D^{-1}]$$

It is clear that f is surjective and is uniquely defined with respect to the universal property. It remains to check that f is injective. First, show that for any element in $D^{-1}R \otimes M$, it can be written as:

$$\frac{1}{d} \otimes m$$

for some $d \in D$ and $m \in M$. Let's say $f(1/d \otimes m) = m/d = 0$. Then $sm = 0$ for some $s \in D$. Then:

$$\frac{1}{d} \otimes m = \frac{s}{ds} \otimes m = \frac{1}{ds} \otimes sm = \frac{1}{ds} \otimes 0 = 0$$

showing that f is injective, and hence an R-module isomorphism, completing the proof.

As a direct conssequence, $D^{-1}R$ is a *flat R-module*.

Corollary 1.4.11: Localization Commuting With Tensor

Let M and N be R-modules and $D \subseteq R$ multiplicatively closed. Then there is a unique isomorphism:

$$\varphi : M[D^{-1}] \otimes_{D^{-1}R} N[D^{-1}] \to (M \otimes_R N)[D^{-1}] \qquad (m/d) \otimes (n/s) \mapsto (m \otimes n)/ds$$

In particular, if $\mathfrak{p}$ is any prime ideal:

$$M_\mathfrak{p} \otimes_{R_\mathfrak{p}} N_\mathfrak{p} \cong (M \otimes_R N)_\mathfrak{p}$$

***Proof*:**
By proposition 1.4.10, localizing is base change along $R \to D^{-1}R$, so $M[D^{-1}] = D^{-1}R \otimes_R M$. Associativity of the tensor product then gives

$$M[D^{-1}] \otimes_{D^{-1}R} N[D^{-1}] \cong D^{-1}R \otimes_R (M \otimes_R N) = (M \otimes_R N)[D^{-1}],$$

and the prime-ideal case is the special case $D = R \setminus \mathfrak{p}$.

(Some more cool results from Micheal's p. 43)

1.4.3 Local Properties

Localization allows us to diminish our problem to a ring much closer to a field, in particular if we want to study a family of prime ideals contained in some prime ideal p. In fact, there are many results about rings that can be deduced by studying rings *locally*, namely by studying rings at each prime ideal[1]. This motivates the following concept:

Definition 1.4.12: Local Properties

Let P be a property of a ring (ex. commutative, free, etc.). Then a property P of a ring R (resp. of an R-module M) is called a *local property* if it respects the following:

$$R \text{ (resp. } M) \text{ has property } P \Leftrightarrow R_p \text{ (resp. } M_p) \text{ has property } P \text{ for each prime ideal } p \subseteq R$$

In otherwords, a property is local if we can move back and forth between studying any ring (resp. module) localized at a prime. Here are some examples of local properties:

Proposition 1.4.13: Zero Module Is A Local Property

Let M be an R-module. Then the following are equivalent:

1. $M = 0$
2. $M_p = 0$ for all prime ideals $p \subseteq R$
3. $M_m = 0$ for all maximal ideal $m \subseteq R$

***Proof*:**
Certainy (1) $\Rightarrow$ (2) $\Rightarrow$ (3). For (3) $\Rightarrow$ (1), let $0 \neq x \in M$ be a nonzero element of an R-module M with $0 \neq 1$. Let $I = \text{Ann}(x)$. Then I is contained in some maximal ideal m: $I \subseteq m$. Consider M_m. Since $M_m = 0$, $x/1 = 0$, implying $rx = 0$ for some $r \in R - m$. Howver $\text{Ann}(x) \subseteq m$, meaning no such element can exist, a contradiction, showing that $M = 0$ as we sought to show.

Another property that only needs to be verified locally is the injectivity/surjectivity of a function:

Proposition 1.4.14: Injectivity/Surjectivity/Isomorphism Is A Local Property

Let $\varphi : M \to N$ be an R-module homomorphism. Then the following are equivalent:

1. φ is an injection/surjection/isomorphism
2. $\varphi_p : M_p \to N_p$ is injection/surjection/isomorphism for all prime ideals $p \subseteq R$
3. $\varphi_m : M_m \to N_m$ is injection/surjection/isomorphism for all maximal ideals $m \subseteq R$

***Proof*:**
The proof shall be doen for injectivity, surjectivitiy follows easly and isomorphism is combining the two.

(1) $\Rightarrow$ (2) Since $0 \to M \xrightarrow{\varphi} N$ is exact, by proposition 1.4.9, $0 \to M_p \xrightarrow{\varphi_p} N_p$ is exact, hence φ_p is injective for any prime ideal $p \subseteq R$.

(2) $\Rightarrow$ (3) Every maximal idael is prime

[1] This observation is at the basis of interpreting rings as geometric objects

(3) $\Rightarrow$ (1) Let $K = \ker(\varphi)$. Then the sequence $0 \to K \to M \to N$ is exact, and hence proposition 1.4.9 $0 \to K_m \to M_m \to N_m$ is exact, therefore $K'_m \cong \ker(\varphi_m) = 0$ since φ_m is injective. Thus, by proposition 1.4.13 φ is injective.

to show surjectivity, simply reverse the arrows.

Corollary 1.4.15: Exactness Is A Local Property

Let $A \xrightarrow{f} B \xrightarrow{g} C$ be a sequence of R-module homomorphisms. Then it is exact if and only if $A_\mathfrak{m} \xrightarrow{f_\mathfrak{m}} B_\mathfrak{m} \xrightarrow{g_\mathfrak{m}} C_\mathfrak{m}$ is exact for every maximal ideal $\mathfrak{m} \subseteq R$.

***Proof*:**
Exactness at B says the homology $H = \ker g / \mathrm{im} f$ vanishes. Localization is exact (proposition 1.4.9), so it commutes with kernels, images, and quotients, giving $H_\mathfrak{m} = \ker g_\mathfrak{m} / \mathrm{im} f_\mathfrak{m}$. By proposition 1.4.13, $H = 0$ iff $H_\mathfrak{m} = 0$ for every maximal $\mathfrak{m}$, which is exactness of the localized sequences.

A final example that will be useful is that faltness is a local property:

Proposition 1.4.16: Flatness Is A Local Property

Let M be an R-module. Then the following are equivalent:

1. M is a flat R-module
2. M_p is a flat R_p-module for each prime ideal $p \subseteq R$
3. M_m is a flat R_m-module for each maximal ideal $m \subseteq R$

***Proof*:**
(1) $\Rightarrow$ (2) comes from pervious lemmas and (2) $\Rightarrow$ (3) is immediate. For (3) $\Rightarrow$ (1), let $N \to P$ be any R-module homomorphism and $m \subseteq R$ any maximal idea. Then:

$$
\begin{aligned}
N \to P \text{ is injective } &\Rightarrow N_m \to P_m \text{ is injective} && \text{proposition 1.4.14} \\
&\Rightarrow N_m \otimes_{R_m} M_m \to P_m \otimes_{R_m} M_m \text{ is injective} && \text{flatness property} \\
&\Rightarrow (N \otimes_R M)_m \to (P \otimes_R M)_m \text{ is injective} && \text{corollary 1.4.11} \\
&\Rightarrow N \otimes_R M \to P \otimes_R M \text{ is injective} && \text{proposition 1.4.14}
\end{aligned}
$$

Hence, M is flat.

Proposition 1.4.17: Integral Closure Is A Local Property

Let R be an integral domain. Then the following are equivalent:

1. R is integrally closed
2. R_p is integrally closed for each prime ideal $p \subseteq R$
3. R_m is integrally closed for each maximal ideal $m \subseteq R$

***Proof*:**
exercise.

The following is a way of proving results locally:

Proposition 1.4.18: Local Properties On Modules

Let R be a subring of a field k, and let M be an R-module contained in a k-vector space V (equiv. the map $M \to M \otimes_R k$ is injective). Then:

$$M = \bigcap_{\mathfrak{m}} M_{\mathfrak{m}} = \bigcap_{\mathfrak{p}} M_{\mathfrak{p}}$$

***Proof*:**
Certainly $M \subseteq \cap_{\mathfrak{m}} M_{\mathfrak{m}}$. For the other direciton, take $x \in \cap_{\mathfrak{m}} M - \mathfrak{m}$. For each maximal ideal $\mathfrak{m}$, $x = m/s$ for some $m \in M$ and $s \in R - \mathfrak{m}$. Hence $sx \in M$. Define the ideal

$$I_a = \{a \in R \mid ax \in M\}$$

Then $as \in I_a$. But by construction $s \notin \mathfrak{m}$,hence $I_a \nsubseteq \mathfrak{m}$. But then it must be that $I_a = R$, hence $x = 1 \cdot x \in M$, the giving the result

Extending to prime ideals is left as an exercise.

Corollary 1.4.19: Local Properties Of Ideals

Let R be an integral domain. Then every ideal $I \subseteq R$ is equal to the intersection of its localizations at the maximal ideals of R, and also to the intersections of its localization at the prime ideals of R

Hence, to show a property of an ideal I in an integral domain R, it suffices to show that the properly holds for all its localization $I_{\mathfrak{p}}$ and it is preserved under intersections.

Exercise 1.4.2

1. Show that M is projective if and only if each M_p is free.

1.4.4 Local Rings

Recall that the localization of $\mathbb{Z}$ at any prime ideal (p), $\mathbb{Z}_{(p)}$, has a unique maximal ideal. It was left as an exercise to see whether any localization at a prime, R_p left R with only a single maximal ideal. This is indeed the case, meaning there is a unique field associated to R_p, letting us study R in terms of the collection of field induced by R_p. Thus, for any integral domain R, we can always produce a collection of rings that only have a single maximal ideal $\mathfrak{m}_\mathfrak{p}$. Once we interpret a ring as a geometric object $\operatorname{Spec}(R)$, we shall see that the rings $R_\mathfrak{p}$ representing the local properties of $\operatorname{Spec}(R)$, not unlike how we have stalks at a point in Differential Geometry. Due to this, it is worth-while studying rings with a unique maximal ideal:

Definition 1.4.20: Local Ring

Let R be a commutative ring. Then R is a *local ring* if it has a unique maximal ideal.

Every field is a local ring, its unique maximal ideal being (0). The definition is enlightening: if R is local with maximal ideal $\mathfrak{m}$ and $x \notin \mathfrak{m}$, then x must be a unit, since otherwise (x) is a proper ideal and so lies in a maximal ideal, forcing $x \in \mathfrak{m}$. Conversely, if the non-units of R form an ideal then that ideal is the unique maximal ideal, so R is local.

Example 1.4.21: Local Rings

1. Every field is trivially a local ring.
2. $\mathbb{Z}/p^n\mathbb{Z}$ is a local ring with $n > 1$ with maximal ideal $\overline{p}$
3. if $f \in k[x]$ is irreducible then $k[x]/(f(x)^n)$ is a local ring with maximal ideal $\overline{(f(x))}$
4. (geometric example) Let M be a smooth manifold (real or complex). Then define $\mathcal{O}_{M,x}$ to be the germ of smooth functions at x, that is, two smooth functions $f, g \in C^\infty(M)$ are in the same equivalence relation $f \sim g$ (i.e. $[f] = [g]$ are the same germ) if there exists some open set $U \subseteq M$ containing x such that $f|_U = g|_U$. Clearly, $\mathcal{O}_{M,x}$ is a ring (using the ring operation of the codomain $\mathbb{R}$ or $\mathbb{C}$). The maximal ideal of $\mathcal{O}_{M,x}$ are those functions that are 0 at x

In fact, this last example will have a direct parallel in scheme theory, see [1].

Proposition 1.4.22: Local Ring Equivalences

Let R be a commutative ring with $1 \neq 0$. Then the following are equivalent:

1. R is a local ring.
2. the sum of any two non-units is a non-unit
3. for every $x \in R$, either x is a unit or $1 - x$ is a unit
4. if a finite sum is a unit, then one of its terms is a unit

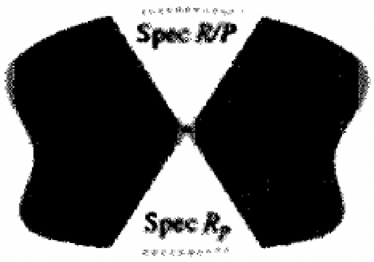
***Proof*:**
(1) $\Rightarrow$ (2): if R is local with maximal ideal $\mathfrak{m}$, an element is a non-unit iff it generates a proper ideal iff it lies in $\mathfrak{m}$, so the non-units are exactly $\mathfrak{m}$, which is closed under addition.

(2) $\Rightarrow$ (4): if a finite sum is a unit but no term is, then every term is a non-unit, so by (2) the sum is a non-unit, a contradiction.

(4) $\Rightarrow$ (3): apply (4) to $x + (1 - x) = 1$.

(3) $\Rightarrow$ (1): let $\mathfrak{m}$ be the set of non-units; it absorbs multiplication by R. If $x, y \in \mathfrak{m}$ had a unit sum $u = x + y$, then $1 = u^{-1}x + u^{-1}y$, so by (3) one of $u^{-1}x, u^{-1}y$ is a unit, making x or y a unit, a contradiction. Hence $\mathfrak{m}$ is closed under addition and is an ideal. Every proper ideal consists of non-units, hence lies in $\mathfrak{m}$, so $\mathfrak{m}$ is the unique maximal ideal and R is local.

Using localization and quotienting, we have a large amount of control over the prime ideals we want to work with: given a (commutative) ring R, if we take R_p, we eliminates all ideals except those contained in p, while R/p eliminates except those contained not contained in p (i.e. containing p). This can be visualized like so:



Hence, R_p/q^e (resp. $(R/p)_{\overline{q}}$) will only keep all the ideals between p and q, and if $p = q$, then the result is a field, known as the *residue field*:

Definition 1.4.23: Residue Field

Let R be a commutative local ring with maximal ideal m. Then R/m is called the *residue field*.

The quotient $R/\mathfrak{p}$ and the ring $R_\mathfrak{p}$ are canonically related: the ring $R_\mathfrak{p}$ can be thought of as the localization of $R/\mathfrak{p}$:

Proposition 1.4.24: Localization and Quotient Commute

Let $\mathfrak{p} \subseteq R$ be prime ideal of a ring R, and let $\mathfrak{m}_\mathfrak{p}$ be the unique maximal ideal of $R_\mathfrak{p}$. Then localizing at $\mathfrak{p}$ and then taking the field of fractions is isomorphic to taking quotient at $\mathfrak{p}$ and then localizing (at $\overline{fp} = \overline{0}$) Then there is a canonical embedding:

$$R/\mathfrak{p} \hookrightarrow R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}$$

that maps $R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}$ into the field of fractions of $R/\mathfrak{p}$

This may be visualized as the following diagram:

$$\begin{tikzcd}
 & R/\mathfrak{p} \arrow[dr, "\text{localize}"] & \\
R \arrow[ur, "\text{quotient}"] \arrow[dr, "\text{localize}"'] & & K(R/\mathfrak{p}) \cong R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \\
 & R_{\mathfrak{p}} \arrow[ur, "\text{quotient}"'] &
\end{tikzcd}$$

***Proof*:**
exercise

1.4.5 Nakayama's Lemma

(Atiyah Michael in Comm alg has some interesting results that feel are worth putting in here: p.21 on fin. gen modules)

As mentioned at the beginning, localization let's us get geometric information on a local level. Nakayama's lemma will be one way we translate that information to a global level. In this section, we will restrict to finitely generated modules (the infinite case is treated in the exercises).

Let's say M is a finitely generated module over R. If we were to localize M at a prime, we would get a local ring $R[D^{-1}]$. For simplicity, let's denote this ring G. As we've shown, G has only one maximal ideal m. Modding G by m, we get the field G/m. M over G/m is a vector-space, and so it has a basis. The heart of Nakayama's lemma is that this can be reversed: Using the basis of M over G/m, we can find a basis for M over G!

We will guild this up in three steps, which are collectively called Nakayama's Lemma:

Definition 1.4.25: Jacobson Radical

Let R be a ring. Then the *jacobson radical* of R, denoted $\text{Jac}(R)$ is the intersection fo all maximal ideals of R:

$$\text{Jac}(R) = \bigcap_{\substack{M \subseteq R \\ \text{maximal}}} M$$

(analogous to Fratini subgroup)

Proposition 1.4.26: Properties Of Jacobson Radical

Let $J \subseteq R$ be the Jacobson radical of a commutative ring R. Then:

1. If $I \subsetneq R$ is a proper ideal, then the ideal generated by I and J is also proper: $(I, J) \subsetneq R$
2. If R is Artinian, then the Jacobson radical is nilpotent: $J^n = 0$ for some n
3. $x \in J$ if and only if $1 - rx$ is a unit for every $r \in R$

Proof:

1. If I is maximal, then $J \subseteq I$ and we're done, and if not then $I \subseteq M$ for some maximal ideal so $(I, J) \subseteq M \subsetneq R$
2. Let J be the jacobson radical. By the D.C.C. the following chain stabalizes
$$J \supseteq J^2 \supseteq \cdots \supseteq J^n = J^{n+1} = \cdots$$
Consider the set of all annihilators of J^n:
$$I = \{x \in R \mid xJ^n = 0\}$$
I claim that $I = (1)$, which will show that $1J^n = J^n = 0$ (i.e. J is nilpotent). For the sake of contradiction, let's say it is not I. Then pick $I' \supsetneq I$ that is minimal (i.e. there is no ideal between I' and I). Such an ideal exists by the D.C.C. condition. Then by minimality, it must be that I' is generated by adding just one more element, that is
$$I' = I + Rx \qquad x \in R,\ x \notin I$$
Thus, I'/I is a simple R-module by minimality (i.e. has no other ideals in it). However, it is easy to see that any simple R-module is of the form R/m for some maximal ideal m. Thus,
$$I'/I \cong R/m$$
multiplying both sidses by m, we "kill" the right hand side and get
$$mI' \subseteq I$$
Thus $JI' \subseteq I$., so $xJ \in I$, and so by definition of I,
$$xJ^{n+1} = 0$$
but by assumption $xJ^{n+1} = J^n$, but that implies $x \in I$. However, $x \notin I$, a contradiction.
3. Doing the contra-positive, let's say $x \notin \text{Jac}(R)$. Then $x \notin M'$ for some maximal ideal M'. By maximality, $M + (x) = (1)$, so $y + rx = 1$, implying $1 - rx = y \in M'$, showing that $1 - rx$ cannot be a unit (this does not show "for all $r \in R$"). Conversely, if $1 - rx$ is not a unit, $1 - rx \in M$ for some M. However, $1, rx \notin M$, since if they were then $1 \in M$. Thus, $rx \notin \text{Jac}(R)$ for all r, but then $x \notin \text{Jac}(R)$.

Lemma 1.4.27: Nakayama's Lemma I

Let M be a finitely generated R-module, and suppose $\alpha \subseteq R$ is an ideal that is contained in all maximal ideals of R (i.e. $\alpha \subseteq \text{Jac}(R)$). Then:
$$\text{if } \alpha M = M \text{ then } M = \{0\}$$

***Proof*:**
We will proceed by induction on the number of generators of M. If $n = 1$, then $\langle x_1 \rangle = M$. If $\alpha M = M$, we have that:

$$x_1 = a_1 x_1 \text{ for some } a_1 \in \alpha$$

we get that $(1 - a_1)x_1 = 0$, meaning $1 - a_1$ annihilates x_1.

However, $(1 - a_1)$ is in fact a unit (as will be shown in a moment). Since it's a unit, there exists a b such that

$$0 = b \cdot 0 = b(1 - a_1)x_1 = x_1$$

Thus it must be that $x_1 = 0$, showing that $M = \{0\}$. So it's left to show that $1 - a_1$ is a unit. Let's say $(1 - a_1)$ was a proper ideal. Then it must lie in some maximal ideal. Since $a_1 \in \alpha$ and α is a subset of all maximal ideals, it must be that $1 - \alpha_1 + \alpha_1 = 1$ is in a maximal ideal, however that would make the ideal a non-proper ideal, a contradiction.

For the case of $n = k$, let $x_1, x_2, ..., x_k$ generate M, and suppose $\alpha M = M$. Then

$$x_k = a_1 x_1 + a_2 x_2 + \cdots + a_k x_k$$

so $(1 - a_k)x_k = a_1 x_1 + \cdots + a_{k-1} x_{k-1} \in \langle x_1, \ldots, x_{k-1} \rangle$. As in the base case $1 - a_k$ is a unit, hence $x_k \in \langle x_1, \ldots, x_{k-1} \rangle$, so $M = \langle x_1, \ldots, x_{k-1} \rangle$ is generated by $k-1$ elements and still satisfies $\alpha M = M$. By the induction hypothesis $M = \{0\}$, as we sought to show

If we restrict to $R = A$ being a local ring and $\alpha = m$ the maximal ideal of A, we will have the first step in constructing a basis for M over A:

Lemma 1.4.28: Nakayama's Lemma II

Let M be a finitely generated A-module over a local ring A with maximal ideal m, and let F be a submodule of M. Then:

$$\text{if } M = F + mM \text{ then } M = F$$

***Proof*:**
Take $M/F = (F + mM)/F = mM/F$ so that $M/F = mM/F$. Then by lemma 1.4.27, $M/F = \{0\}$, so that means that M moded by F makes everything 0, and so

$$M = F$$

Finally, we put these two together to create create a basis for a module over a local ring using a basis for a smaller module!

Lemma 1.4.29: Nakayama's Lemma III

Let M be a A-module over a local ring A and m the maximal ideal of A. If $x_1, x_2, ..., x_n$ generated M/mM (M mod mM), then they also generators for M

***Proof*:**
Let $F = \langle x_1, x_2, ..., x_n \rangle \subseteq M$ be the submodule generated by the x_i. Then

$$M = F + mM$$

Then by lemma 1.4.28, $M = F$, and so M has $\langle x_1, x_2, ..., x_n \rangle$ as generators, as we sought to show

Nakayama's lemma let's us reverse some constructions. As we saw in **??**, a [finitely generated?] projective module over a PID is free. Using Nakayama's lemma, we can show that a finitely generated module over a local ring is also free!

Theorem 1.4.30: Projective Module over Local Ring

Let M be a finitely generated projective module over a local ring A. Then M is free. In particular, If if $\{x_1, ..., x_n\}$ are elements of M who's residue class $\{\overline{x_1}, ..., \overline{x_1}\}$ form a basis for M/mM over A/m, then $x_1, ..., x_n$ is a *basis* for M over A.

Furthermore, if $\overline{x_1}, ..., \overline{x_n}$ are linearly independent over A/m, then this collection can be completed to a basis for M over A

***Proof*:**
First, Note that $M/\mathfrak{m}M$ over $A/\mathfrak{m}$ is indeed a vector space with action $(a + \mathfrak{m})(m + \mathfrak{m}M) = (am + \mathfrak{m}M)$ being well-defined.

Since $M/\mathfrak{m}M$ is a finite-dimensional $A/\mathfrak{m}$-vector space, choose $x_1, \ldots, x_n \in M$ lifting a basis of it (so by Nakayama's lemma III they generate M). Let F be a free module with basis $e_1, e_2, ..., e_n$, and consider the homomorphism $f : F \to M$, $f(e_i) = x_i$. We'll show that f is in fact an isomorphism. Since $x_1, ..., x_n$ generate M mod $\mathfrak{m}M$, by Nakayama's lemma III, they generate M, and so f is surjective. Next, since M is projective, f splits, i.e., $F = P_0 \oplus P_1$ where $P_0 = \ker(f)$ and P_1 and $P_1 \cong M$ through f. We'll show $P_0 = 0$.

Take some $\sum a_i x_i \in \ker(f)$ so that $\sum a_i x_i = 0$. Then reducing to the mod, we have $\sum a_i x_i + \mathfrak{m}M = 0 + \mathfrak{m}M = \sum (a_i \mathfrak{m}) \cdot (x_i + \mathfrak{m}M) = 0 + \mathfrak{m}M$. Since the $\overline{x}_i$ form a basis for $M/\mathfrak{m}M$ over $A/\mathfrak{m}A$, it must be that $a_i \in \mathfrak{m}$. Thus, we have $\sum a_i e_i \in \mathfrak{m}F$. Therefore:

$$\ker(f) \subseteq \mathfrak{m}F = \mathfrak{m}\ker(f) \oplus \mathfrak{m}M$$

Thus, $\mathfrak{m}\ker(f) = \ker(f)$. Since $\ker(f)$ is the direct summand of finitely generated modules, it is itself finitely generated. Thus, by Nakayama's lemma, $\ker(f) = 0$, and so f is an isomorphism! Therefore, we have $F \cong M$, and so M is free!

Notice that this immediately implies that the $x_1, ..., x_n$ are linealy independent since they form a basis for M over A (since f is an isomorphism and $f(e_i) = x_i$). Finally, if we extend $\overline{x}_1, ..\overline{x}_r$ to $\overline{x}_1, ..., \overline{x}_r, ...\overline{x}_n$ to be a basis for $M/\mathfrak{m}M$, then we just showed they will indeed form a basis for M.

A cool result of this theorem is a way to show some properties f based on f on the respective local modules. Let M be module over a local ring A. Then there is an associated module $M(\mathfrak{m}) = M/\mathfrak{m}M$

Then $\_\_\_(\mathfrak{m})$ induces a functor mapping M to $M(\mathfrak{m})$ and

$$f : M \to N \quad \mapsto \quad f_{(\mathfrak{m})} : M(\mathfrak{m}) \to N(\mathfrak{m})$$

Notice that if f is surjective, then $f_{(\mathfrak{m})}$ is surjective. The magic of Nakayama's lemma is that the converse is true too!

Proposition 1.4.31: Local Homomorphisms

Let M, N be finitely generated modules over a local ring A and $f : M \to N$ be an A-module homomorphism. Then:

1. f is surjective if and only if $f_{(\mathfrak{m})}$ is surjective
2. Assume f is injective. Then if $f_{(\mathfrak{m})}$ is surjective, f is an isomorphism
3. If M and N are free and $f_{(\mathfrak{m})}$ is injective, then f is injective; and f is an isomorphism if and only if $f_{(\mathfrak{m})}$ is an isomorphism

***Proof*:**

1. Let $f : M \to N$ be an R-module homomorphism and let the inducd $f_{(m)}$ be surjective so that $f(M \bmod \mathfrak{m}M) = N \bmod \mathfrak{m}N$. Since N is finitely generated, so is any quotient of it, and so $N \bmod \mathfrak{m}N$ is finitely generated by $\{\overline{x}_1, \overline{x}_2, ..., \overline{x}_n\}$ By Nakayama's lemma, $\{x_1, x_2, ..., x_n\}$ generated N. Since $x_1, ..., x_n \in f(M)$, f is also surjective

   More consisely, since $f_{(\mathfrak{m})}$ is surjective, $N = \mathrm{im}(f) + \mathfrak{m}N$. Thus, by Nakayama's lemma, $N = \mathrm{im}(f)$.

2. By part (1), surjectivity of $f_{(\mathfrak{m})}$ makes f surjective; together with the assumed injectivity of f, this makes f an isomorphism.

3. Assume M and N are free and $f_{(\mathfrak{m})}M/\mathfrak{m}M \to N/\mathfrak{m}N$ is injective. Since M is free, let $\{x_1, ..., x_n\} \subseteq M$ so that $\{\overline{x}_1, ..., \overline{x}_n\}$ is a basis for $M/\mathfrak{m}M$. Then by theorem 1.4.30, $x_1, ..., x_n$ forms a basis for M. Suppose now that $f(a_1x_1 + \cdots + a_nx_n) = 0$. By definition of $f_{(\mathfrak{m})}$, $f_{(\mathfrak{m})}(\overline{a}_1\overline{x}_1 + \cdots + \overline{a}_n\overline{x}_n) = \overline{0}$. Since $f_{(\mathfrak{m})}$ is injective, we have $\overline{a}_1\overline{x}_1 + \cdots + \overline{a}_n\overline{x}_n = 0$, and since $\overline{x}_1, ..., \overline{x}_n$ is a basis, we have that the coefficients are all $\overline{0}$, that is $a_i \in \mathfrak{m}$.

   To show that $\ker(f) = 0$, we'll use Nakayama's lemma, so it suffices to show that $\ker(f) \subseteq \mathfrak{m}\ker(f)$. By what we've just shown, we have $\ker(f) \subseteq \mathfrak{m}M$. If $f_{(\mathfrak{m})}$ were surjective, then this proof is rather quick. By part (1), we have that f is surjective. Thus the following short exact sequence splits:

   $$0 \to \ker(f) \to M \to N \to 0$$

   since N is free, So $M \cong \ker(f) \oplus N$, and $\ker(f) \subseteq \mathfrak{m}M \cong \ker(f) \oplus N$, and so $\ker(f) \subseteq \mathfrak{m}\ker(f)$, and so by Nakayama's lemma $\ker(f) = 0$.

   If $f_{(\mathfrak{m})}$ is not surjective, then we need to do a little more work. We'll take advantage of the basis lifting property we proved in the previous theorem. Let $\{\overline{x}_1, ..., \overline{x}_n\}$ be a basis for $M/\mathfrak{m}M$ over $A/\mathfrak{m}A$ which induces a basis $\{x_1, .., x_n\}$ for M over A.

$\langle f_{(\mathfrak{m})}(\overline{x}_1), ..., f_{(\mathfrak{m})}(\overline{x}_n)\rangle = \mathrm{im}(f_{(\mathfrak{m})})$ is a submodule of $N/\mathfrak{m}N$ which have that generating set as a basis (since $f_{(\mathfrak{m})}$ is injective). Thus, if:

$$\overline{a}_1 f_{(\mathfrak{m})}(\overline{x}_1) + \cdots + \overline{a}_n f_{(\mathfrak{m})}(\overline{x}_n) = 0$$

then since f is a homomorphism, we have

$$f_{(\mathfrak{m})}(\overline{a}_1\overline{x}_1 + \cdots + \overline{a}_1\overline{x}_1) = 0$$

and since these elements form a basis and f is injective, $\overline{a}_i = 0$ for all i. Now, extend this to a basis for $N/\mathfrak{m}N$ over $A/\mathfrak{m}A$:

$$\{f_{(\mathfrak{m})}(\overline{x}_1), ..., f_{(\mathfrak{m})}(\overline{x}_n), \overline{y}_1, ..., \overline{y}_m\}$$

Since N is free, this basis lifts to a basis for N over A, in particular:

$$\{f(x_1), ..., f(x_n), y_1, ..., y_m\}$$

since $f_{(\mathfrak{m})}(\overline{x}_i) = f_{(\mathfrak{m})}(x_i + \mathfrak{m}M) = f(x_i) = \mathfrak{m}N$. With this, we can now get our conclusion: since the $f(x_i)$ are linearly independent, we have that:

$$a_1 f(x_1) + \cdots + a_n f(x_n) = f(a_1x_1 + \cdots + a_nx_n) = 0$$

then it must be that the $a_i = 0$ for all the i, but then this is the condition for a function to be injective, and so f is injective, as we sought to show

Proposition 1.4.32: Projective Modules Determined by Their Reduction

Let P and P' be finitely generated projective modules over a local ring A. Then $P \cong P'$ if and only if $P/\mathfrak{m}P \cong P'/\mathfrak{m}P'$.

***Proof*:**
By theorem 1.4.30, P and P' are free, with ranks $\dim_{A/\mathfrak{m}}(P/\mathfrak{m}P)$ and $\dim_{A/\mathfrak{m}}(P'/\mathfrak{m}P')$; these ranks agree iff $P/\mathfrak{m}P \cong P'/\mathfrak{m}P'$, in which case $P \cong P'$. The converse is immediate.

Recall that if R is a finite integral domain, then it is necessarily a field (**??**). Naturally, this is not necessarily the case for infinite rings ($\mathbb{Z}$ being the prototypical example). We may thus take $Q = \mathrm{Frac}(R)$, and consdier the R-module Q

Corollary 1.4.33: Infinite Generation Of Ring Of Fractions

Let R be an integral domain that is not a field, and $Q = \mathrm{Frac}(R)$ its field of fractions. Then Q, considered as an R-module, is not finitely generated.

***Proof*:**
Suppose Q were finitely generated as an R-module. Since R is not a field, choose a nonzero maximal ideal $\mathfrak{m}$. As every nonzero element of R is already a unit in Q, we have $Q_{\mathfrak{m}} = Q$, still finitely generated over the local ring $R_{\mathfrak{m}}$. For any $0 \neq a \in \mathfrak{m}$ and $q \in Q$ we have $q = a(q/a)$ with

$a \in \mathfrak{m}R_\mathfrak{m}$, so $\mathfrak{m}R_\mathfrak{m} \cdot Q = Q$. By lemma 1.4.27, $Q = 0$, contradicting $R \subseteq Q$.

Recall that when dealing with infinite functions, we may have some strange results, like a surjection $f : M \to M$ that is not an injection. Nakayama's lemma insures we get some sanity here:

Corollary 1.4.34: Surjective Implies Injective

Let M be a finitely generated R-module. Then if $f : M \to M$ is a surjective R-module homomorphism, it is injecive.

Note that no Noetherian assumption on R or M is required.

***Proof*:**
Make M a module over $R[x]$ by letting x act as f, i.e. $x \cdot m = f(m)$. Surjectivity of f gives $(x)M = xM = M$. Since M is a finitely generated $R[x]$-module with $(x)M = M$, the determinant trick (the Cayley-Hamilton form of Nakayama's lemma) yields $q(x) \in (x)$ with $(1 - q(x))M = 0$. Write $q(x) = x\,h(x)$. For $m \in \ker f$ we have $xm = f(m) = 0$, hence $m = q(x)m = h(x)(xm) = 0$. Thus $\ker f = 0$ and f is injective.

2

Primary Decomposition and Associated Primes

The integer n factors uniquely into prime powers, and the ideal $(n) \subseteq \mathbb{Z}$ inherits the factorization: writing $n = p_1^{a_1} \cdots p_k^{a_k}$ gives $(n) = (p_1^{a_1}) \cap \cdots \cap (p_k^{a_k})$, a finite intersection of ideals each built from a single prime. Over a general ring the unique factorization of elements fails, and the factorization of ideals into primes fails with it. What survives, in every Noetherian ring, is the weaker statement that an ideal is a finite intersection of *primary* ideals, the ideal-theoretic analogue of a prime power. This is the content of the Lasker-Noether theorem, and it is the decomposition facet of the finiteness thread: the same Noetherian hypothesis that produced Hilbert's basis theorem produces here the existence and finiteness that make the theory run.

The primes attached to such a decomposition are not artifacts of the factorization but invariants of the ideal, its *associated primes*. They split into two kinds. The minimal ones record the irreducible components and are uniquely determined; the *embedded* ones sit inside those components and can shift from one decomposition to another. Read through the correspondence between primes and points, the minimal primes will become the irreducible components of a variety and the embedded primes its embedded components, a reading developed downstream in [9]. Here the account is purely ring-theoretic, and its natural home is a module: the associated primes $\operatorname{Ass}(M)$ and the support $\operatorname{Supp}(M)$ are the invariants the later chapters, and the geometry and homological volumes, reach back for.

2.1 Primary Ideals

A prime ideal $\mathfrak{p}$ is one for which $xy \in \mathfrak{p}$ forces $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. The primary ideals relax the second alternative to a power.

Definition 2.1.1: Primary Ideal

A proper ideal $\mathfrak{q} \subsetneq R$ is *primary* if, whenever $xy \in \mathfrak{q}$ with $x \notin \mathfrak{q}$, some power y^n lies in $\mathfrak{q}$.

Passing to the quotient makes the condition transparent: $\mathfrak{q}$ is primary exactly when $R/\mathfrak{q} \neq 0$ and every zero-divisor of $R/\mathfrak{q}$ is nilpotent. Indeed $xy \in \mathfrak{q}$ with $x \notin \mathfrak{q}$ says that $\bar{y}$ kills the nonzero element $\bar{x}$ of $R/\mathfrak{q}$, and $y^n \in \mathfrak{q}$ says $\bar{y}$ is nilpotent.

Proposition 2.1.2: The Radical of a Primary Ideal

If $\mathfrak{q}$ is primary then $\mathfrak{p} = \sqrt{\mathfrak{q}}$ is prime, and it is the smallest prime containing $\mathfrak{q}$. One says $\mathfrak{q}$ is $\mathfrak{p}$-*primary*.

***Proof*:**
The radical of any ideal is the intersection of the primes containing it, so $\sqrt{\mathfrak{q}}$ is contained in every prime over $\mathfrak{q}$; it suffices to show it is prime. Suppose $xy \in \sqrt{\mathfrak{q}}$, say $(xy)^m \in \mathfrak{q}$, and $x \notin \sqrt{\mathfrak{q}}$. Then no power of x lies in $\mathfrak{q}$, so from $x^m y^m \in \mathfrak{q}$ and primarity applied to $x^m \cdot y^m$ we get $(y^m)^k \in \mathfrak{q}$ for some k, whence $y \in \sqrt{\mathfrak{q}}$.

The radical being prime does *not*, by itself, make an ideal primary; the converse needs the radical to be maximal.

Lemma 2.1.3: Ideals with Maximal Radical are Primary

If $\sqrt{\mathfrak{q}} = \mathfrak{m}$ is maximal, then $\mathfrak{q}$ is primary. In particular every power $\mathfrak{m}^n$ of a maximal ideal is $\mathfrak{m}$-primary.

***Proof*:**
In $R/\mathfrak{q}$ the image of $\mathfrak{m}$ is the nilradical, since $\mathfrak{m} = \sqrt{\mathfrak{q}}$, and it is also maximal. Thus $R/\mathfrak{q}$ has a single prime ideal, so every element is either a unit or nilpotent, and every zero-divisor is nilpotent. For the last claim, $\sqrt{\mathfrak{m}^n} = \mathfrak{m}$.

Example 2.1.4: Primary and Non-Primary Ideals

1. In $\mathbb{Z}$ the primary ideals are (0) and the prime powers (p^n); this is the factorization $(n) = \bigcap (p_i^{a_i})$ read one prime at a time.
2. In $R = k[x,y]$ the ideal $\mathfrak{q} = (x, y^2)$ is primary: $R/\mathfrak{q} \cong k[y]/(y^2)$, whose only zero-divisors are the multiples of $\bar{y}$, all nilpotent. Its radical is (x,y), and $(x,y)^2 \subsetneq \mathfrak{q} \subsetneq (x,y)$, so a primary ideal need not be a power of its radical prime.
3. A power of a prime need not be primary. In $R = k[x,y,z]/(xy - z^2)$ let $\mathfrak{p} = (\bar{x}, \bar{z})$, which is prime because $R/\mathfrak{p} \cong k[y]$ is a domain. Then $\bar{x}\bar{y} = \bar{z}^2 \in \mathfrak{p}^2$, yet $\bar{x} \notin \mathfrak{p}^2$ and $\bar{y} \notin \sqrt{\mathfrak{p}^2} = \mathfrak{p}$, so $\mathfrak{p}^2$ is not primary although its radical $\mathfrak{p}$ is prime.

2.2 Primary Decomposition

Definition 2.2.1: Primary Decomposition

A *primary decomposition* of an ideal I is an expression

$$I = \mathfrak{q}_1 \cap \mathfrak{q}_2 \cap \cdots \cap \mathfrak{q}_n$$

with each $\mathfrak{q}_i$ primary. It is *minimal* if the radicals $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$ are pairwise distinct and no $\mathfrak{q}_i$ contains the intersection of the others. An ideal admitting a primary decomposition is called *decomposable.*

Any primary decomposition can be trimmed to a minimal one, once we know that primary ideals with a common radical combine.

Lemma 2.2.2: Intersections of $\mathfrak{p}$-Primary Ideals

A finite intersection $\mathfrak{q} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_m$ of $\mathfrak{p}$-primary ideals (same radical $\mathfrak{p}$) is again $\mathfrak{p}$-primary.

***Proof*:**
Radicals commute with finite intersections, so $\sqrt{\mathfrak{q}} = \bigcap \sqrt{\mathfrak{q}_i} = \mathfrak{p}$. Suppose $xy \in \mathfrak{q}$ with $y \notin \sqrt{\mathfrak{q}} = \mathfrak{p}$. For each i, $xy \in \mathfrak{q}_i$ and $y \notin \sqrt{\mathfrak{q}_i}$, so no power of y lies in $\mathfrak{q}_i$; primarity of $\mathfrak{q}_i$ then gives $x \in \mathfrak{q}_i$. Hence $x \in \mathfrak{q}$, and $\mathfrak{q}$ is primary.

Given any primary decomposition, group the $\mathfrak{q}_i$ by their radical and replace each group by its intersection, which lemma 2.2.2 keeps primary; then discard any term containing the intersection of the rest. The result is minimal, so *every decomposable ideal has a minimal primary decomposition.* The point of the next two sections is that the data surviving this trimming is intrinsic to I.

2.3 Associated Primes

The invariants are cleanest for modules, where they will also serve the later theory of depth and support.

Definition 2.3.1: Associated Prime

Let M be an R-module. A prime $\mathfrak{p}$ is an *associated prime* of M if $\mathfrak{p} = \mathrm{Ann}(x)$ for some $x \in M$; equivalently, if $R/\mathfrak{p}$ embeds in M. The set of associated primes is written $\mathrm{Ass}(M)$. The associated primes of an ideal I mean those of the module R/I.

The two forms match because a copy of $R/\mathfrak{p}$ inside M is generated by an element whose annihilator is $\mathfrak{p}$, and conversely $\mathrm{Ann}(x) = \mathfrak{p}$ makes $r \mapsto rx$ an embedding $R/\mathfrak{p} \hookrightarrow M$.

Lemma 2.3.2: Zero-Divisors and Annihilators

Let R be Noetherian and $M \neq 0$ a finitely generated module. A maximal element of the family $\{\mathrm{Ann}(x) : 0 \neq x \in M\}$ is prime, so it lies in $\mathrm{Ass}(M)$; in particular $\mathrm{Ass}(M) \neq \emptyset$.

***Proof*:**
The family is nonempty and, R being Noetherian, has a maximal member $\mathfrak{p} = \mathrm{Ann}(x)$. It is proper since $x \neq 0$. If $ab \in \mathfrak{p}$ with $b \notin \mathfrak{p}$, then $bx \neq 0$ and $\mathrm{Ann}(bx) \supseteq \mathfrak{p} + (a) \supseteq \mathfrak{p}$; maximality forces $\mathrm{Ann}(bx) = \mathfrak{p}$, so $a \in \mathrm{Ann}(bx) = \mathfrak{p}$. Thus $\mathfrak{p}$ is prime.

The associated primes are exactly the radicals appearing in a minimal decomposition.

Theorem 2.3.3: First Uniqueness Theorem

Let R be Noetherian and let I have a minimal primary decomposition $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ with radicals $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. Then

$$\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\} = \mathrm{Ass}(R/I),$$

and in particular the set of radicals does not depend on the chosen minimal decomposition.

***Proof*:**
Work in R/I; annihilators there are the ideals $(I : x)/I$ for $x \in R$, and $\sqrt{(I : x)}$ measures which components miss x. For $x \in R$ one computes $(I : x) = \bigcap_i (\mathfrak{q}_i : x)$, and $(\mathfrak{q}_i : x) = R$ when $x \in \mathfrak{q}_i$ while $\sqrt{(\mathfrak{q}_i : x)} = \mathfrak{p}_i$ when $x \notin \mathfrak{q}_i$ (primarity: if $y(x) \in \mathfrak{q}_i$ with $x \notin \mathfrak{q}_i$ then $y \in \mathfrak{p}_i$). Hence

$$\sqrt{(I : x)} = \bigcap_{x \notin \mathfrak{q}_i} \mathfrak{p}_i$$

If this radical is itself prime it must equal some $\mathfrak{p}_i$, since a prime containing a finite intersection contains one of the factors and each $\mathfrak{p}_i$ here contains the intersection. Thus every prime of the form $\sqrt{(I : x)}$ is one of the $\mathfrak{p}_i$. Conversely, minimality provides for each i an element x_i lying in every $\mathfrak{q}_j$ with $j \neq i$ but not in $\mathfrak{q}_i$, and then $\sqrt{(I : x_i)} = \mathfrak{p}_i$. The primes among $\sqrt{(I : x)}$ are therefore exactly $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$. For $\mathrm{Ass}(R/I) \subseteq \{\mathfrak{p}_i\}$: if $\mathfrak{p} \in \mathrm{Ass}(R/I)$ then $\mathfrak{p} = (I : x)/I$ is an annihilator, and being prime it equals its radical $\sqrt{(I : x)}$, hence some $\mathfrak{p}_i$. For the reverse inclusion, fix i and use minimality to pick $x_i \in \bigcap_{j \neq i} \mathfrak{q}_j$ with $x_i \notin \mathfrak{q}_i$; then $(I : x_i) = (\mathfrak{q}_i : x_i)$, which is $\mathfrak{p}_i$-primary (for $x \notin \mathfrak{q}$ with $\mathfrak{q}$ being $\mathfrak{p}$-primary, $(\mathfrak{q} : x)$ lies between $\mathfrak{q}$ and $\mathfrak{p}$, and $ab \in (\mathfrak{q} : x)$ with $a \notin (\mathfrak{q} : x)$ gives $(ax)b \in \mathfrak{q}$, $ax \notin \mathfrak{q}$, so $b \in \mathfrak{p}$). Thus the cyclic submodule $R\bar{x}_i \cong R/(\mathfrak{q}_i : x_i)$ of R/I has all its zero-divisors in $\mathfrak{p}_i$; a maximal annihilator in it is prime (lemma 2.3.2), lies in $\mathfrak{p}_i$, and, containing $(\mathfrak{q}_i : x_i)$, contains $\sqrt{(\mathfrak{q}_i : x_i)} = \mathfrak{p}_i$, so it equals $\mathfrak{p}_i$. Since annihilators in a submodule are annihilators in R/I, this gives $\mathfrak{p}_i \in \mathrm{Ass}(R/I)$, and hence equality.

2.4 The Lasker-Noether Theorem

Existence rests on two Noetherian facts: that ideals are finite intersections of irreducible ones, and that over a Noetherian ring irreducible means primary. Call a proper ideal *irreducible* if it is not the intersection of two strictly larger ideals.

Lemma 2.4.1: Irreducible Implies Primary

In a Noetherian ring every irreducible ideal is primary.

***Proof*:**
Passing to R/I, it suffices to show that the zero ideal, if irreducible, is primary; that is, that a zero-divisor of a ring in which (0) is irreducible is nilpotent. Let $xy = 0$ with $y \neq 0$. The annihilators $\mathrm{Ann}(x) \subseteq \mathrm{Ann}(x^2) \subseteq \cdots$ stabilize, say $\mathrm{Ann}(x^n) = \mathrm{Ann}(x^{n+1})$. Then $(x^n) \cap (y) = 0$: if $ax^n = by$ then $ax^{n+1} = bxy = 0$, so $a \in \mathrm{Ann}(x^{n+1}) = \mathrm{Ann}(x^n)$ and $ax^n = 0$. Since (0) is irreducible and $(y) \neq 0$, we must have $(x^n) = 0$, so x is nilpotent.

Theorem 2.4.2: Lasker-Noether

In a Noetherian ring every proper ideal is a finite intersection of irreducible ideals, and hence is decomposable. Consequently every proper ideal has a minimal primary decomposition.

***Proof*:**
Let Σ be the set of ideals that are *not* finite intersections of irreducible ideals. If $\Sigma \neq \emptyset$ it has a maximal member I by the ascending chain condition. Now I is not irreducible (an irreducible ideal is such an intersection, of length one), so $I = J \cap K$ with $J, K \supsetneq I$. By maximality J and K are each finite intersections of irreducibles, hence so is I, contradicting $I \in \Sigma$. Thus $\Sigma = \emptyset$. Each irreducible ideal is primary by lemma 2.4.1, giving a primary decomposition, which is trimmed to a minimal one by grouping common radicals and discarding redundant terms.

The finiteness of the associated primes is now immediate.

Corollary 2.4.3: Finiteness of Associated Primes

Over a Noetherian ring, $\mathrm{Ass}(R/I)$ is finite for every ideal I; more generally $\mathrm{Ass}(M)$ is finite for every finitely generated module M.

***Proof*:**
The finitely generated case includes every R/I, so we prove it. Two standard facts suffice. First, a short exact sequence $0 \to M' \to M \to M'' \to 0$ gives $\mathrm{Ass}(M) \subseteq \mathrm{Ass}(M') \cup \mathrm{Ass}(M'')$: if $\mathfrak{p} = \mathrm{Ann}(x)$ then $Rx \cong R/\mathfrak{p}$, and either $Rx \cap M' \neq 0$, so a nonzero element of the domain $R/\mathfrak{p}$ lies in M' and (being torsion-free) still has annihilator $\mathfrak{p}$, whence $\mathfrak{p} \in \mathrm{Ass}(M')$; or $Rx \cap M' = 0$, so Rx injects into M'' and $\mathfrak{p} \in \mathrm{Ass}(M'')$. Second, applying lemma 2.3.2 repeatedly to split off a submodule $\cong R/\mathfrak{p}$ and pass to the quotient produces, by the ascending chain condition, a finite filtration $0 = M_0 \subsetneq \cdots \subsetneq M_r = M$ with $M_j/M_{j-1} \cong R/\mathfrak{p}_j$. Subadditivity along it gives $\mathrm{Ass}(M) \subseteq \{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\}$, a finite set; for $M = R/I$ with I proper, theorem 2.3.3 identifies it precisely as the radicals $\{\mathfrak{p}_i\}$.

2.5 Support and the Isolated Components

The associated primes live inside a coarser invariant, the support, whose minimal members are the geometrically visible components.

Two consequences are used constantly and are recorded here, the first because it is invoked by name later in this book and in [5] without ever having been stated.

Proposition 2.5.1: Prime Avoidance

Let I be an ideal of a ring R and $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ prime ideals. If $I \subseteq \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_n$ then $I \subseteq \mathfrak{p}_i$ for some i. Equivalently, if I is contained in no $\mathfrak{p}_i$, then I contains an element lying in no $\mathfrak{p}_i$.

***Proof*:**
Induct on n, the case $n = 1$ being trivial. Discarding redundant terms, assume the covering is irredundant, so that for each i there is $a_i \in I$ with $a_i \notin \mathfrak{p}_j$ for $j \neq i$; if some a_i also avoided $\mathfrak{p}_i$ we would be done, so assume $a_i \in \mathfrak{p}_i$ for every i. Consider

$$b \;=\; a_1 + a_2 a_3 \cdots a_n$$

It lies in I. If $b \in \mathfrak{p}_1$ then, since $a_1 \in \mathfrak{p}_1$, the product $a_2 \cdots a_n \in \mathfrak{p}_1$, so some $a_j \in \mathfrak{p}_1$ with $j \geq 2$, contradicting the choice of a_j. If $b \in \mathfrak{p}_i$ for some $i \geq 2$ then, since $a_2 \cdots a_n \in \mathfrak{p}_i$, we get $a_1 \in \mathfrak{p}_i$, again a contradiction. So b lies in I and in no $\mathfrak{p}_i$, contradicting the assumed covering; hence the covering was not irredundant and induction finishes the proof.

Corollary 2.5.2: The Zero-Divisors Are the Union of the Associated Primes

Let R be Noetherian and $M \neq 0$ a finitely generated module. Then

$$\{\, z \in R : z \text{ is a zero-divisor on } M \,\} \;=\; \bigcup_{\mathfrak{p} \in \operatorname{Ass}(M)} \mathfrak{p}$$

In particular an element lying outside every associated prime of M is a non-zero divisor on M.

***Proof*:**
If $\mathfrak{p} \in \operatorname{Ass}(M)$ then $\mathfrak{p} = \operatorname{Ann}(x)$ for some $x \neq 0$, so every element of $\mathfrak{p}$ kills x and is a zero-divisor. Conversely let $zx = 0$ with $x \neq 0$, so $z \in \operatorname{Ann}(x)$. The family of annihilators $\operatorname{Ann}(y)$ for $0 \neq y \in M$ that contain $\operatorname{Ann}(x)$ is nonempty and, R being Noetherian, has a maximal element $\operatorname{Ann}(y_0)$. It is maximal in the whole family as well: an annihilator strictly containing it also contains $\operatorname{Ann}(x)$, so lies in the subfamily. By lemma 2.3.2 it is therefore prime, so $\operatorname{Ann}(y_0) \in \operatorname{Ass}(M)$, and it contains z.

Definition 2.5.3: Support

The *support* of an R-module M is $\operatorname{Supp}(M) = \{\mathfrak{p} \in \operatorname{Spec} R : M_{\mathfrak{p}} \neq 0\}$.

Proposition 2.5.4: Support of a Finitely Generated Module

If M is finitely generated then $\operatorname{Supp}(M) = V(\operatorname{Ann} M)$. In particular $\operatorname{Supp}(R/I) = V(I)$.

***Proof*:**
For finitely generated $M = (x_1, \ldots, x_m)$, the localization $M_\mathfrak{p}$ vanishes iff each generator dies, that is iff $\operatorname{Ann}(x_i) \not\subseteq \mathfrak{p}$ for all i. Since $\mathfrak{p}$ is prime, a containment $\bigcap_i \operatorname{Ann}(x_i) \subseteq \mathfrak{p}$ would force some $\operatorname{Ann}(x_i) \subseteq \mathfrak{p}$ (a prime containing a finite intersection contains a factor); so $M_\mathfrak{p} = 0$ iff $\operatorname{Ann}(M) = \bigcap_i \operatorname{Ann}(x_i) \not\subseteq \mathfrak{p}$. Thus $M_\mathfrak{p} \neq 0$ iff $\operatorname{Ann}(M) \subseteq \mathfrak{p}$, which is $\mathfrak{p} \in V(\operatorname{Ann} M)$.

Proposition 2.5.5: Associated Primes Sit in the Support

For finitely generated M over a Noetherian ring, $\operatorname{Ass}(M) \subseteq \operatorname{Supp}(M)$, and the two sets have the same minimal elements. For an ideal I these common minimal elements are the minimal primes over I.

***Proof*:**
If $\mathfrak{p} = \operatorname{Ann}(x) \in \operatorname{Ass}(M)$ then $R/\mathfrak{p} \hookrightarrow M$ localizes to $(R/\mathfrak{p})_\mathfrak{p} = \kappa(\mathfrak{p}) \neq 0 \hookrightarrow M_\mathfrak{p}$, so $\mathfrak{p} \in \operatorname{Supp}(M)$. For the minimal elements: $\operatorname{Supp}(M) = V(\operatorname{Ann} M)$ by proposition 2.5.4, whose minimal elements are the minimal primes over $\operatorname{Ann}(M)$. Each such minimal prime $\mathfrak{p}$ has $M_\mathfrak{p} \neq 0$ finitely generated over the Noetherian local ring $R_\mathfrak{p}$, so $\operatorname{Ass}(M_\mathfrak{p}) \neq \emptyset$ by lemma 2.3.2; but $\mathfrak{p}$ is minimal in $\operatorname{Supp}(M)$, so $\mathfrak{p}R_\mathfrak{p}$ is the only prime of $R_\mathfrak{p}$ in the support, forcing $\operatorname{Ass}(M_\mathfrak{p}) = \{\mathfrak{p}R_\mathfrak{p}\}$ and hence $\mathfrak{p} \in \operatorname{Ass}(M)$ (associated primes localize, as in corollary 2.6.2). So minimal elements of $\operatorname{Supp}(M)$ lie in $\operatorname{Ass}(M)$, and being minimal in the larger set they are minimal in $\operatorname{Ass}(M)$; the reverse inclusion of minimal elements is immediate from $\operatorname{Ass}(M) \subseteq \operatorname{Supp}(M)$.

The associated primes thus divide in two: the *isolated* (or minimal) primes, which are the minimal primes over I, and the *embedded* primes, which properly contain another associated prime. The isolated components are intrinsic; the embedded ones are not.

Theorem 2.5.6: Second Uniqueness Theorem

Let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ be a minimal primary decomposition with $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. If $\mathfrak{p}_i$ is isolated (minimal among the $\mathfrak{p}_j$), the component $\mathfrak{q}_i$ is uniquely determined by I: it is the contraction of $IR_{\mathfrak{p}_i}$ to R. Embedded components need not be unique.

***Proof*:**
Localize at an isolated prime $\mathfrak{p} = \mathfrak{p}_i$. For $j \neq i$, $\mathfrak{p}_j \not\subseteq \mathfrak{p}$ by isolation, so $\mathfrak{q}_j$ meets $R \setminus \mathfrak{p}$ and $\mathfrak{q}_j R_\mathfrak{p} = R_\mathfrak{p}$; for the index i, $\mathfrak{q}_i R_\mathfrak{p}$ is $\mathfrak{p}R_\mathfrak{p}$-primary with contraction $\mathfrak{q}_i$ (see proposition 2.6.1). Intersecting, $IR_\mathfrak{p} = \mathfrak{q}_i R_\mathfrak{p}$, so $\mathfrak{q}_i = IR_\mathfrak{p} \cap R$ depends only on I and $\mathfrak{p}$. Non-uniqueness of embedded components is exhibited by example 2.5.7.

Example 2.5.7: Non-Uniqueness of Embedded Components

In $R = k[x, y]$ the ideal $I = (x^2, xy)$ has radical (x) and the two minimal primary decompositions

$$I = (x) \cap (x, y)^2 = (x) \cap (x^2, y),$$

both with radicals (x) and (x, y). The isolated (x)-component is (x) in each, as theorem 2.5.6 requires. The embedded (x, y)-component is $(x, y)^2 = (x^2, xy, y^2)$ in the first and (x^2, y) in the second, so it is genuinely not determined by I.

2.6 Localization and Primary Decomposition

The behavior under localization used above is the following, and it is what recovers the isolated components.

Proposition 2.6.1: Primary Ideals under Localization

Let $S \subseteq R$ be multiplicative and $\mathfrak{q}$ a $\mathfrak{p}$-primary ideal.

1. If $S \cap \mathfrak{p} \neq \emptyset$ then $S^{-1}\mathfrak{q} = S^{-1}R$.
2. If $S \cap \mathfrak{p} = \emptyset$ then $S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary and its contraction to R is $\mathfrak{q}$.

***Proof*:**
If $s \in S \cap \mathfrak{p}$ then $s^n \in \mathfrak{q}$ for some n, and s^n is a unit in $S^{-1}R$, giving (1). For (2), assume $S \cap \mathfrak{p} = \emptyset$. Since localization is exact, $S^{-1}R/S^{-1}\mathfrak{q} \cong \bar{S}^{-1}(R/\mathfrak{q})$, the localization of $R/\mathfrak{q}$ at the image $\bar{S}$ of S. The zero-divisors of $R/\mathfrak{q}$ are nilpotent, hence lie in $\mathfrak{p}/\mathfrak{q}$; as $S \cap \mathfrak{p} = \emptyset$, the elements of $\bar{S}$ avoid $\mathfrak{p}/\mathfrak{q}$ and are non-zero-divisors of $R/\mathfrak{q}$. A zero-divisor of $\bar{S}^{-1}(R/\mathfrak{q})$ then has a zero-divisor of $R/\mathfrak{q}$ for numerator, hence is nilpotent; so $S^{-1}\mathfrak{q}$ is primary, with radical $S^{-1}\mathfrak{p}$. Its contraction is $\mathfrak{q}$: if $sx \in \mathfrak{q}$ with $s \in S$, then $s \notin \mathfrak{p} = \sqrt{\mathfrak{q}}$ forces $x \in \mathfrak{q}$ by primarity.

Corollary 2.6.2: Associated Primes Localize

For a multiplicative set S and finitely generated M over a Noetherian ring,

$$\mathrm{Ass}(S^{-1}M) = \{\, S^{-1}\mathfrak{p} : \mathfrak{p} \in \mathrm{Ass}(M),\ \ \mathfrak{p} \cap S = \emptyset \,\}$$

Consequently, if $I = \bigcap \mathfrak{q}_i$ is a minimal decomposition, localizing at an isolated prime $\mathfrak{p}_i$ deletes every component except $\mathfrak{q}_i$, recovering $\mathfrak{q}_i = IR_{\mathfrak{p}_i} \cap R$.

***Proof*:**
For $\supseteq$: an embedding $R/\mathfrak{p} \hookrightarrow M$ with $\mathfrak{p} \cap S = \emptyset$ localizes, by exactness (proposition 1.4.9), to $S^{-1}R/S^{-1}\mathfrak{p} \hookrightarrow S^{-1}M$, and $S^{-1}\mathfrak{p}$ is prime, so $S^{-1}\mathfrak{p} \in \mathrm{Ass}(S^{-1}M)$. For $\subseteq$: let $P = \mathrm{Ann}(x/s) \in \mathrm{Ass}(S^{-1}M)$ and $\mathfrak{p} = P \cap R$, a prime with $\mathfrak{p} \cap S = \emptyset$. As R is Noetherian, $\mathfrak{p} = (a_1, \ldots, a_k)$; each $a_j/1 \in P$ kills x/s, so $t_j a_j x = 0$ in M for some $t_j \in S$, and with $t = t_1 \cdots t_k$ we get $\mathfrak{p} \subseteq \mathrm{Ann}(tx)$. Conversely, if $r \in \mathrm{Ann}(tx)$ then $r \cdot (x/s) = rtx/(ts) = 0$, so $r \in P \cap R = \mathfrak{p}$; hence

$\mathfrak{p} = \mathrm{Ann}(tx) \in \mathrm{Ass}(M)$ and $P = S^{-1}\mathfrak{p}$. The recovery statement is proposition 2.6.1(2) applied to $S = R \setminus \mathfrak{p}_i$, exactly as in the proof of theorem 2.5.6.

Read geometrically, $\mathrm{Supp}(R/I) = V(I)$ decomposes into irreducible pieces indexed by the minimal primes over I, and the isolated primary components cut out those pieces with their natural scheme structure while the embedded primes mark where that structure thickens; this translation is the subject of [9], which cites the present chapter for the algebra.

3

Completion

The next type of construction we'll do is called a *completion.* These will be in the appropriate sense "stronger" then localizations, which will result in them being more restrictive, though also more powerful. A completion of a ring in many ways parallel's the completion of the rationals numbers $\mathbb{Q}$ to the real numbers $\mathbb{R}$. In fact, defining the appropriate metric, this process is a special case of the cauchy-sequence completion of the rational numbers to the reals. Namely, the completion of a ring is the equivalence class of all Cauchy-sequences which converge to the same point. Due to the topological background needed for this perspective, we shall relegate the equivalence of what shall be explicated in this section and Cauchy-sequences to **??**.

There were rings that we have observed earlier that are completions of simpler rings; we will use completions to construct rings like the ring of *formal power series* $R[[x]]$ from $R[x]$, the ring $R[x_1, x_2, ...]$, the ring with *infinitely many monomial* from $R[x_1, ..., x_n]$, and the *p-adic integers* $\mathbb{Z}_p$ from $\mathbb{Z}$..

Completions will be handy for us in many ways. One important use that we will see when looking into algebraic geometry is looking at the "singularities" of schemes (or more particularly algebraic curves). Singularities will be points that don't be had as nicely as most other points, for example in the following image (which is the zero set of the equation $y^2 - x^2\,(x + 1) = 0$) does not behave well at $(0, 0)$:

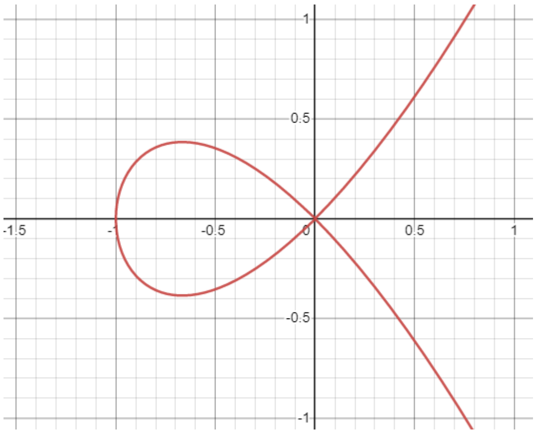


Figure 3.1: zero set of $y^2 - x^2(x+1) = 0$

This may seem similar to the geometric notion of localization, and this is indeed the case. Indeed, in the case of $\mathbb{Z}_{(p)}$ the p-adic integers are the completion of this ring, namely $\mathbb{Z}_p$ is the completion of $\mathbb{Z}_{(p)}$.

Completions shall also play a very important role in algebraic number theory. In this book, a more "classical" approach to algebraic number theory was taken, which shall give the reader great insights on the fundamental of algebraic number theory. However, a large part of *class number theory* shall rely on special completions of rings.

In this section, topological notion's will be introduced, and so an acquaintance of the material in **??** is recommended. It should be noted that the topologies that will be discussed will be very different from that of Euclidean topologies, a new notion of size that is rather different from Euclidean distance shall be developed. The reader that is familiar with the construction of the real numbers using Cauchy sequences shall find it much easier to follow this section.

3.1 Constructing Completions

We will first define what is a "sequence". Let's say we have a collection of countable rings $R_1, R_2, ..., R_n$ and homomorphisms:

$$\cdots \xrightarrow{f_3} R_3 \xrightarrow{f_2} R_2 \xrightarrow{f_1} R_1$$

Then a sequence, or a *coherent sequence* is an element $(x_1, x_2, ...) \in \prod_{i=1}^{\infty} R_i$ where:

$$f_i(x_{i+1}) = x_i$$

If we take the special case of $I \subseteq R$ being an ideal and

$$\cdots \xrightarrow{f_3} R/(I^3) \xrightarrow{f_2} R/(I^2) \xrightarrow{f_1} R/I$$

then we will define a form of completion from these maps. For ease of notation, the sequence is usually written in the reverse order, as the reader shall understand once examples will be presented

Definition 3.1.1: Completion Of A Ring

Let R be a ring and $I \subseteq R$ be an ideal, and:

$$R/I \xleftarrow{f_1} R/(I^2) \xleftarrow{f_2} R/(I^3) \leftarrow \cdots$$

Then:

$$\hat{R} := \left\{ (a_1, a_2, ...,) \in \prod_{i=1}^{\infty} R_i \;\middle|\; f_i(a_{i+1}) = a_i \right\}$$

which is often denoted as:

$$\varprojlim R/I^n := \hat{R}$$

and is called the *limit* (or in older textbooks the *inverse limit* or *projective limit*). The collection $\{R/I^n\}$ is called the *inverse system*, with the function usually taken as implicit

A couple of remarks are in order:

1. There is a natural topology and a metric in which these elements can be thought of as Cauchy sequences and $\hat{R}$ is indeed the completion of R. If $\{R/\mathfrak{a}^n R\}$ is an inverse system with ideal $\mathfrak{a} \subseteq R$, and any $x \in R$, define a neighborhood basis (or local basis) $\{x + \mathfrak{a}^n R\}_{n=1}^{\infty}$. Then this forms a basis for a topology. This topology is called the $\mathfrak{a}$-*adic topology*, U is open in this topology if and only if for each $x \in U$, there exists a positive integer n such that

   $$x + \mathfrak{a}^n R \subseteq U$$

   Then R is Hausdorff if and only if $\cap \mathfrak{a}^n R = 0$. If this is the case, define the metric $d(x, y) = 2^{-n}$ whenever $x \neq y$, where n is the largest integer with $x - y \in \mathfrak{a}^n R$ (equivalently $x - y \in \mathfrak{a}^n R \setminus \mathfrak{a}^{n+1} R$). The constant 2 does not matter, it can be replaced with any $C > 1$ and it will give the same topology.

   This exact same process can be repeated on an arbitrary R-module, where the inverse system would be $\{M/a^n M\}$, and for a given ideal $I \subseteq R$, we would define the I-adic topology on M.

2. This is a special case of a more general construction of a *limit* in the category of **Ring**. Namely, it is the limit of a special form of the diagram:

   $$\cdots \to \bullet \to \bullet \to \bullet$$

   where each arrow has the coherence condition stated above.

3. The transition maps $R/\mathfrak{a}^{n+1} \to R/\mathfrak{a}^n$ of the systems built from powers of an ideal are surjective; such inverse systems are called *surjective systems*.

4. If R is a Noetherian integral domain and $I \subsetneq R$ is a proper ideal, then $\bigcap_n I^n = 0$, and hence $R \subseteq \hat{R}$. For those who shall read **??**, this shows that inegral Noetherian domains are Hausdorff in the I-adic topology.

Example 3.1.2: Completion of Rings

1. Let $R = k[x]$ and $I = (x)$. Then:
$$\varprojlim(k[x]/(x)^n) = k[[x]]$$
To see this, consider:
$$\cdots k[x]/(x^3) \to k[x]/(x^2) \to k[x]/(x) = k$$
Notice that in each $k[x]/(x^k)$, can be thought of the collection of all polynomial of degree up to $k-1$. Hence, a coherent sequence $(a_1, a_2, ...)$ can be thought of as an "infinitely long polynomial" and would usually be denoted
$$\sum_k^\infty a_k x^k$$
Similarly, the reader may verify that $\widehat{k[x,y]} = k[[x,y]]$ with $I = (x,y)$

2. Let $R = \mathbb{Z}$ and $I = (10)$. Then $\varprojlim \mathbb{Z}/(10)^n$ is represented by all infinitely long numbers. Just like before, each $\mathbb{Z}/(10)^k$ can be thought of as as holding information up to the $k-1$th position. So if we take:
$$Z/(10) \leftarrow Z/(100) \leftarrow \mathbb{Z}/(1000) \leftarrow$$
and construct a coherent chain, we may chose elements like 5, 25, 425, ... and so on. The reader may see that this would represent an infinitely long number. The resulting ring is often denoted $\mathbb{Z}_{10}$ and is called the 10-adic integers. Any element of $\mathbb{Z}_{10}$ could formally be represented as
$$\sum_k^\infty a_k 10^k$$

3. Let $R_n = k[x_1, ..., x_n]$ with transition maps setting the last variable to 0. The limit $\varprojlim R_n$ strictly contains the polynomial ring $k[x_1, x_2, \ldots]$: it also has elements of infinite support, such as the coherent sequence $(x_1,\, x_1 + x_2,\, \ldots)$ representing $\sum_i x_i$

Completions are a way to construct rings. It would be nice if exactness interact nicely with it. If $\{A_n\}$, $\{B_n\}$, $\{C_n\}$ are three inverse systems where:

$$\begin{tikzcd}
0 \arrow[r] & A_{n+1} \arrow[r, "\psi"] \arrow[d, "\alpha"] & B_{n+1} \arrow[r, "\varphi"] \arrow[d, "\beta"] & C_{n+1} \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & A'_n \arrow[r, "\psi'"] & B'_n \arrow[r, "\varphi'"] & C'_n \arrow[r] & 0
\end{tikzcd}$$

Then we'll denote it as $0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0$. Then we naturally get the induced homomorphisms:
$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$$
however this is not always exact. The following proposition shows us when it is:

Proposition 3.1.3: Limits and Exactness for inverse Systems

Let $0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0$ be an exact sequence of inverse systems. Then:

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n$$

is always exact. If $\{A_n\}$ is a surjective system, then:

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$$

is always exact

Hence, when working with completions induced by surjective systems, the completion functor $\widehat{(-)}$ preserves injective and surjective functions[1].

***Proof*:**
This is an application of Snakes lemma. Let $A = \prod_{n=1}^{\infty} An$, and

$$d^A : A \to A \qquad d^A(a_n) = a_n - f_{n+1}(a_{n+1})$$

Then by construction

$$\ker d^A = \varprojlim A_n$$

We may define B, C via d^B and d^C respectively. Now, as we have an exact sequence of inverse systems, we get the following diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \big\downarrow{\scriptstyle d^A} & & \big\downarrow{\scriptstyle d^B} & & \big\downarrow{\scriptstyle d^C} & & \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

Then by **??** we get:

$$0 \to \ker d^A \to \ker d^B \to \ker d^C \to \mathrm{coker} d^A \to \mathrm{coker} d^B \to \mathrm{coker} d^C \to 0$$

Finally, if the system is a surjective system, we can solve inductively the equation:

$$x_n - f_{n+1}(x_{n+1}) = a_n$$

to show that d^A is surjective, completing the proof.

The following result shall be stated as it gives a very fascinating result, however for the full background the reader should read **??**:

[1] In **??**, this shall imply that the completion functor has trivial cohomology, while general inverse systems shall have non-trivial cohomology

Corollary 3.1.4: Completion of Group Exact Sequences

Let $0 \to G' \to G \xrightarrow{p} G'' \to 0$ be an exact sequence of groups. Let G have a topology given by the subgroups $\{G_n\}$, and induce the topology on G', G'' via the subgroups $\{G' \cap G_n\}$ and $\{p(G_n)\}$. Then:

$$0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$$

is exact

***Proof*:**
Apply proposition 3.1.3 to

$$0 \to \frac{G'}{G' \cap G_n} \to \frac{G}{G_n} \to \frac{G''}{pG_n} \to 0$$

Corollary 3.1.5: Quotient in Completion

Let $\hat{G}_n \leqslant \hat{G}$. Then:

$$\hat{G}/\hat{G}_n \cong G/G_n$$

***Proof*:**
The quotient G/G_n carries the discrete topology (its induced filtration is eventually 0), so it equals its own completion. Applying the exact completion functor (proposition 3.1.3) to $0 \to G_n \to G \to G/G_n \to 0$ gives $0 \to \hat{G}_n \to \hat{G} \to G/G_n \to 0$, whence $\hat{G}/\hat{G}_n \cong G/G_n$.

As a consequence, in particular interest is the case of $\mathbb{Z}_{10}$ where

$$\mathbb{Z}_{10} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_5$$

Hence $\mathbb{Z}_{10}$ is decomposeable to smaller rings. These rings are also in many ways nicer: notice that as a consequence of the above isomorphism that $\mathbb{Z}_{10}$ has zero divisors (any product of two non-trivial rings has zero-divisors)[2], showing that in general the completion may introduce zero-divisors, while $\mathbb{Z}_2$ and $\mathbb{Z}_5$ do not introduce zero-divisors. Furthermore, $\mathbb{Z}_p$ introduces more units (this was more evident with $k[[x]]$ where the units were explicitly found in **??**). For this reason, we would concentrate on the p-adic integers $\mathbb{Z}_p$. Any element of $\mathbb{Z}_p$ could be represented as

$$\sum_k^\infty a_k p^k$$

We may generalize the result for finding units in completions:

Lemma 3.1.6: Units In Completion

Let R be a ring and $\hat{R}$ a completion of R via p. Then if $u \in R/p^n$ is a unit, $u \in R/p^{2n}$ is a unit.

[2] Can you find two explicit elements?

***Proof*:**
Let u be a unit so $1 + us \in p^n$. Squaring, we get:

$$1 + u(2s + us^2) \in p^{2n}$$

Hence, it is a unit in R/p^{2n}.

The reader should use this to show that u is a unit in $\hat{R}$. This also generalizes **??**, for this directly shows that any power series with nonzero constant is a unit. In the case where $p = \mathfrak{m}$ is a maximal ideal (as in the case for p-adic integers and $k[x]$), we get the following result:

Lemma 3.1.7: Completion At Maximal Ideals

Let R be a ring and $\mathfrak{m}$ a maximal ideal. Then $\varprojlim R/\mathfrak{m}^n$ is a local ring, that is, it contains a single maximal ideal.

***Proof*:**
Let $z = (z_n) \in \varprojlim R/\mathfrak{m}^n$ have nonzero residue $z_1 \in R/\mathfrak{m}$. Since $\mathfrak{m}/\mathfrak{m}^n$ is the unique maximal ideal of $R/\mathfrak{m}^n$, each z_n has nonzero residue mod $\mathfrak{m}$ and so is a unit there, and the inverses z_n^{-1} are coherent; hence z is a unit. Thus the non-units are exactly $\ker(\varprojlim R/\mathfrak{m}^n \to R/\mathfrak{m})$, an ideal, so $\varprojlim R/\mathfrak{m}^n$ is local.

Thus, there is a natural map:

$$R_{\mathfrak{m}} \to \varprojlim R/\mathfrak{m}^n$$

where $R_{\mathfrak{m}}$ is the localization. Recall that in $R_{\mathfrak{m}}$, the elements are constructed by taking formal inverses, hence completion at a maximal ideal shows a "dual" way of creating units by finding them in some way "at infinity". Note that it is important that $\mathfrak{m}$ be maximal, for this will fail if an ideal is only prime (however, one may localize the ring at the prime ideal to make it maximal and then take the completion).

When working with localizaiton, we saw that a localization of an integral domain is an integral domain. This is no longer true for completion:

Example 3.1.8: Completion Looses Integral Domain
Let

$$R = \frac{k[x, y]}{(y^2 - x^3 - x^2)}$$

with k a field of characteristic 0. Then completing this ring, since completion is exact, we get:

$$\hat{R} = \frac{k[[x, y]]}{(y^2 - x^3 - x^2)} \cong \frac{k[[x, y]]}{(y - x\sqrt{1 + x})(y + x\sqrt{1 + x})}$$

where $\sqrt{1 + x}$ is the short-hand for the power-series associated to $\sqrt{1 + x}$. Then it is easy to see that this ring has zero-divisors. This shall eventually be made rather clear when a geometric perspective is taking. The ring R shall in a concrete sense have the curve $y^2 = x^3 + x^2$ associated to it:

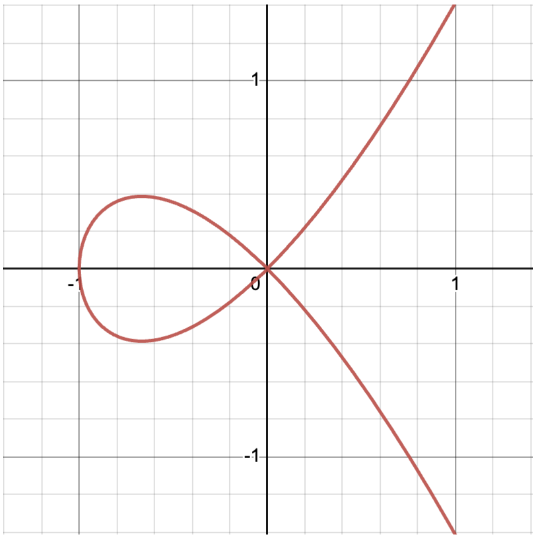


Figure 3.2: $y^2 = x^3 + x^2$

This curve shall be shown to be *irreducible* in a concrete way. The localization shall represent this curve minus some finite amount of points. The completion at (x, y) shall represent "zooming in" at the point (x, y), at which point geometric object shall roughly correspond to:

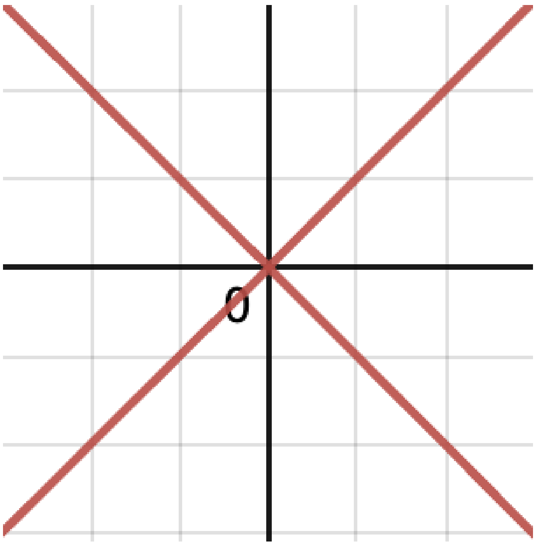


Figure 3.3: Approximate Geometric picture corresponding to $\varprojlim R/(x, y)^n$

Then this shall be shown to be decomposable in a rigorous sense into two components (the two strokes of the "x" in the above picture).

Definition 3.1.9: Separated and Complete

Let M be an R-module with the $\mathfrak{a}$-adic filtration, and let $\iota\colon M \to \hat{M} = \varprojlim M/\mathfrak{a}^n M$ be the natural map to its completion. Then M is

- *separated* if ι is injective, equivalently $\bigcap_n \mathfrak{a}^n M = 0$;
- *complete* if ι is surjective;

and *separated and complete* (usually abbreviated to just *complete*) if ι is an isomorphism, i.e. $M \cong \hat{M}$. A ring R is complete for the $\mathfrak{a}$-adic topology if it is so as a module over itself.

The two conditions genuinely differ: over a non-Noetherian ring a module can be complete without being separated. The completion $\hat{M}$ is always separated and complete, and it is the universal such object.

Proposition 3.1.10: Universal Property of Completion

Let R carry the $\mathfrak{a}$-adic topology with completion $\iota\colon R \to \hat{R}$. Then $\hat{R}$ is separated and complete, and ι is universal among continuous homomorphisms into separated complete rings: for every separated complete ring S and continuous homomorphism $f\colon R \to S$ there is a unique continuous homomorphism $\hat{f}\colon \hat{R} \to S$ with $\hat{f} \circ \iota = f$.

***Proof*:**
Continuity of f gives, after reindexing, $f(\mathfrak{a}^n) \subseteq \mathfrak{b}^n$ for a defining ideal $\mathfrak{b}$ of S, so f induces compatible maps $R/\mathfrak{a}^n \to S/\mathfrak{b}^n$. Passing to limits yields $\hat{f}\colon \hat{R} \to \hat{S} = S$ (using that S is separated and complete), and $\hat{f} \circ \iota = f$ holds on each quotient. Uniqueness follows because $\iota(R)$ is dense in $\hat{R}$ and S is separated.

Taking $\mathfrak{a}^n M$ as the filtration, a homomorphism $f\colon M \to N$ satisfies $f(\mathfrak{a}^n M) \subseteq \mathfrak{a}^n N$, so it induces maps on the quotients and lifts to $\hat{f}\colon \hat{M} \to \hat{N}$; completion is thus a functor.

3.2 Filtrations

The following construction is a more general version of an inverse system that drops many of the algebraic properties and is a more "pure topological" definition. It is being presented here for its use in the study of graded algebras, as well as it's use when defining *spectral sequences* in **??**

Definition 3.2.1: Filtration

Let M be an R-module. Then a *filtration* of M is an infinite chain:

$$M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \supseteq \cdots$$

where each M_n is a submodule of M. The chain is usually denoted (M_n). If furthermore $\mathfrak{a} \subseteq R$ is an ideal, then an $\mathfrak{a}$-filtration of M is a filtration st

$$\mathfrak{a}M_n \subseteq M_{n+1} \qquad \forall n \in \mathbb{N}$$

An $\mathfrak{a}$-filtration (M_n) is called *stable* (or $\mathfrak{a}$-*stable*) if there exists an N such that $\mathfrak{a}M_n = M_{n+1}$ for all $n \geq N$.

An $\mathfrak{a}$-stable filtration can be thought of as the generalization of a graded module which is in some way finite. The following lemma will be important to compare two $\mathfrak{a}$-stable filtrations:

Lemma 3.2.2: Comparing Stable $\mathfrak{a}$-Filtartions

Let $(M_n), (M'_n)$ be stable $\mathfrak{a}$-filtrations of M. Then they have bounded difference, that is, there exists an integer n_0 such that $M_{n+n_0} \subseteq M'_n$ and $M'_{n+n_0} \subseteq M_n$ for all $n \geq 0$.

This will imply that all stable $\mathfrak{a}$-filtrations determine the same topology on M (The $\mathfrak{a}$-topology)

***Proof*:**
It suffices to compare each stable $\mathfrak{a}$-filtration with the $\mathfrak{a}$-adic filtration $(\mathfrak{a}^n M)$. For any $\mathfrak{a}$-filtration, $\mathfrak{a}M_n \subseteq M_{n+1}$ gives $\mathfrak{a}^n M \subseteq M_n$ by induction. Stability provides an N with $M_{n+1} = \mathfrak{a}M_n$ for $n \geq N$, so $M_{n+N} = \mathfrak{a}^n M_N \subseteq \mathfrak{a}^n M$. Thus $\mathfrak{a}^n M \subseteq M_n$ and $M_{n+N} \subseteq \mathfrak{a}^n M$; doing the same for (M'_n) with some N' and combining, with $n_0 = \max(N, N')$, gives $M_{n+n_0} \subseteq M'_n$ and $M'_{n+n_0} \subseteq M_n$.

The key use of this lemma comes when defining the α-adic topology given in **??**, namely all stable $\mathfrak{a}$-filtrations produce the same topology.

Exercise 3.2.1

1. Show that $\mathbb{Z}_p \cap \mathbb{Q} \cong \mathbb{Z}_{(p)}$
2. Show that $\hat{\hat{R}} \cong \hat{R}$

3.3 More on Graded Algebras

Graded algebras were initially explored in **??**. In this section, we expand further on graded-algebras, explore their relation to the Neotherian property, and look at some important associated graded algebras given an ideal I or module M. This exploration shall bear fruitful results that will be important in the development of *projective algebraic geometry* and *homological algebra*.

First, recall that a ring R is a graded ring if $R = \bigoplus_i R_i$ where $R_i R_j \subseteq R_{i+j}$, which makes R into an R_0-algebra, and each R_i is an R_0-module. A graded R-module is a module $M = \bigoplus_0^\infty M_i$ such that $R_m M_n \subseteq M_{m+n}$. If M is also a graded ring, then M is an R-graded algebra. Elements of M_n are called *homogeneous*, and an ideal $I \subseteq R$ is called homogeneous if $I = \bigoplus_i^\infty (R_i \cap I)$. A ring or R-module homomorphism is called graded if it respects the graded structure ($f(M_n) \subseteq N_n$). Each graded ring has an ideal $R_+ = \bigoplus_{n>0} R_n$. The elements of R_n for each n are called the *homogeneous elements*.

Proposition 3.3.1: Noetherian Graded Rings

Let R be a graded ring. Then the following are equivalent:

1. R is Noetherian
2. R_0 is Noetherian and R is a finitely generated R_0-algebra

***Proof*:**

(1) $\Rightarrow$ (2) As $R_0 \cong R/R_+$, R_0 is Noetherian. As $R_+ \subseteq R$ is an ideal of R, it is finitely generated, say $x_1, ..., x_s$. Then these elements can be used to construct homogeneous s element, let's use the same label and say they are of order $k_1, ..., k_s$ (all necessarily order greater than 0). Let $R' \subseteq R$ be the ring generated as an R_0 algebra over the homogeneous elements $x_1, ..., x_s$. Then I claim that $R = R'$, which will the proof. First, let' show $A_n \subseteq A'$ using induction. It is clear if $n = 0$. For $n > 0$, taking $y \in A_n$, as $y \in A_+$, y is a linear combination of the x_i

$$y = \sum_i^s a_i x_i$$

with $a_i \in A_{n-k_i}$ (and $A_m = 0$ of $m < 0$). Then by the inductive hypothesis, each a_i is a polynomial in the x's with coefficients in A_0. Hence, by the principle of induction, the same holds for y, we have that $A_n \subseteq A'$. As $A' \subseteq A$ by definition, we have $A = A'$, as we sought to show

(2) $\Rightarrow$ (1) Follow's from HBT

Any ring R induces a graded ring by taking $\mathfrak{a} \subseteq R$ and taking $R^* = \bigoplus_n^\infty \mathfrak{a}^n$. Similarly, any R-module M induces a graded R^*-module $M^* = \bigoplus_n M_n$ where M_n is the $\mathfrak{a}$-filtration. The "canonical" grading comes from $S = R[x_1, ..., x_n]$ with $\mathfrak{a} = (x_1, ..., x_n)$ which will produce the homogeneous grading of S.

In the case where R is Noetherian and $\mathfrak{a}$ is finitely generated (say by $x_1, ..., x_s$), then R^* is a quotient of the polynomial ring $R[x_1, ..., x_s]$ (with the x_i in degree 1), hence Noetherian by proposition 3.3.1.

Proposition 3.3.2: Topology of Noetherian Graded Module

Let R be a Noetherian ring, M a finitely generated R-module, (M_n) an $\mathfrak{a}$-filtration of M. Then the following are equivalent:

1. M^* is a finitely generated R^*-module
2. The filtration (M_n) is stable

***Proof*:**
This is about creating the right conditions to see the filter is stable. Take $Q_n = \bigoplus_{r=0}^n M_r$. As each M_r is finitely generated, Q_n is finitely generated. Then Q_n forms a subgroup of M^* but not in general a R^*-submodule. We may however use it to generate one:

$$M_n^* = M_0 \oplus \cdots M_n \oplus \mathfrak{a}M_n \oplus \mathfrak{a}^2 M_n \oplus \cdots \oplus \mathfrak{a}^r M_n \oplus \cdots$$

Now, as Q_n is finitely generated as an R-module, M_n^* is finitely generated as an R^*-module, and the M_n^* form an ascending chain whose unions is M^*.

Finally, we now have that as R^* is Noetherian ,M^* is finitely generated as an R^*-module if and only if the chain stops $(M^* = M_{n_0}^*)$ if and only if the filtration is stable.

Corollary 3.3.3: Graded Noetherian then Component Noetherian

Let R be graded and Noetherian, and suppose M is a finitely generated graded R-module. Then M_n is a finitely-generated R_0-module for all $n \in \mathbb{N}$

***Proof*:**
Let

$$d_j = \deg m_j$$

Then every element of $m \in M_n$ can be written as:

$$m = \sum_j f_j(x) m_j \qquad f_j(x) \in R$$

where f_j is homogeneous of degree $n - d_j$ (and is zero if $n < d_j$). Hence, M_n is finitely generated as an R_0-module generated by all $g_j(x)m_j$ where $g_j(x)$ is a monomial in x_i of total degree $n - d_j$, as we sought to show.

3.4 Artin-Rees Lemma

The Artin-Rees lemma is a central result that will allow us to compare the induced and $\mathfrak{a}$-adic topology on a submodule, giving us insight on their topologies.

Theorem 3.4.1: Artin-Rees Lemma

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, and (M_n) a stable $\mathfrak{a}$-filtration of M. Then if $M' \subseteq M$ is a submodule of M, $(M' \cap M_n)$ is a stable $\mathfrak{a}$-filtration of M'

***Proof*:**
As $\mathfrak{a}(M' \cap M_n) \subseteq \mathfrak{a}M' \cap \mathfrak{a}M_n \subseteq M' \cap M_{n+1}$, $(M' \cap M_n)$ is an $\mathfrak{a}$-filtration, and so it defines a graded R^*-module which is a submodule of M^*, and so it is finitely generated, letting us apply by proposition 3.3.2 to completing the proof.

Corollary 3.4.2: Graded Sub-module, Computational Shortcut

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, and $M' \subseteq M$ a submodule. Then there exists a k such that

$$(\mathfrak{a}^n M) \cap M' = \mathfrak{a}^{n-k}\left((\mathfrak{a}^k M) \cap M'\right)$$

for all $n \geq k$

***Proof*:**
By the Artin-Rees lemma (theorem 3.4.1), $(\mathfrak{a}^n M \cap M')$ is a stable $\mathfrak{a}$-filtration of M', so there is a k with $\mathfrak{a}(\mathfrak{a}^n M \cap M') = \mathfrak{a}^{n+1} M \cap M'$ for all $n \geq k$; iterating gives $\mathfrak{a}^n M \cap M' = \mathfrak{a}^{n-k}(\mathfrak{a}^k M \cap M')$ for $n \geq k$.

Theorem 3.4.3: Submodule Adic Topology Equivalence

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, $M' \subseteq M$ a submodule. Then the filtrations

$$\mathfrak{a}^n M' \qquad \text{and} \qquad (\mathfrak{a}^n M) \cap M'$$

have bounded difference, hence the $\mathfrak{a}$-topology of M' coincides with the induced topology given by the $\mathfrak{a}$-topology of M

***Proof*:**
Direct result of corollary 3.4.2 and theorem 3.4.1.

Corollary 3.4.4: Module Completion Preserves Exactness

Let
$$0 \to M' \to M \to M'' \to 0$$
be a short exact sequence of finitely generated R-modules over a Noetherian ring R. Then if $\mathfrak{a} \subseteq R$ is an ideal, the $\mathfrak{a}$-adic completion functor preserves exactness:
$$0 \to \hat{M}' \to \hat{M} \to \hat{M}'' \to 0$$

***Proof*:**
Theorem 3.4.3 provides the condition needed to use corollary 3.1.4.

Next, as we have a natural homomorphism $R \to \hat{R}$, we can think of $\hat{R}$ as an R-algebra, and hence we have an extension of scalars of an R-module M given to an $\hat{R}$-module $\hat{R} \otimes_R M$. This gives a natural sequence of maps that we may want to understand:
$$\hat{R} \otimes_R M \to \hat{R} \otimes_R \hat{M} \to \hat{R} \otimes_{\hat{R}} \hat{M} = \hat{M}$$
In general, these maps are neither injective nor surjective. Under nicer conditions, we have a better control:

Proposition 3.4.5: Module Completion as Tensor Product

Let R be any ring. Then if$M$ is finitely generated,
$$\hat{R} \otimes_R M \twoheadrightarrow \hat{M}$$
is surjective. If furthermore R is Neotherian, then:
$$\hat{R} \otimes_R M \cong \hat{M}$$

***Proof*:**
Let $F \cong R^n$ so that $M \cong F/N$ for appropriate N. The conclusion can be drawn from the following commutative diagram:
$$\begin{tikzcd}
 & \hat{R} \otimes_R N \arrow[r] \arrow[d, "\gamma"] & \hat{R} \otimes_R F \arrow[r] \arrow[d, "\beta"] & \hat{R} \otimes_R M \arrow[r] \arrow[d, "\alpha"] & 0 \\
0 \arrow[r] & \hat{N} \arrow[r] & \hat{F} \arrow[r, "\delta"] & \hat{M} \arrow[r] & 0
\end{tikzcd}$$
where the top line is given by the right-exactness of the tensor product, and the bellow line is given by corollary 3.1.4. Note that as completion commutes with finite direct sums (use corollary 3.1.4), we have that $\hat{R} \otimes_R F \cong \hat{F}$, and so α is surjective, giving us the first part. If R is Noetherian, hen N is finitely generated and so γ is surjective, and so we get exactness of the bottom line. Then by chasing the diagram we will get hat α is injective, and hence an isomorphism, completing the proof.

From this, we get that when R is Noetherian, the extension of scalar functor $\widehat{(-)}$ is equivalent to the functor $\hat{R} \otimes_R (-)$ from the category of R-modules to the category of $\hat{R}$-modules is exact. As a consequence:

Corollary 3.4.6: Flatness of Completion

Let R be a Neotherian ring, $\mathfrak{a} \subseteq R$ an ideal, $\hat{R}$ the $\mathfrak{a}$-adic completing of R. Then $\hat{R}$ is a flat R-algebra.

***Proof*:**
direct result of above discussion.

Proposition 3.4.7: Properties of Adic Completion

Let R be Noetherian and $\hat{R}$ its $\mathfrak{a}$-adic completion. Then:

1. $\hat{\mathfrak{a}} = \hat{R}\mathfrak{a} = \hat{R} \otimes_R \mathfrak{a}$
2. $\widehat{\mathfrak{a}^n} = \hat{\mathfrak{a}}^n$
3. $\mathfrak{a}^n/\mathfrak{a}^{n+1} \cong \hat{\mathfrak{a}}^n/\hat{\mathfrak{a}}^{n+1}$
4. $\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{R}$

***Proof*:**
exercise (or see Atiyah p.109)

We shall show the last one. By (2) and the fact that $\hat{\hat{R}} \cong \hat{R}$, we see that $\hat{R}$ is complete in the $\mathfrak{a}$-adic topology. Hence for any $x \in \hat{\mathfrak{a}}$,

$$(1-x)^{-1} = 1 + x + x^2 + \cdots$$

converges in $\hat{R}$, and so $1-x$ is a unit. Then this implies that $\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{R}$, completing the proof.

Let us next characterize the kernel of the map $M \to \hat{M}$

Theorem 3.4.8: Krull's Theorem

LEt R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, and $\hat{M}$ the $\mathfrak{a}$-completion of M, Then the kernel

$$E = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$$

given by the map $M \to \hat{M}$ consist of all $x \in M$ that are annihilated by some element $1 + \mathfrak{a}$.

***Proof*:**
It suffices to show that $(1-\alpha)E = 0$. IT is clear that if $(1-\alpha)x = 0$

$$x = \alpha x = \alpha^2 x = \cdots \in \bigcap_{n=1}^{\infty} \mathfrak{a}^n M = E$$

Hence we shall show the converse. By the Artin-Rees consequence (corollary 3.4.2) applied to $E \subseteq M$, there is a k with $\mathfrak{a}^n M \cap E = \mathfrak{a}^{n-k}(\mathfrak{a}^k M \cap E)$ for $n \geq k$. Since $E \subseteq \mathfrak{a}^n M$ for all n, the left-hand side is E and the right-hand side is $\mathfrak{a}^{n-k}E$, so $E = \mathfrak{a}^{n-k}E \subseteq \mathfrak{a}E$, giving $\mathfrak{a}E = E$. As M is finitely generated over the Noetherian ring R, E is finitely generated, so the determinant form of Nakayama's lemma yields $\alpha \in \mathfrak{a}$ with $(1-\alpha)E = 0$, completing the proof.

Proposition 3.4.9: Localization is A Subring of Completion

Let R be Noetherian and let $S = 1 + \mathfrak{a}$. Then

$$S^{-1}R \hookrightarrow \hat{R}$$

***Proof*:**
The map $R \to \hat{R}$ and $R \to S^{-1}R$ have the same kernel, and for any $\alpha \in \hat{\mathfrak{a}}$ we have

$$(1-\alpha)^{-1} = 1 + \alpha + \alpha^2 + \cdots$$

converges in $\hat{A}$, and hence all S become a unit in $\hat{A}$. By the universal property of $S^{-1}A$, there is a natural homomorphism $S^{-1}A \to \hat{A}$ which is injective by theorem 3.4.8.

The Noetherian condition is necessary, an easy example lies within the realm of analysis:

Example 3.4.10: Localization and Completion Without Noetherian Property
Take R to be the collection of smooth functions on $\mathbb{R}$, $C^\infty(\mathbb{R})$. Let $\mathfrak{a}$ be the ideal of all f which vanish at 0. This ideal is maximal as $R/\mathfrak{a} \cong \mathbb{R}$. Then $\mathfrak{a}$ is generated by x (if you know some multi-variable calculus prove that any f that vanishes at 0 can be written as $f(x) = xg(x)$ for g smooth, it is a variation of Hadamard's lemma), and $E = \bigcap_{n=1}^{\infty} \mathfrak{a}^n$ is the set of all $f \in R$ such that all derivatives vanish at the origin. On the other hand, f is annihilated by some element $1 + \alpha$ for $\alpha \in \mathfrak{a}$ iff f vanishes identically in some neighborhood of 0. Now, the function e^{-1/x^2} has vanishing derivative at 0, but is not identically zero near 0. Hence the kernels of the $R \to \hat{R}$ and $\mathbb{R} \to S^{-1}R$ with $S = 1 + \mathfrak{a}$ do not coincide.

This can also be a way of showing that R cannot be Noetherian.

Corollary 3.4.11: Krull Theorem, Noetherian Domain

Let R be a Noetherian domain and $(1) \neq \mathfrak{a} \subseteq R$ an ideal. Then

$$\bigcap \mathfrak{a}^n = 0$$

***Proof*:**
By Krull's theorem (theorem 3.4.8), $\bigcap \mathfrak{a}^n$ consists of the x annihilated by some $1+\alpha$ with $\alpha \in \mathfrak{a}$. Since $\mathfrak{a}$ is proper, $-1 \notin \mathfrak{a}$, so $1+\alpha \neq 0$; as R is a domain, $(1+\alpha)x = 0$ forces $x = 0$, whence $\bigcap \mathfrak{a}^n = 0$.

Corollary 3.4.12: Krull Theorem, Hausdorff Topology

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal contained in the Jacobson radical, and M a finitely generated R-module. Then the $\mathfrak{a}$-topology of M is Hausdroff, i.e.

$$\bigcap \mathfrak{a}^n M = 0$$

***Proof*:**
Since $\mathfrak{a} \subseteq \mathrm{Jac}(R)$, every element of $1+\mathfrak{a}$ is a unit (proposition 1.4.26). By Krull's theorem (theorem 3.4.8), each element of $\bigcap \mathfrak{a}^n M$ is annihilated by such a unit, hence is 0.

Corollary 3.4.13: Krull Theorem, Kernel Into Localization

Let R be a Noetherian ring, $\mathfrak{p} \subseteq R$ a prime ideal. Then the intersection of all $\mathfrak{p}$-primary ideals of A is the kernel of $A \to A_\mathfrak{p}$

***Proof*:**
Write $K = \ker(A \to A_\mathfrak{p}) = \{a \in A \mid sa = 0 \text{ for some } s \notin \mathfrak{p}\}$. Every $\mathfrak{p}$-primary ideal $\mathfrak{q}$ contains K: if $sa = 0 \in \mathfrak{q}$ with $s \notin \mathfrak{p} = \sqrt{\mathfrak{q}}$, then $a \in \mathfrak{q}$; hence $K \subseteq \bigcap\{\mathfrak{p}\text{-primary}\}$. Conversely the symbolic powers $\mathfrak{p}^{(n)} = (\mathfrak{p}^n A_\mathfrak{p}) \cap A$ are $\mathfrak{p}$-primary, and since $\mathfrak{p}A_\mathfrak{p}$ lies in the Jacobson radical of $A_\mathfrak{p}$, corollary 3.4.12 gives $\bigcap_n \mathfrak{p}^n A_\mathfrak{p} = 0$, whence $\bigcap_n \mathfrak{p}^{(n)} = K$. Thus $\bigcap\{\mathfrak{p}\text{-primary}\} \subseteq \bigcap_n \mathfrak{p}^{(n)} = K$, and the two inclusions give equality.

3.5 Associated Graded Rings: Noetherian Closed under Completion

The following will let us show another way of characterizing complete rings, and let us prove that the completion of a Noetherian ring is Noetherian.

Definition 3.5.1: Associated Graded Ring

Let R be a ring and $\mathfrak{a} \subseteq R$ an ideal. Then the *associated graded ring* is

$$G_{\mathfrak{a}}(R) = \bigoplus_{n=0}^{\infty} \mathfrak{a}^n/\mathfrak{a}^{n+1}$$

where $\mathfrak{a}^0 = R$. If unambiguous, $G(R)$ will be written instead of $G_{\mathfrak{a}}(R)$. For an R-module M with an $\mathfrak{a}$-filtration (M_n), the graded $G(R)$-module is given by:

$$G(M) = \bigoplus_{n=0}^{\infty} M_n/M_{n+1} = \bigoplus_{n=0}^{\infty} G_n(M)$$

Multiplication is certainly well-defined: for each $x_n \in \mathfrak{a}^n$, $\overline{x}_n$ is the image of x_n in $\mathfrak{a}^n/\mathfrak{a}^{n+1}$. Then $\overline{x}_m\overline{x}_m$ is $\overline{x_m x_n}$ which is the image of $x_m x_n$ in $\mathfrak{a}^{m+n}/\mathfrak{a}^{m+n+1}$ (if the reader is not convinced, check to see if this is well-defined).

Proposition 3.5.2: Properties of Associated Graded Rings

Let R be a Noetherian ring and $\mathfrak{a} \subseteq R$ an ideal. Then:

1. $G_{\mathfrak{a}}(R)$ is Noetherian
2. $G_{\mathfrak{a}}(R)$ is isomorphic to $G_{\hat{\mathfrak{a}}}(\hat{R})$ as graded rings
3. If M is a finitely generated R-module and (M_n) is a stable $\mathfrak{a}$-filtration of M, then $G(M)$ is a finitely generated graded $G_{\mathfrak{a}}(R)$-module

***Proof*:**

1. $G_{\mathfrak{a}}(R)$ is generated as an algebra over the Noetherian ring $R/\mathfrak{a}$ by the images of generators of $\mathfrak{a}$ in $\mathfrak{a}/\mathfrak{a}^2$, hence is Noetherian by HBT (theorem 1.2.1).
2. By proposition 3.4.7(3), $\mathfrak{a}^n/\mathfrak{a}^{n+1} \cong \hat{\mathfrak{a}}^n/\hat{\mathfrak{a}}^{n+1}$ in each degree, and these degreewise isomorphisms respect multiplication, giving $G_{\mathfrak{a}}(R) \cong G_{\hat{\mathfrak{a}}}(\hat{R})$.
3. Pick N with $\mathfrak{a}M_n = M_{n+1}$ for $n \geq N$ (stability). Then $G(M)$ is generated over $G_{\mathfrak{a}}(R)$ by the finitely many pieces M_n/M_{n+1} with $n \leq N$, each finitely generated over $R/\mathfrak{a}$ (corollary 3.3.3); hence $G(M)$ is finitely generated over $G_{\mathfrak{a}}(R)$.

The most important case of an associated graded ring is the one where the ideal is generated by a regular sequence: there the associated graded ring is as simple as it could be, a polynomial ring, and the geometric consequence is that the normal cone of a regularly embedded subscheme is a vector bundle rather than a general cone.

Theorem 3.5.3: Associated Graded Ring of a Regular Sequence

Let A be a Noetherian ring and let $x_1, \ldots, x_d \in A$ be a regular sequence ([5, Regular Sequence], a co-requisite of this book: x_i is a nonzerodivisor on $A/(x_1, \ldots, x_{i-1})$ for each i, and $A/(x_1, \ldots, x_d) \neq 0$). Let $\mathfrak{a} = (x_1, \ldots, x_d)$. Then the $(A/\mathfrak{a})$-algebra homomorphism

$$(A/\mathfrak{a})[T_1, \ldots, T_d] \longrightarrow G_{\mathfrak{a}}(A), \qquad T_i \longmapsto \overline{x_i} \in \mathfrak{a}/\mathfrak{a}^2$$

is an isomorphism of graded rings. In particular $\mathfrak{a}^n/\mathfrak{a}^{n+1}$ is a free $A/\mathfrak{a}$-module on the monomials of degree n in the $\overline{x_i}$.

***Proof*:**
Surjectivity. The ideal $\mathfrak{a}^n$ is generated by the degree-n monomials in the x_i, so $\mathfrak{a}^n/\mathfrak{a}^{n+1}$ is generated as an $A/\mathfrak{a}$-module by their classes, which are the images of the degree-n monomials in the T_i. Surjectivity in each degree follows.

Injectivity, by induction on d. For $d = 0$ the ideal is zero and both sides are A.

Suppose $d \geq 1$ and the statement holds for shorter regular sequences. Let $F \in (A/\mathfrak{a})[T_1, \ldots, T_d]$ be homogeneous of degree n with $F(\overline{x}) \in \mathfrak{a}^{n+1}$; the claim is that all coefficients of F vanish in $A/\mathfrak{a}$. Lift the coefficients to A and write F for a lift, so $F(x_1, \ldots, x_d) \in \mathfrak{a}^{n+1}$.

Group F by its T_d-degree, $F = \sum_j T_d^j G_j(T_1, \ldots, T_{d-1})$ with G_j homogeneous of degree $n - j$. Reducing modulo x_d kills every term with $j > 0$, so $G_0(\overline{x_1}, \ldots, \overline{x_{d-1}}) \in \overline{\mathfrak{a}}^{n+1}$ in $A/(x_d)$, where $x_1, \ldots, x_{d-1}$ is again a regular sequence and generates $\overline{\mathfrak{a}}$. By the inductive hypothesis applied in $A/(x_d)$, every coefficient of G_0 lies in $\mathfrak{a} + (x_d) = \mathfrak{a}$, which is the required vanishing for the $j = 0$ part.

Hence $F - G_0 = T_d \cdot H$ for a homogeneous H of degree $n-1$, and $x_d \cdot H(x) = F(x) - G_0(x) \in \mathfrak{a}^{n+1}$. Since $G_0(x) \in \mathfrak{a}^{n+1}$ we get $x_d H(x) \in \mathfrak{a}^{n+1}$. Now $\mathfrak{a}^{n+1} = x_d\mathfrak{a}^n + \mathfrak{a}'^{\,n+1}$ where $\mathfrak{a}'$ is generated by $x_1, \ldots, x_{d-1}$; writing $x_d H(x) = x_d a + b$ with $a \in \mathfrak{a}^n$ and $b \in \mathfrak{a}'^{\,n+1}$ gives $x_d(H(x) - a) = b$. Because x_d is a nonzerodivisor modulo $\mathfrak{a}'$ and $b \in \mathfrak{a}'^{\,n+1} \subseteq \mathfrak{a}'$, this forces $H(x) - a \in \mathfrak{a}^n$, hence $H(x) \in \mathfrak{a}^n$. The inductive hypothesis in degree $n-1$, applied to H, then gives that all coefficients of H lie in $\mathfrak{a}$, completing the induction and the proof.

Lemma 3.5.4: Associated Graded Ideal and Completion Map Relation

Let $\varphi : A \to B$ be a homomorphism of filtered groups ($\varphi(A_n) \subseteq B_n$), and let $G(\varphi) : G(A) \to G(B)$, $\hat{\varphi}\hat{A} \to \hat{B}$ be the induced homomorphisms of the associated graded group and complete group respectively. Then:

1. $G(\varphi)$ is injective $\Rightarrow$ $\hat{\varphi}$ is injective
2. $G(\varphi)$ is surjective $\Rightarrow$ $\hat{\varphi}$ is surjective

***Proof*:**
The key is to look at the following commutative diagram of exact sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_n/A_{n+1} & \longrightarrow & A/A_{n+1} & \longrightarrow & A/A_n & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle G_n(\varphi)} & & \downarrow{\scriptstyle \alpha_{n+1}} & & \downarrow{\scriptstyle \alpha_n} & & \\ 0 & \longrightarrow & B_n/B_{n+1} & \longrightarrow & B/B_{n+1} & \longrightarrow & B/B_n & \longrightarrow & 0 \end{array}$$

which by the snake lemma gives us the usual:

$$0 \to \ker G_n(\varphi) \to \ker \alpha_{n+1} \to \ker \alpha_n \to \text{coker}G_n(\varphi) \to \text{coker}\alpha_{n+1} \to \text{coker}\alpha_n \to 0$$

Now, by induction on n, we see that:

1. $\ker \alpha_n = 0$
2. $\text{coker}\alpha_n = 0$

taking the limit of the homomorphisms α_n and applying proposition 3.1.3 completes the proof.

Proposition 3.5.5: Partial Converse of proposition 3.5.2(3)

Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal, M an R-module, and (M_n) an $\mathfrak{a}$-filtration of M. Suppose R is complete in the $\mathfrak{a}$-topology, M is Hausdorff in its filtration topology ($\bigcap_n M_n = 0$), and $G(M)$ is a finitely generated $G(R)$-module.

Then M is a finitely generated R-module.

***Proof*:**
As $G(M)$ is finitely generated, pick some finite set of generators. Split them up into the homogeneous components, say ζ_i (with $1 \leq i \leq \nu$) where ζ_i has degree $n(i)$, and is the image of some $x_i \in M_{n(i)}$. Let F^i be the module with the stable $\mathfrak{a}$-filtration given by $F^i_K = \mathfrak{a}^{k+n(i)}$ and let $F = \bigoplus_{i=1}^r F^i$. Then the mapping that takes the generator 1 of each F^i to x_i gives a homomorphism of filtered groups

$$\varphi : F \to M$$

Hence it induces a homomorphism of $G(A)$-modules. By construction this map is surjective. Hence by lemma 3.5.4 $\hat{\varphi}$ is surjective. Now consider the diagram:

$$\begin{array}{ccc} F & \xrightarrow{\varphi} & M \\ {\scriptstyle \alpha}\downarrow & & \downarrow{\scriptstyle \beta} \\ \hat{F} & \xrightarrow{\hat{\varphi}} & \hat{M} \end{array}$$

As F is free and $A = \hat{A}$, α is an isomorphism. Next, as M is Hausdorff, β is injective. . As $\hat{\varphi}$ is surjective, φ is surjective, and so $x_1, ..., x_r$ generate M as an A-module completing the proof.

Corollary 3.5.6: Partial Converse in Noetherian Case

Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal, M an R-module, and (M_n) an $\mathfrak{a}$-filtration of M. Suppose R is complete in the $\mathfrak{a}$-topology, M is Hausdorff in its filtration topology ($\bigcap_n M_n = 0$), and $G(M)$ is a finitely generated $G(R)$-module.

Then if $G(M)$ is a Noetherian $G(R)$-module, then M is a Noetherian R-module.

***Proof*:**
If we show that every submodule $M' \subseteq M$ is finitely generated, we're done. Let $M'_n = M' \cap M_n$. Then (M'_n) is an $\mathfrak{a}$-filtration of M', and the embedding $M'_n \to M_N$ induces an injective homomorphism:

$$M'_n/M'_{n+1} \to M_n/M_{n+1}$$

Hence we gt a embedding of $G(M')$ in $G(M)$, $G(M)$ is Noetherian, $G(M')$ is finitely generated, and so M' is Hausdroff. As $\cap M'_n \subseteq \cap M_N = 0$, we get that M' is finitely generated, as we sought to show.

Theorem 3.5.7: Noetherian Closed Under Completion

Let R be a Noetherian ring and $\mathfrak{a} \subseteq R$ an ideal. Then the $\mathfrak{a}$-completion $\hat{R}$ of R is Noetherian

***Proof*:**
As:

$$G_{\mathfrak{a}}(R) = G_{\hat{\mathfrak{a}}}(\hat{R})$$

is Noetherian, apply corollary 3.5.6 to the complete ring $\hat{R}$ and taking $M = \hat{R}$.

Corollary 3.5.8: Power Series Ring is Noetherian

Let R be a Noetherian ring, and let $S = R[[x_1, ..., x_n]]$ be the power series ring in n variables. Then S is Noetherian.

***Proof*:**
The ring $R[x_1, ..., x_n]$ is Noetherian by Hilbert's Basis Theorem, and S is the completion for the $(x_1, ..., x_n)$-adic topology.

Completion also leaves the dimension of a Noetherian local ring unchanged, which is how dimension is often computed by passing to the simpler complete ring.

Corollary 3.5.9: Completion Preserves Dimension

Let $(R, \mathfrak{m})$ be a Noetherian local ring with $\mathfrak{m}$-adic completion $\hat{R}$. Then $\dim \hat{R} = \dim R$.

***Proof*:**
By proposition 3.5.2(2), $G_{\mathfrak{m}}(R) \cong G_{\hat{\mathfrak{m}}}(\hat{R})$ as graded rings, and corollary 3.1.5 gives $R/\mathfrak{m}^n \cong \hat{R}/\hat{\mathfrak{m}}^n$; hence the Hilbert-Samuel functions $n \mapsto \ell(R/\mathfrak{m}^n)$ and $n \mapsto \ell(\hat{R}/\hat{\mathfrak{m}}^n)$ coincide. Since the Krull dimension of a Noetherian local ring is the degree of its Hilbert-Samuel polynomial (chapter 7), $\dim R = \dim \hat{R}$.

3.6 The Cohen Structure Theorem

Completion gives the local structure of a ring coordinates: a complete Noetherian local ring containing its residue field is a quotient of a power series ring, the algebraic form of a formal coordinate chart around a point.

Theorem 3.6.1: Cohen Structure Theorem, Equicharacteristic Case

Let $(R, \mathfrak{m})$ be a complete Noetherian local ring containing a field k with $R/\mathfrak{m} = k$. Then $R \cong k[[x_1, \ldots, x_n]]/I$ for an ideal I, where $n = \dim_k \mathfrak{m}/\mathfrak{m}^2$ is the minimal number of generators of $\mathfrak{m}$.

***Proof*:**
Choose $y_1, \ldots, y_n \in \mathfrak{m}$ whose residues form a k-basis of $\mathfrak{m}/\mathfrak{m}^2$; by Nakayama's lemma (lemma 1.4.27) they generate $\mathfrak{m}$. The composite $k \hookrightarrow R \twoheadrightarrow R/\mathfrak{m} = k$ is the identity, making R a k-algebra, so we may define

$$\varphi\colon k[[x_1, \ldots, x_n]] \to R, \qquad \sum_{\alpha} c_\alpha x^\alpha \mapsto \sum_{\alpha} c_\alpha y^\alpha,$$

the sum on the right converging because $y^\alpha \in \mathfrak{m}^{|\alpha|}$ and R is $\mathfrak{m}$-adically complete (definition 3.1.9); φ is a k-algebra homomorphism. For surjectivity, reduce modulo $\mathfrak{m}^m$. Each graded piece $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ is spanned over k by the degree-j monomials in the residues $\bar{y}_i$, since multiplication gives a surjection $\mathrm{Sym}^j(\mathfrak{m}/\mathfrak{m}^2) \twoheadrightarrow \mathfrak{m}^j/\mathfrak{m}^{j+1}$ and $\mathfrak{m}/\mathfrak{m}^2$ is spanned by the $\bar{y}_i$. Hence $R/\mathfrak{m}^m$, whose composition factors are the $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ for $j < m$, is spanned over k by monomials in the y_i of degree $< m$, all in the image of φ; so φ is surjective modulo every $\mathfrak{m}^m$. As $R = \varprojlim R/\mathfrak{m}^m$ and $k[[x_1, \ldots, x_n]]$ is complete, φ is surjective. Thus $R \cong k[[x_1, \ldots, x_n]]/I$ with $I = \ker \varphi$, and $n = \dim_k \mathfrak{m}/\mathfrak{m}^2$ is minimal by the choice of the y_i.

The hypothesis $R/\mathfrak{m} = k$ is the equicharacteristic case with trivial residue extension, the form deformation theory over a field uses. In general Cohen's theorem first produces a *coefficient field* $k \hookrightarrow R$ lifting the residue field (equicharacteristic case) or, in mixed characteristic, a complete discrete valuation coefficient ring; that mixed-characteristic theory, through Witt vectors, is developed in [6].

3.7 Weierstrass Preparation and Division

The Cohen structure theorem realizes a complete local ring as a quotient of a power series ring; the *Weierstrass* theorems describe the internal structure of the power series ring itself, showing that division by a sufficiently generic series behaves like polynomial division. Fix a complete local ring $(R, \mathfrak{m})$ (for instance $R = k[[x_1, \dots, x_{n-1}]]$) and one further variable x.

Definition 3.7.1: Regular Power Series

A power series $f = \sum_{i \geq 0} a_i x^i \in R[[x]]$ is *regular of order* d in x if $a_0, \dots, a_{d-1} \in \mathfrak{m}$ and a_d is a unit; equivalently, its image in $(R/\mathfrak{m})[[x]]$ is x^d times a unit.

Theorem 3.7.2: Weierstrass Division Theorem

Let $(R, \mathfrak{m})$ be complete and $f \in R[[x]]$ regular of order d in x. Then every $g \in R[[x]]$ can be written uniquely as

$$g = qf + r, \qquad q \in R[[x]], \quad r \in R[x], \ \deg r < d$$

Theorem 3.7.3: Weierstrass Preparation Theorem

Let $(R, \mathfrak{m})$ be complete and $f \in R[[x]]$ regular of order d in x. Then $f = u\,\omega$ uniquely, where $u \in R[[x]]$ is a unit and

$$\omega = x^d + b_{d-1}x^{d-1} + \cdots + b_0, \qquad b_i \in \mathfrak{m},$$

is a monic *Weierstrass polynomial.*

***Proof*:**
Preparation follows from division: apply theorem 3.7.2 to $g = x^d$, writing $x^d = qf + r$ with $\deg r < d$; reducing modulo $\mathfrak{m}$ shows q is a unit and $\omega := x^d - r$ is a Weierstrass polynomial, so $f = u\omega$ with $u = q^{-1}$. Division itself is proved by successive approximation: writing $f = P + x^d U$ with $P = \sum_{i<d} a_i x^i \in \mathfrak{m}[x]$ and $U = a_d + a_{d+1}x + \cdots$ a unit, the operator returning the low-degree remainder of $g - (\text{quotient part})f$ is a contraction for the $\mathfrak{m}$-adic topology, and completeness of R makes the resulting series for q and r converge; uniqueness holds because a Weierstrass polynomial is a non-zero-divisor. See Zariski-Samuel or Eisenbud, Ch. 7.

Iterating in $x_1, \dots, x_n$ pins down the local structure of $k[[x_1, \dots, x_n]]$: it is a regular local ring of dimension n in which, after a generic linear change of coordinates, ideals are generated by Weierstrass polynomials, the algebraic counterpart of resolving a formal germ into a finite branched cover of a coordinate subspace. This underlies formal and analytic local geometry, intersection multiplicities, and Puiseux expansions used downstream.

4

Ring Extension and Integral Extension

In this section, we generalize the notion of extension from fields to commutative rings. Just like how $k \subseteq F$ can be seen as a field extension, if we let k and R be rings instead, we will et the accompying notion of ring extension:

4.1 Ring and Integral Extensions

Definition 4.1.1: Ring Extension

Let S be a commutative ring and $R \subseteq S$ a subring. Then we say that S is a *ring extension* of R. Equivalently, if there exists an injective ring homomorphism $\varphi : R \to S$, then S is a ring extension of R.

Unlike fields, we must specify the homomorphism to be injective, since ring homomorphisms need not be injective. The notation S/R is also used just as in field, though more scarcely. Just like how any field extension F/k makes F into a k-vector-space and we have special k-homomorphisms letting k-commute between the homomorphism, a ring extension S/R have an analogue: in fact, a k-homomorphism is *precisely* a k-algebra homomorphism!

Recall that any *R-algebra* can be defined as a ring homomorphism $\varphi : R \to S$, *provided* that $\varphi(R) \subseteq Z(S)$. Since we are working with commutative rings, this is *always* the case, and so we can always thing of a ring extension S over R is an R-algebra! This is why we focus on *commutative rings*. There is a similar notion of ring extension for non-commutative which is closer to the notion of group extension (i.e. a short exact sequence extending the object), however ring extensions don't break down into nice building blocks since kernels of rings are not rings, which means we would have to extend a ring R by an abelian group (i.e. two-sided ideal) I. This very much changes how we would study rings. However, we can always study the structure of any commutative ring S by how behaves as an R-algebra similarly to how we studied fields, which is what we'll do in these chapters.

4.1.2 Advanced Intuition Integral extensions shall correspond to *proper maps* of morphisms. Non-integral extensions shall never be proper, thus geometrically exhibiting greatly different behaviour. In particular, taking base change if necessary, the image of closed sets (essentially zero-sets of polynomials) need not be closed.

Perhaps the first notion we might want to genelize is the equivalent of an *algebraic element*, since they capture the relations within a ring extension with respect to the base ring. The following is the exactly this generlization:

Definition 4.1.3: Integral Elements

Let $R \subseteq S$.

1. An element $s \in S$ is called an *integral over* R if s is the root of a monic polynomial over $R[x]$:
$$s^n + r_{n-1}s^{n-1} + \cdots + r_1 s + r_0 = 0 \qquad r_i \in R$$
2. S is called *integral over* R or an *integral extension of* R if every element of S is integral over R. The integral closure of an integral domain is also called the *normalization of* R. Sometimes, the integral extension will be written as S/R
3. The set $\overline{R}^S$ containing all the elements of S that are integral over R is called the *integral closure of* R
4. If $R = \overline{R}^S$, then R is said to be *integrally closed over* S.
5. If R is an integral domain that is integrally closed in it's field of fractions F, $\overline{R}^F$, then R is called *normal domain.*

The fact that the polynomial it satisfies has to be monic can be thought of as not wanting to add relations that produce "units" or unlit-like elements (ex. $3x - 1 = 0$ implies there is a new element that behaves like $1/3$). For a more specific motivation, we can think of the following: If $f(x) \in R[x]$ is a polynomial of minimal degree with root r, and $h(x) \in R[x]$ is another monic polynomials with minimal degree with root r, then we would want that $h(x)|f(x)$, just like for minimal polynomials over fields. In the process of doing so, we would require the coefficient of the leading terms to divide. Since they can be arbitrary, we would require the coefficients of these polynomial to be units, or just 1, in order for the division to follow through. It follows that it is a natural condition that all polynomials be monic for an integral extension.

Another motivation comes from Gauss's lemma: over a UFD R with fraction field K, a primitive polynomial of $R[x]$ is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$. Monic polynomials are primitive, so factorization of monic polynomials over R mirrors factorization over K; this parallel between $R[x]$ and $K[x]$ becomes important in [3].

Integral elements share many of the same properties as algebraic elements:

Proposition 4.1.4: Integral Element Equivalences

Let $R \subseteq S$ be a subring of S. Then the following are equivalent:

1. s is integral over R
2. $R[s]$ is a finitely generated R-module
3. The element s is in T for some subring T where $R \subseteq T \subseteq S$, that is a finitely generated R-module

***Proof*:**

$1 \Rightarrow 2$ Recall that $R[s]$ can be expressed as all R-linear combinations of powers of s. Since s is integral over R:

$$s^n + r_n s^{n-1} + \cdots + r_1 s + r_0 = 0$$

in particular:

$$s^n = -(r_n s^{n-1} + \cdots + r_1 s + r_0)$$

It follows that for all $m \geq n$, s^m can be expressed in terms of $1, s, \ldots, s^{n-1}$ (for the same reasons as we've shown for algebraic elements). Hence:

$$R[s] = R + Rs + \cdots + Rs^{n-1}$$

Notice here how if the polynomial was not monic, we might not have been able to isolae s^n, causing complications.

$2 \Rightarrow 3$ Let $T = R[s]$. Then the reuslt immediately follows

$3 \Rightarrow 1$ We take advantage of rational canonical forms and Cramer's rule (remember that if R is an integarl domain, we can take it's field of fractions, and the resulting matri multiplication remains the same). Since T is finitely generated, let $v_1, v_2, ..., v_n$ be a finite generating set. Since $s \in T$ and T is a ring, $sv_i \in T$ for all $1 \leq i \leq n$. Thus:

$$sv_i = \sum_j a_{ij} v_j$$

Giving us the equation n:

$$\sum_j (\delta_{ij} s - a_{ij}) v_j = 0 \qquad 1 \leq i \leq n$$

where δ_{ij} is one when $i = j$ and 0 otherwise (i.e. the Kronecker delta). Letting B be the matrix such that the entries are $B_{ij} = \delta_{ij} s - a_{ij}$ and letting v be the $n \times 1$ matrix whose entries are $v_1, ..., v_n$, we get:

$$Bv = 0$$

Thus, by Cramer's rule:

$$(\det B) v_i = 0 \qquad 1 \leq i \leq n$$

Now, since 1 is an R-linear combination of $v_1, v_2, ..., v_n$,

$$\det B = (\det B)1 = (\det B)\sum_i a_i v_i = \sum a_i((\det B)v_i) = 0$$

i.e. $\det B = 0$. Now, B is by definition $B = sI - A$ (where $A_{ij} = a_{ij}$). Thus, s is the root of a monic polynomial $\det(xI - A) \in R[x]$, namely the characteristic polynomial of A, and so is the root of a monic polynomial with coefficients in R, giving (1)

Corollary 4.1.5: Transitivity of Integrality

Let $R \subseteq S \subseteq T$ be rings. If T is integral over S and S is integral over R, then T is integral over R.

***Proof*:**
Let $t \in T$, a root of a monic $x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in S[x]$. Each a_i is integral over R, so in the tower $R \subseteq R[a_0] \subseteq \cdots \subseteq R[a_0, \ldots, a_{n-1}] =: R'$ every step is module-finite (proposition 4.1.4), hence R' is a finitely generated R-module. As t is integral over R', $R'[t]$ is module-finite over R', hence over R; by proposition 4.1.4 (applied to the subring $R'[t]$), t is integral over R.

Corollary 4.1.6: Integral Elements Form Rings

Let $R \subseteq S$ be a subring of S, and $s, t \in S$. Then:

1. If s and t are integral over R, so is $s \pm t$ and st
2. $\overline{R}^S$ is a subring of S containing R, and is integrally closed[a]

[a]If we were working over rng's, $\overline{R}^S$ need not be integrally closed

***Proof*:**
Let s, t be integral over R, sy that by proposition 4.1.4 $R[s]$ and $R[t]$ are finitely generated R-modules, say:

$$R[s] = Rs_1 + \cdots + Rs_n$$
$$R[t] = Rt_1 + \cdots + Rt_m$$

Then:

$$R[s,t] = Rs_1t_1 + \cdots + Rs_it_j + \cdots + Rs_nt_m$$

is a ring containing $s \pm t$ and st that is also a finitely generated R-module, so $s \pm t$ and st are integral over R. Thus $\overline{R}^S$ is a subring of S containing R; it is integrally closed in S, since any $s \in S$ integral over $\overline{R}^S$ is integral over R by transitivity (corollary 4.1.5), hence lies in $\overline{R}^S$.

In particular, taking the integral closure is idempotent: $\overline{(\overline{R}^S)}^S = \overline{R}^S$, since integrality is transitive (corollary 4.1.5).

Note that integral closure is dependent on the ring over which we are closing; we do not have the equivalent notion of "integral closure of R" without specifying a ring over which it is integrally closed (Why?)

Corollary 4.1.7: Module-Finite versus Finite-Type-and-Integral

Let S be an R-algebra. Then S is a finitely generated R-module if and only if S is of finite type over R and integral over R.

Notice how if S is not integral, then it is possible that S is of finite type over R but not finitely generated over R, the most cannonical example being $R[x]$ over R (since $R[x]$ is finitely generated as an R-algebra by 1 and x, but $1, x, x^2, \ldots$ is an infinite set that is a basis for $R[x]$)

***Proof*:**
If S is a finitely generated R-module it is of finite type, and every element is integral (proposition 4.1.4). Conversely, let $S = R[s_1, \ldots, s_n]$ be of finite type with each s_i integral over R. In the tower

$$R \subseteq R[s_1] \subseteq R[s_1, s_2] \subseteq \cdots \subseteq R[s_1, \ldots, s_n] = S$$

each step $R[s_1, \ldots, s_{i-1}][s_i]$ is module-finite, since s_i is integral over the smaller ring (proposition 4.1.4); a module-finite extension of a module-finite extension is module-finite, so S is a finitely generated R-module.

Before giving examples, we'll show that we have been studying special types of integral closures when we worked with field exensions in **??**:

Proposition 4.1.8: Integral Extension over Fields

Let S/R be an integral extension where S is an integral domain. Then R is a field if and only if S is a field

Note that if S was not an integral domain or commutative, this theorem does not work: Take $M_2(k)$ over any field. Then $M_2(k)$ is a 4-dimensional k-algebar, but certianly not a field (it contains zero-divisors, and is not commutative).

***Proof*:**

($\Rightarrow$) Let R be a field and take $0 \neq s \in S$. Since s is integral over R, we have:

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0 \qquad a_i \in R$$

Without loss of generality, let $a_0 \neq 0$; otherwise, factor an s, and using the fact that S is an integral domain, we see that either $s = 0$ or $s^{n-1} + \cdots + a_1 = 0$, and since $s \neq 0$, then we simply choose $s^{n-1} + \cdots + a_1 = 0$ to be our equation. Thus, moving a_0 and factoring s we get:

$$s(s^{n-1} + \cdots + a_1) = -a_0$$

Since R is a field, $\frac{-1}{a_0} \in R$, so:

$$s \cdot \big((-1/a_0)(s^{n-1} + \cdots + a_1)\big) = 1$$

showing that s has a multiplictative inverse. Since s was arbitrary, we have that S is a field.

($\Leftarrow$) Let S be a field and take $0 \neq r \in R$ (automatically implying that R is an integral domain). Since $r \in R$, $r \in S$, and so $r^{-1} \in S$. Since S is an integral extension of R:

$$r^{-n} + a_{n-1}r^{-(n-1)} + \cdots + a_1 r^{-1} + a_0 = 0$$

Multiply the entire expression by r^{n-1}, and move all the terms except the r^{-1} terms to the right hand side to get:

$$r^{-1} = -(a_{n-1} + \cdots + a_1 r^{n-2} + a_0 r^{n-1})$$

which is a linear combination of elements in R, showing that $r^{-1} \in R$, and so R is a field.

Thus, any algebraic field extension is an integral extension.

Example 4.1.9: Integral Extensions

1. As we've just shown, all algebraic extensions of fields are integral extensions.

2. Let R be a Unique Factorization Domain and let $F = \text{Frac}(R)$. Then $\overline{R}^F = R$. The $\supseteq$ direction is obvious, so consider the $\subseteq$ direciton: suppose that $a/b \in \overline{R}^F$ with $\gcd(a, b) = 1$. Then by definition
$$\left(\frac{a}{b}\right)^n + r_{n-1}\left(\frac{a}{b}\right)^{n-1} + \cdots + r_1\frac{a}{b} + r_0 = 0$$
Then:
$$a^n = b(-r_{n-1}a^{n-1} - \cdots - r_1 ab^{n-2} - r_0 b^{n-1})$$
so $b \big| a^n$, and hence $b \big| a$. Since $\gcd(a, b) = 1$, it must thus be that b is a unit, and hence we have that $a/b \in R$, showing that $\overline{R}^F = R$. Hence, all UFD's are normal domains. This should make intuitive sense, since any element $a/b \in \text{Frac}(R)$ would satisfy the equation $bx - a = 0$, which is not-monic.

   More generally, any polynomial ring over a UFD R, $R[x_1, ..., x_n]$ is integrally closed over it's field of fractions $R(x_1, ..., x_n)$ ([8]), showing every polynomial ring over an Unique Factorization Domain is a normal domain.

3. Let S be integral over R and $I \subseteq S$ be an ideal. Then S/I is integral over $R/(R \cap I)$. To see this, notice that a monic polynomial over R where $s \in S$ is a root, when reduced modulo I gives a monic polynomial over $R/(R \cap I)$ where $\overline{s} \in S/I$ is a root.

4. Let k be any field and take $k[x, y]$ (which is integrally closd since it's a UFD). Take the prime ideal $(x^2 - y^3) \subseteq k[x, y]$ (see Dummit and Foote, §9.1, Exercise 14). Consider the integral domain
$$R = k[x, y]/(x^2 - y^3) = k[\overline{x}, \overline{y}]$$
Notice that R is *not* normal: the element $\overline{x}/\overline{y}$ is the solution to the equation
$$\left(\frac{\overline{x}}{\overline{y}}\right)^3 - \overline{x} = 0$$

but evidently $\overline{x}/\overline{y} \notin R$. Thus, R cannot be a Unique Factorization Domain.

If you wish to practice, another example is $\mathbb{Z}[\sqrt{-3}]$. It's integral closure over it's field of fraction's is:

$$\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$$

Corollary 4.1.10: Normalization Commutes with Localization

Let R be a domain with fraction field K, $D \subseteq R$ a multiplicative set, and $\overline{R}$ the integral closure of R in K. Then, computing integral closures inside K, $\overline{D^{-1}R} = D^{-1}\overline{R}$.

***Proof*:**
If $x \in \overline{R}$ satisfies $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ with $a_i \in R$, the same relation holds over $D^{-1}R$, so $D^{-1}\overline{R} \subseteq \overline{D^{-1}R}$. Conversely, if $y \in K$ is integral over $D^{-1}R$, then clearing the denominators in its integral relation shows sy is integral over R for some $s \in D$, so $sy \in \overline{R}$ and $y = (sy)/s \in D^{-1}\overline{R}$. In particular R is normal iff $R_{\mathfrak{m}}$ is normal for every maximal ideal $\mathfrak{m}$.

Locality shows where to compute an integral closure but not whether the answer is manageable. The integral closure of a Noetherian domain in its fraction field can fail to be a finite module, and can even fail to be Noetherian, so finiteness is a genuine theorem and not a formality. For the rings that geometry actually produces, finitely generated algebras over a field, it holds; this is what makes the normalization of a variety a variety again rather than an unwieldy limit.

Theorem 4.1.11: Finiteness of Integral Closure

Let k be a field, let A be a domain that is finitely generated as a k-algebra, and let L be a finite field extension of $\mathrm{Frac}(A)$. Then the integral closure of A in L is a finite A-module. In particular it is a Noetherian ring and is finitely generated as a k-algebra, and taking $L = \mathrm{Frac}(A)$, the normalization $\overline{A}$ of A is a finite A-module.

***Proof*:**
The argument is given in full for k perfect, which is the case every consumer in this series needs; the imperfect case is discussed after the proof.

Step 1: reduce to a polynomial ring. By Noether normalization (theorem 7.1.1) there is a polynomial subalgebra $B = k[x_1, \ldots, x_d] \subseteq A$ with A finite, hence integral, over B. An element of L integral over A is then integral over B by transitivity of integral dependence, and conversely $B \subseteq A$, so the integral closures of A and of B in L coincide; it suffices to prove the theorem for B. Note that $\mathrm{Frac}(B) \subseteq \mathrm{Frac}(A) \subseteq L$ with both extensions finite, so $L/\mathrm{Frac}(B)$ is finite. The ring B is a unique factorization domain and hence normal (example 4.1.9).

Step 2: arrange separability. Because k is perfect, the finitely generated field extension L/k is separably generated: some subset of any finite generating set is a transcendence basis $x_1, \ldots, x_d$ for which L is a finite separable extension of $k(x_1, \ldots, x_d)$. This is proved in *Core Algebra* [8]. Choosing the Noether normalization of Step 1 compatibly with such a basis, we may assume

$L/\mathrm{Frac}(B)$ is separable.

Step 3: the trace form pins the integral closure inside a free module. Let C denote the integral closure of B in L and put $F = \mathrm{Frac}(B)$. Choose an F-basis $e_1, \ldots, e_n$ of L consisting of elements integral over B; this is possible because any F-basis can be scaled into C, an element of L being algebraic over F and hence integral after multiplication by a suitable element of B clearing denominators in its minimal polynomial. Since L/F is separable, the trace form is nondegenerate and there is a dual basis $e_1^*, \ldots, e_n^*$ of L over F with $\mathrm{tr}_{L/F}(e_i e_j^*) = \delta_{ij}$, by *Core Algebra* [8].

Let $c \in C$ and write $c = \sum_j \lambda_j e_j^*$ with $\lambda_j \in F$. Then $\lambda_j = \mathrm{tr}_{L/F}(c\,e_j)$. Now $c\,e_j$ is a product of elements integral over B, hence integral over B, and the trace of an element integral over B lies in B: the trace is, up to sign, a coefficient of the characteristic polynomial of multiplication by that element, which is a power of its minimal polynomial, and the minimal polynomial of an integral element has coefficients in B because B is integrally closed in F. So $\lambda_j \in B$ for every j, and

$$C \;\subseteq\; Be_1^* + \cdots + Be_n^*$$

Step 4: conclude. The right-hand side is a free B-module of rank n, and B is Noetherian by Hilbert's basis theorem, so its submodule C is a finite B-module. Since B is a finitely generated k-algebra, C is a finitely generated k-algebra and a Noetherian ring, as we sought to show.

4.1.12 Note: The Imperfect Case Step 2 is the only place perfectness is used, and it cannot be avoided there: over an imperfect field a finitely generated extension need not be separably generated (*Core Algebra* [8] records a counterexample), so the trace form can degenerate and the argument above stops. The theorem is nonetheless true over any field. The standard route replaces k by a finite purely inseparable extension over which the situation becomes separably generated, and then descends; see Matsumura [10], or [11, Tag 0335], where the result appears in the form that a finite type algebra over a field is a Nagata ring, hence Japanese, which is exactly the finiteness asserted here. No book in this series currently consumes the imperfect case.

4.2 Extension and Contraction of Ideals

Probably the biggest difference between fields and rings is that ideals are nontrivial. Thus, it is meaningful to ask what happens to ideals in (general) ring extensions. Recall that if $f : R \to S$ is a ring homomorphism, and $J \subseteq S$ is an ideal, $f^{-1}(J)$ is an ideal. However, if $I \subseteq R$ is an ideal, $f(I)$ need not be an ideal (take $\mathbb{Z} \hookrightarrow \mathbb{Q}$ with any non-trivial ideal of $\mathbb{Z}$). The next best thing would be to define the smallest ideal of S containing $f(I)$:

Definition 4.2.1: Extension and Contraction

Let $\varphi : R \to S$ be a homomorphism between commutative rings and let $I \subseteq R$, $J \subseteq S$.

1. The *extension* of I to S is the ideal $\varphi(I)S$, commonly denoted I^e
2. The *contraction* of J in R is the ideal $\varphi^{-1}(J)$, commonly denoted J^c

In the case where φ is the natural inclusion $\iota : R \to S$, then the extension of I is IS, and the contraction of J is $J \cap R$. If $I \subseteq R$, there are multiple extension of R, say S/R and T/R, then the extension of I is sometiems denoted I_S or I_T to emphasize to what ring I is being extended too S or T respectively.

Proposition 4.2.2: Properties of Extensions and Contractions

1. $I \subseteq IS \cap R$, or more generally that I is contained in the contraction of its extension to S ($I \subseteq I^{ec}$)
2. $(J \cap R)S \subseteq J$, or more genrally that J is contained in the extension of its contraction in R ($J \supseteq J^{ce}$)
3. $I^e = I^{ece}$
4. $J^c = J^{cec}$
5. If J is prime, J^c is prime. If I is prime, I^e need not be prime (consider $\mathbb{Z} \hookrightarrow \mathbb{Q}$). If $\varphi : R \to S$ is surjective and I is a prime ideal containing $\ker \varphi$, then I^e is prime.
6. If J is maximal, J^c need not be maximal, and similarly if I is maximal then I^e need not be maximal.

***Proof*:**
(1) holds since $I \subseteq \varphi^{-1}(\varphi(I)S) = I^{ec}$, and (2) since $I^{ce} = \varphi(\varphi^{-1}(J))S \subseteq J$. Applying (1) to I^e gives $I^e \subseteq I^{ece}$, and applying $(-)^e$ to (2) gives $I^{ece} \subseteq I^e$, so (3) holds; (4) is dual. For (5): $J^c = \varphi^{-1}(J)$ is prime as a preimage of a prime; over $\mathbb{Z} \hookrightarrow \mathbb{Q}$ the nonzero prime (p) has $(p)^e = \mathbb{Q}$, not a proper ideal; and if φ is surjective with prime $I \supseteq \ker \varphi$, then $I^e = \varphi(I)$ and $S/I^e \cong R/I$ is a domain. (6) is $\mathbb{Z} \hookrightarrow \mathbb{Q}$ again, where (0) is maximal in $\mathbb{Q}$ but not in $\mathbb{Z}$.

Lemma 4.2.3: Extension And Contraction With Common Operations

Let $\mathfrak{a}_1, \mathfrak{a}_2 \subseteq A$ be ideals of A and $\mathfrak{b}_1, \mathfrak{b}_2 \subseteq B$ be ideals of B. Then:

$$\begin{aligned}
(\mathfrak{a}_1 + \mathfrak{a}_2)^e &= \mathfrak{a}_1^e + \mathfrak{a}_2^e & (\mathfrak{b}_1 + \mathfrak{b}_2)^c &\supseteq \mathfrak{b}_1^c + \mathfrak{b}_2^c \\
(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e &\subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e & (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c &= \mathfrak{b}_1^c \cap \mathfrak{b}_2^c \\
(\mathfrak{a}_1 \mathfrak{a}_2)^e &= \mathfrak{a}_1^e \mathfrak{a}_2^e & (\mathfrak{b}_1 \mathfrak{b}_2)^c &\supseteq \mathfrak{b}_1^c \mathfrak{b}_2^c \\
(\mathfrak{a}_1 : \mathfrak{a}_2)^e &\subseteq (\mathfrak{a}_1^e : \mathfrak{a}_2^e) & (\mathfrak{b}_1 : \mathfrak{b}_2)^c &\subseteq (\mathfrak{b}_1^c : \mathfrak{b}_2^c) \\
\sqrt{\mathfrak{a}}^e &\subseteq \sqrt{\mathfrak{a}^e} & \sqrt{\mathfrak{b}}^c &= \sqrt{\mathfrak{b}^c}
\end{aligned}$$

***Proof*:**
Each identity is routine from the definitions of extension and contraction (Atiyah-Macdonald, Chapter 1); the inclusions that are not equalities are strict already for $\mathbb{Z} \hookrightarrow \mathbb{Q}$.

If $\varphi : R \to S$ is a ring homomorphism, we may decompose φ into the following:

$$R \xrightarrow{p} \varphi(R) \xhookrightarrow{j} S$$

where p maps R surjectively onto $\varphi(R)$, and j injects into S. Relating the ideals of R to the ideals of $\varphi(R)$ is very easy:

Proposition 4.2.4: Correspondence Between Ideals And Image Of Ideals

Let $\varphi : R \to S$ be a ring homomorphism. Then there exists a bijective correspondence:

$$\{I \subseteq \varphi(R)\} \underset{\varphi(-)}{\overset{(-)^c}{\rightleftarrows}} \{\ker(\varphi) \subseteq J \subseteq R\}$$

Furthermore, prime ideals of $\varphi(R)$ correspond to prime ideals of R containing $\ker(\varphi)$.

***Proof*:**
Since $\varphi(R) \cong R/\ker\varphi$, this is the correspondence theorem for the quotient $R \twoheadrightarrow R/\ker\varphi$: ideals of $R/\ker\varphi$ correspond to ideals of R containing $\ker\varphi$ via image and preimage, and primes correspond to primes.

On the other hand, the correspondence between $\varphi(R)$ and S is much harder, and is covered in section 4.3 when S/R is an integral extension. As shown in proposition 4.2.2, every prime ideal "lies over" another prime ideal. This motivates the following vocuabilary:

Definition 4.2.5: Lying Over

Let $I \subseteq S$ be an ideal and S/R an integral extension. If $J = I \cap R$, I is said to be *lying over* J.

Example 4.2.6: Extending ideals
Take $\varphi : \mathbb{Z} \to \mathbb{Z}[i]$ to be the inclusion. Then a prime ideal $(p) \subseteq \mathbb{Z}$ may or may not remain prime when extended to $\mathbb{Z}[i]$. Using Fermat's theorem of sums of squares (**??**), it can be deduced that:

1. $(2)^e = (1+i)^2$
2. If $p \equiv 1 \mod 4$, then p^e is the product of two distinct prime ideals (ex. $(5)^e = (2+i)(2-i)$)
3. if $p \equiv 3 \mod 4$, then p^e is prime in $\mathbb{Z}[i]$

In general, it is rather hard to determine when $\varphi(p)$ is prime when p is prime. If $\varphi : R \to S$ is an integarl extension, then the situation becomes much better, however it will require the notion of *localization* which shall be covered in the next section. One special case is note mentioning:

Proposition 4.2.7: Maximal Ideals Lying Over

Let S/R be an integral extension and $\mathfrak{m} \subseteq S$ a prime ideal. Then $\mathfrak{m} \cap R$ is maximal if and only if $\mathfrak{m}$ is maximal.

***Proof*:**
Consider the integral domain $S/\mathfrak{m}$. Then $S/\mathfrak{m}$ is an integral extension of $R/(\mathfrak{m} \cap R)$. Then by proposition 4.1.8, $S/\mathfrak{m}$ is a field if and only if $R/(\mathfrak{m} \cap R)$ is a field. But then $\mathfrak{m}$ is maximal if and only if $\mathfrak{m} \cap R$ is maximal.

4.3 Integral Extensions and Prime Ideals

We return to the question of the extension of ideals, in particular prime ideals, given S/R. The going-down half rests on the following, where integrality over an *ideal* I means being a root of a monic $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with all $a_i \in I$.

Lemma 4.3.1: Integral over an Ideal

Let R be a normal domain with fraction field K, let $S \supseteq R$ be a domain, and let $I \subseteq R$ be an ideal. If $x \in S$ is integral over I, then x is algebraic over K and the non-leading coefficients of its minimal polynomial over K lie in $\sqrt{I}$.

***Proof*:**
The integral relation shows x is algebraic over K; let $m(t) = t^r + c_{r-1}t^{r-1} + \cdots + c_0$ be its minimal polynomial over K. In a splitting field the conjugate roots $x = x_1, \ldots, x_r$ of m each satisfy the same relation with coefficients in I, so all x_j are integral over I; hence so are the c_i, being (up to sign) elementary symmetric functions of the x_j. Each $c_i \in K$ is integral over R, so $c_i \in R$ since R is normal; and $c_i \in R$ integral over I satisfies $c_i^m + b_{m-1}c_i^{m-1} + \cdots + b_0 = 0$ with $b_j \in I$, giving $c_i^m \in I$, i.e. $c_i \in \sqrt{I}$.

Theorem 4.3.2: Integral Extension Prime Correspondence (Cohen-Seidenberg Theorem)

Let S be an integral extension of the integral domain R

1. (**Lying-over Property**) Let $\mathfrak{p} \subseteq R$ be a prime ideal. Then there is a prime ideal $P \subseteq S$ such that $\mathfrak{p} = P \cap R$. Furthermore, $\mathfrak{p}$ is maximal if and only if P is maximal.
2. (**Going-up Property**) Let $p_1 \subseteq p_2 \subseteq \cdots \subseteq p_n$ be a chain of prime ideals of R, and suppose there exists a chain $q_1 \subseteq q_2 \subseteq \cdots \subseteq q_m$, $(m < n)$ such that $p_i = q_i \cap R$. Then there exists prime ideals $q_{m+1}, ..., q_n$ that complete the chain: $q_m \subseteq q_{m+1} \subseteq \cdots \subseteq q_n$, $p_i = q_i \cap R$ for all i.
3. (**Going-down Property**) Suppose in addition that S is a domain. If R is a normal domain, $p_1 \supseteq p_2 \supseteq \cdots \supseteq p_n$ be a chain of prime ideals of R, and suppose there exists a chain $q_1 \supseteq q_2 \supseteq \cdots \supseteq q_m$, $(m < n)$ such that $p_i = q_i \cap R$. Then there exists prime ideals $q_{m+1}, ..., q_n$ that complete the chain: $q_m \supseteq q_{m+1} \supseteq \cdots \supseteq q_n$, $p_i = q_i \cap R$ for all i.
4. (**Incomparability property**) Let q_1 and q_2 lie over p. Then either $q_1 = q_2$, or they do not contain each other: $q_1 \not\subseteq q_2$ and $q_2 \not\subseteq q_1$

***Proof*:**

1. We'll take advantage of the corresponence theorem for rings and their localization. Let $D = R - P$. Then as we saw, D is multiplicatively closed, and so $R[D^{-1}]$ is well-defined. Since D is also a multiplicatvely closed set of S, $S[D^{-1}]$ is also well-defined. Furthermore, $S[D^{-1}]/R[D^{-1}]$ is an integral extensions of $R[D^{-1}]$: for any $s \in S[D^{-1}]$, if $s \in S$, clearly S is still integral. If $t = \frac{s}{d}$ for some $s \in S$ and $d \in D$, then letting

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

This equation is still zero if we multiply by d^{-n}:

$$\frac{s^n}{d^n} + \frac{a_{n-1}s^{n-1}}{d^n} + \cdots + \frac{a_1 s}{d^n} + \frac{a_0}{d^n} = 0$$

and shuffling the coefficients we get:

$$\frac{s^n}{d^n} + \frac{da_{n-1}s^{n-1}}{d^{n-1}} + \cdots + \frac{d^{n-1}a_1 s}{d} + \frac{a_0}{d^n} = 0$$

and replacing each $d^{n-i}a_i = b_i$, and $s/d = t$,

$$t^n + b_{n-1}t^{n-1} + \cdots + b_1 t + b_0 = 0$$

showing that t is integral over $R[D^{-1}]$, and so $S[D^{-1}]$ is an integral extension of $R[D^{-1}]$. Let $m \subseteq S[D^{-1}]$ be a maximal ideal. Then by the corresponence of ideals from a ring and it's localization, we have that $m \cap R_p$ is a maximal ideal in R_p, and $m \cap R_p$ is the extension of P to the local ring R_p, so it contracts to p. It might be easier to see if we have a visual:

$$\begin{tikzcd} S \arrow[r, "\pi"] & S[D^{-1}] \\ R \arrow[u, "\iota"] \arrow[r, "\pi"] & R[D^{-1}] = R_p \arrow[u, "\iota"] \end{tikzcd}$$

The maximal ideal m is maximal in $S[D^{-1}]$, it's pre-image is a maximal ideal of $R[D^{-1}]$, and the pre-image of that map is p, showing that $p = m \cap R$, as we sought to show.

2. This can be reducd to showing that if $p_1 \subseteq p_2$ and q_1 lies above p_1 such that $p_1 = q_1 \cap R$, then there exists a q_2 that lies above p_2 ($q_2 = p_2 \cap R$) and $q_1 \subseteq q_2$. First, let $\overline{S} = S/q_1$ and $\overline{R} = R/p_1$. Then $\overline{S}/\overline{R}$ is an integral extension by similar reasoning to what we've shown in part 1. By the ideal-correspondence for the quotient $R \to R/p_1$ (proposition 4.2.4), p_2/p_1 is a prime ideal of $\overline{R}$. Then by part (1), we have that there exists an ideal $q_2/q_1 \subseteq \overline{S}$ that lies abouve p_2/p_1, in particular:

$$q_2/q_1 \cap \overline{R} = p_2/p_1$$

At this point we are essentially done: taking the pre-image of q_2/q_1, we get q_2, and by the correspondence theorem we get

$$q_2 \cap R = p_2$$

3. It suffices to find, given $p_1 \supseteq p_2$ in R and q_1 over p_1, a prime $q_2 \subseteq q_1$ over p_2. Localize S at the multiplicative set $S \setminus q_1$; it is enough that $p_2 S_{q_1} \cap R = p_2$, for then a prime of S_{q_1} over p_2 contracts to the desired $q_2 \subseteq q_1$. The inclusion $\supseteq$ is clear. For $\subseteq$, take $x \in p_2 S_{q_1} \cap R$ and write $x = y/s$ with $y \in p_2 S$, $s \in S \setminus q_1$. Now $y \in p_2 S$ is integral over p_2, so by lemma 4.3.1 its minimal polynomial over $K = \mathrm{Frac}(R)$ has non-leading coefficients in $\sqrt{p_2} = p_2$: say $y^r + u_{r-1}y^{r-1} + \cdots + u_0 = 0$ with $u_i \in p_2$. Then $s = y/x$ has minimal polynomial $s^r + v_{r-1}s^{r-1} + \cdots + v_0$ with $v_i = u_i/x^{r-i} \in R$ (as s is integral over the normal R), and $x^{r-i}v_i = u_i \in p_2$. Were $x \notin p_2$, each $v_i \in p_2$, so $s^r = -(v_{r-1}s^{r-1} + \cdots + v_0) \in p_2 S \subseteq q_1$, forcing $s \in q_1$, a contradiction. Hence $x \in p_2$, giving $p_2 S_{q_1} \cap R = p_2$.

4. For the sake of contradiction, let $q_1 \subseteq q_2$, and let both lie over p: $q_1 \cap R = q_2 \cap R = p$. Consider $\overline{S} = S/q_1$ and $\overline{R} = R/p$. Then as we've shown earlier, $\overline{S}/\overline{R}$ is an integral extension, and $q_2/q_1 \subseteq \overline{S}$ is a prime ideal. Consider:

$$(q_2/q_1) \cap \overline{R} = (q_2 \cap R)/(R \cap q_1) = (R \cap q_1)/(R \cap q_1) = \{\overline{0}\}$$

Now $\overline{S}$ is a domain integral over the domain $\overline{R}$, and the prime q_2/q_1 meets $\overline{R}$ only in $\{0\}$. If $q_2/q_1 \neq 0$, pick $0 \neq \bar{x} \in q_2/q_1$ satisfying an integral relation $\bar{x}^n + \overline{a}_{n-1}\bar{x}^{n-1} + \cdots + \overline{a}_0 = 0$ of least degree; then $\overline{a}_0 = -\bar{x}(\bar{x}^{n-1} + \cdots + \overline{a}_1) \in (q_2/q_1) \cap \overline{R} = \{0\}$, so $\overline{a}_0 = 0$, and cancelling $\bar{x}$ (valid in the domain $\overline{S}$) contradicts minimality. Hence $q_2/q_1 = \{0\}$, i.e. $q_2 = q_1$.

Exercise 4.3.1

1. Let $I \subseteq R$ be an ideal and R a ring. Then if we consider integral extension of *rngs*, show that I^{n+1}/I^n is an integral extension.

2. Let $I \subseteq R$, $f : R \to S$ an integral extension and $I^e \subseteq S$ the extension of S. Then:

$$S/I^e \cong S \otimes_R R/I$$

5

Valuation Ring and Local Rings

Recall from the theory of absolute values ([6]) that an absolute value defines a notion of size on a field. On a finite field the only absolute value is the trivial one ($|x| = 1$ for $x \neq 0$), and by Ostrowski's theorem the only absolute values on $\mathbb{Q}$ up to equivalence are the p-adic $|-|_p$ and the usual $|-|_\infty$. These were defined using a constant c^{-m}; the choice of constant does not affect the topology, so what really matters is the exponent m. This section defines a function capturing that exponent, a useful link between geometric and algebraic intuition on certain fields.

5.1 Valuations

Definition 5.1.1: $\mathbb{R}$-Valuation

Let K be a field. Append ∞ to $\mathbb{R}$ and preserve the total ordering by adding the properties that $g \leq \infty$ for all $g \in \mathbb{R}$ and $g + \infty = \infty + g = \infty + \infty = \infty$. Then a function $\nu : K \to \mathbb{R} \cup \{\infty\}$ is called a $\mathbb{R}$-*valuation* if

1. $\nu(K^\times) \subseteq \mathbb{R}$ and $\nu|_{K^\times} : K^\times \to \mathbb{R}$ is a group-homomorphism, that is $\nu(x \cdot y) = \nu(x) + \nu(y)$.
2. $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$
3. $\nu(x) = \infty$ if and only if $x = 0$

It may be more clear that $\mathbb{R}$-valuations come as a generalization of non-Archimedean absolute values from the original definition given by Emil Artin (and hence why these are called valuations). To make sense of this definition, we shall introduce the slightly more general notion of a totally ordered abelian group.

Definition 5.1.2: Totally Ordered [Abelian] Group

Let G be an abelian group. Then G is said to be a *totally ordered group*, or simply an *ordered group* if $\leq$ is a total ordering on G that is translation invariant, that is for all $x, y, z \in G$ where $x \leq y$:

$$zx \leq zy$$

An abelian group admits a translation-invariant total order if and only if it is torsion-free. General ordered groups genuinely extend the real-valued case, but for our purposes valuations into $\mathbb{Z}$ or $\mathbb{R}$ suffice; the general notion mainly serves to connect $\mathbb{R}$-valuations to non-Archimedean absolute values ([6]):

Example 5.1.3: Value Groups

The field of *Puiseux series* $\bigcup_{n\geq 1} k((x^{1/n}))$ carries a valuation with value group $\mathbb{Q}$, given by $\nu(x^a) = a$. More generally, Hahn series over a totally ordered group Γ realize Γ as a value group.

Now, in definition 5.1.1, the $(\mathbb{R}, +)$ can be replaced with $(G, *)$. The following shows how this links to absolute values:

Definition 5.1.4: G-Valuation Equivalent Definition

Let K be a field. Append 0 to G and preserve the total ordering by adding the properties that $0 \leq g$ for all $g \in G$ and $0 * g = g * 0 = 0 * 0 = 0$. Then a function $\nu : K \to G \cup \{0\}$ is called a *G-valuation* if

1. $\nu(K^\times) \subseteq G$ and $\nu|_{K^\times} : K^\times \to G$ is a group-homomorphism, that is $\nu(x \cdot y) = \nu(x) * \nu(y)$.
2. $\nu(x + y) \leq \max\{\nu(x), \nu(y)\}$
3. $\nu(x) = 0$ if and only if $x = 0$

The image of $\nu(K^\times)$ is called the *value group*.

If $G = \mathbb{Z}$, or equivalently in the $\mathbb{R}$-valuation definition, when $v(K^*) \subseteq \mathbb{Z}$, a special name is given:

Definition 5.1.5: Discrete Valuation

Let K be a field. Then a function $\nu : K \to \mathbb{Z} \cup \{\infty\}$ is called a *discrete valuation* if

1. $\nu(x \cdot y) = \nu(x) + \nu(y)$
2. $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$
3. $\nu(x) = \infty$ if and only if $x = 0$

Example 5.1.6: Valuations

1. Given any non-Archimedean absolute value v_p with values in $(\mathbb{R}^+, \cdot)$, we get a $\mathbb{R}^+$ valuation. Hence, each v_p is a valuation on $\mathbb{Q}$ in the equivalent definition, and v_∞, v_0 are valuations on $k(x)$ for any field k.

Given a field k, it may be asked what valuations may be put on it. This is in correspondence with certain sub-rings of k:

Definition 5.1.7: Valuation Ring

Let ν be a valuation on K. Then

$$\mathcal{O}_K = \{x \in K \mid \nu(x) \geq 0\}$$

forms a ring, called the *Valuation Ring.* If ν is a discrete valuation, the ring is called a *discrete valuation ring.*

The reader should verify this is indeed a ring.

Proposition 5.1.8: Valuation Ring Properties

Let k be a field with valuation ν and let $\mathcal{O}_k$ be the valuation ring associated to it. Then:

1. Given $x \in k \setminus \{0\}$, either $x \in \mathcal{O}_k$ or $x^{-1} \in \mathcal{O}_k$ (where both are possible at once). As a consequence, k is a field of fractions of $\mathcal{O}_k$. [a]
2. The group of units is $\mathcal{O}_K^* = \{x \in k \mid \nu(x) = 0\}$ (equiv. $\nu(x) = 1$)
3. $\mathcal{O}_k$ has exactly one maximal ideal $\mathfrak{m}_v = \{x \in k \mid v(x) > 0\} = \mathcal{O}_k \setminus \mathcal{O}_k^*$ (equiv $v(x) < 1$)
4. The ideal of $\mathcal{O}_k$ form a total ordering under $\subseteq$
5. The ring $\mathcal{O}_k$ is integrally closed (i.e. a normal domain)

[a] Note that the either-or implies a general Γ-valuation!

***Proof*:**
(1) For $x \neq 0$, either $\nu(x) \geq 0$ or $\nu(x^{-1}) = -\nu(x) \geq 0$, so $x \in \mathcal{O}_k$ or $x^{-1} \in \mathcal{O}_k$; hence $k = \operatorname{Frac}(\mathcal{O}_k)$. (2) x is a unit iff $x, x^{-1} \in \mathcal{O}_k$ iff $\nu(x) = 0$. (3) By (2) the non-units form $\mathfrak{m}_\nu = \{x \mid \nu(x) > 0\}$, which is an ideal ($\nu(x+y) \geq \min(\nu(x), \nu(y)) > 0$ and $\nu(rx) \geq \nu(x) > 0$); being exactly the non-units, it is the unique maximal ideal.

For 4, first show it on principal ideals, namely if $(a), (b) \subseteq \mathcal{O}_K$, then $(a) \subseteq (b)$ or $(b) \subseteq (a)$. Do this by noticing that either $ab^{-1} \in \mathcal{O}_K$ or $a^{-1}b \in \mathcal{O}_K$. For general ideals, if $\mathfrak{a}$ and $\mathfrak{b}$ are ideals that are not ordered, then there exists $a \in \mathfrak{a} \setminus \mathfrak{b}$ and $b \in \mathfrak{b} \setminus \mathfrak{a}$. Consider $(a), (b)$ which must be ordered namely $(a) \subseteq (b)$ or $(b) \subseteq (a)$. Say $(a) \subseteq (b)$. Then $(a) \cap (b) = (a)$. But then $a \in (b)$ meaning $a = bx$ for some $x \in \mathcal{O}_K$. But as $bx \in \mathfrak{b}$, this would imply $a \notin \mathfrak{a} \setminus \mathfrak{b}$, a contradiction.

We shall show that $\mathcal{O}_K$ is integrally closed. Say that $x \in K$ is integral over $\mathcal{O}_K$, for the sake of contradiction say $x \notin \mathcal{O}_K$. As it is integral:

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0$$

with $a_i \in \mathcal{O}_K$. If $x \notin \mathcal{O}_K$ then $\nu(x) < 0$, so $x^{-1} \in \mathcal{O}_K$; dividing the relation by x^{n-1} gives

$$x = -(a_1 + a_2 x^{-1} + \cdots + a_n x^{-(n-1)}),$$

whose right-hand side lies in $\mathcal{O}_K$ (each $x^{-j} \in \mathcal{O}_K$), forcing $x \in \mathcal{O}_K$, a contradiction, completing the proof.

These properties in fact characterize valuation rings:

Proposition 5.1.9: Characterizing Valuation Ring

Let k be a field, $R \subseteq k$ a sub-ring, and R^* the group of units of R. Then the following are equivalent:

1. R is a valuation ring of k
2. $k = \mathrm{Frac}(R)$ and the ideals of R form a chain under inclusion
3. for every $x \in k \setminus \{0\}$, either $x \in R$ or $x^{-1} \in R$

Then $G = (k \setminus \{0\})/R^*$ is a totally ordered abelian group and

$$\nu_R : x \mapsto xR^*$$

is a valuation on k. The induced valuation ring is R. As a consequence, two valuations on the same field are equivalent if and only if they have the same valuation ring.

***Proof*:**
(1) $\Rightarrow$ (3) is proposition 5.1.8(1), and (1) $\Rightarrow$ (2) its parts (1) and (4). For (3) $\Rightarrow$ (2): (3) gives $k = \mathrm{Frac}(R)$, and any two ideals $\mathfrak{a}, \mathfrak{b}$ are comparable, for if $a \in \mathfrak{a} \setminus \mathfrak{b}$ and $b \in \mathfrak{b}$ then $a/b \notin R$ (else $a \in (b) \subseteq \mathfrak{b}$), so $b/a \in R$ and $b \in (a) \subseteq \mathfrak{a}$, giving $\mathfrak{b} \subseteq \mathfrak{a}$. For (2) $\Rightarrow$ (3): for $x = a/b \in k^\times$ the principal ideals $(a), (b)$ are comparable, so $a/b \in R$ or $b/a \in R$. Given (3), $G = k^\times/R^*$ is totally ordered by $xR^* \leq yR^* \Leftrightarrow y/x \in R$, and $\nu_R(x) = xR^*$ is a valuation whose ring is $\{x \in k \mid \nu_R(x) \geq 0\} = R$; a valuation is determined up to equivalence by its ring.

5.2 Discrete Valuation Ring

Limiting to discrete valuation rings, then there is a smallest positive value s such that:

$$\nu(K^*) = s\mathbb{Z}$$

We can normalize this result and get an equivalent ν where $\nu(K^*) = \mathbb{Z}$ and $\mathcal{O}_K$, $\mathcal{O}_K^*$, and $\mathfrak{p}$ are the same. Then there must be an element such that

$$\nu(\pi) = 1$$

Labelling this element as π, we can use it to show that any $x \in K \setminus \{0\}$ can be written as :

$$x = u\pi^m$$

for a unit $u \in K^\times$ $(\nu(u) = 0)$. To see this, let

$$\nu(x) = m \Rightarrow \nu(x\pi^{-m}) = 0 \Rightarrow u = x\pi^{-m} \in \mathcal{O}_K^*$$

which gives the result, where $m = \nu(x) \in \mathbb{Z}$. This in fact characterizes DVRs:

Proposition 5.2.1: Characterizing DVR

Let R be an integral domain and $k = \mathrm{Frac}(R)$. Then R is the valuation ring of a discrete valuation ν on k if and only if R is a PID with a unique nonzero prime ideal.

***Proof*:**
($\Rightarrow$) Normalize ν so $\nu(k^\times) = \mathbb{Z}$, with uniformizer π ($\nu(\pi) = 1$). Every nonzero $x \in R$ is $u\pi^n$ with u a unit and $n = \nu(x) \geq 0$, so every nonzero ideal equals $(\pi^n) = \mathfrak{m}^n$ for the least n occurring; thus R is a PID with unique nonzero prime $\mathfrak{m} = (\pi)$. ($\Leftarrow$) If R is a PID with unique nonzero prime (π), each nonzero $x \in R$ factors uniquely as $u\pi^n$; setting $\nu(x) = n$ and extending by $\nu(a/b) = \nu(a) - \nu(b)$ gives a discrete valuation on k with valuation ring R.

The DVRs are exactly the "nicest" one-dimensional local rings, characterized among Noetherian local domains of dimension one in several equivalent ways:

Theorem 5.2.2: Characterizations of a DVR

Let $(R, \mathfrak{m})$ be a Noetherian local domain of Krull dimension 1 with fraction field k. The following are equivalent:

1. R is a discrete valuation ring;
2. R is integrally closed (normal);
3. $\mathfrak{m}$ is principal;
4. every nonzero ideal of R is a power of $\mathfrak{m}$;
5. R is a regular local ring, i.e. $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$.

***Proof*:**
(1) $\Rightarrow$ (2) is proposition 5.1.8(5), and (3) $\Leftrightarrow$ (5) holds since, by Nakayama (lemma 1.4.29), $\mathfrak{m}$ is principal iff $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$. For (2) $\Rightarrow$ (3): pick $0 \neq a \in \mathfrak{m}$; as $\dim R = 1$, $\mathfrak{m}$ is the only prime over (a), so $\mathfrak{m}^n \subseteq (a)$ for some least n. Choose $b \in \mathfrak{m}^{n-1} \setminus (a)$ and put $t = a/b \in k$; then $b\mathfrak{m} \subseteq \mathfrak{m}^n \subseteq (a)$ gives $t^{-1}\mathfrak{m} = (b/a)\mathfrak{m} \subseteq R$. If $t^{-1}\mathfrak{m} \subseteq \mathfrak{m}$, the finitely generated faithful module $\mathfrak{m}$ would be stable under t^{-1}, making t^{-1} integral over R (determinant trick) and hence $t^{-1} \in R$ by normality, contradicting $b \notin (a)$. So $t^{-1}\mathfrak{m} = R$, i.e. $\mathfrak{m} = tR$ is principal. (3) $\Rightarrow$ (4): writing $\mathfrak{m} = (\pi)$, Krull's intersection theorem (corollary 3.4.11) gives $\bigcap_n \mathfrak{m}^n = 0$, so any nonzero ideal is $(\pi^n) = \mathfrak{m}^n$ for the largest n with the ideal $\subseteq \mathfrak{m}^n$. (4) $\Rightarrow$ (1): defining $\nu(x) = n$ where $(x) = \mathfrak{m}^n$ gives a discrete valuation on k with valuation ring R.

Example 5.2.3: Discrete Valuation Ring

1. The ring of p-adic integers $\mathbb{Z}_p$ is a discrete valuation ring.
2. For any prime p, $\mathbb{Z}_{(p)}$ the localization of $\mathbb{Z}$ at the prime (p) is a discrete valuation ring.

3. The ring $k[[x]]$ of formal power series is a DVR with unique non-zero prime ideal (x).
4. (If you know some analysis) The ring of convergent power series over $\mathbb{R}$ or $\mathbb{C}$ (those with a positive radius of convergence) is a DVR.
5. (If you know some complex analysis) Let X be a connected Riemann surface and let $M(X)$ be the field of meromorphic functions $X \to \mathbb{C} \cup \{\infty\}$. For each point $p \in X$, define a discrete valuation on $M(X)$ where
$$\nu_p(f) = i$$
where i is the largest number st
$$\frac{f(z)}{(z-p)^i}$$
can be extended to a holomorphic function (i.e. the order of the root/pole).

It shall be seen in algebraic geometry ([1]) that DVRs appear naturally in the study of smooth projective algebraic curves".

Proposition 5.2.4: Representation Of Elements Of a Complete DVR

Let k be a field with *complete* DVR $\mathcal{O}_k$, uniformizer π, and let $\mathcal{T} \subseteq \mathcal{O}_k$ contain 0 and exactly one element of every coset of $\mathfrak{m}$. Then every element $x \in \mathcal{O}_k$ is a unique convergent series
$$\sum_{i \geq 0} r_i \pi^i$$
with coefficients $r_i \in \mathcal{T}$, and every nonzero $x \in k$ is a unique Laurent series $\sum_{i \geq m} r_i \pi^i$ with $r_i \in \mathcal{T}$ for all $i \geq m$ and $r_m \neq 0$

***Proof*:**
By completeness, build the series by repeatedly subtracting coset representatives: let $r_0 \in \mathcal{T}$ represent $x \bmod \mathfrak{m}$, so $x - r_0 \in \mathfrak{m} = (\pi)$; write $x - r_0 = \pi x_1$ and repeat with x_1. The partial sums $\sum_{i<N} r_i \pi^i$ are Cauchy and converge to x, and the r_i are forced as coset representatives, giving uniqueness. The Laurent case follows by factoring out π^m with $m = \nu(x)$.

Valuation rings are the maximal objects in the domination order, which yields a clean description of integral closure.

Proposition 5.2.5: Domination and Existence of Valuation Rings

Let k be a field. Every local subring $A \subseteq k$ is dominated by a valuation ring $\mathcal{O}$ of k (that is, $A \subseteq \mathcal{O}$ and $\mathfrak{m}_{\mathcal{O}} \cap A = \mathfrak{m}_A$). Consequently, for a domain A with fraction field k, the integral closure of A in k is the intersection of the valuation rings of k containing A:
$$\overline{A} = \bigcap_{A \subseteq \mathcal{O}} \mathcal{O},$$
so A is integrally closed iff it is an intersection of valuation rings.

***Proof*:**
Order the local subrings of k dominating A by domination; the union of a chain dominates A, so by Zorn there is a maximal one $\mathcal{O}$. It is a valuation ring: suppose $x \in k$ with $x, x^{-1} \notin \mathcal{O}$. Not both of $\mathfrak{m}_{\mathcal{O}}\mathcal{O}[x]$ and $\mathfrak{m}_{\mathcal{O}}\mathcal{O}[x^{-1}]$ are the unit ideal, for otherwise $1 = \sum_{i \le n} a_i x^i = \sum_{j \le m} b_j x^{-j}$ with $a_i, b_j \in \mathfrak{m}_{\mathcal{O}}$ and n, m minimal, and eliminating the top power of x between the two relations lowers one of the degrees, a contradiction. Say $\mathfrak{m}_{\mathcal{O}}\mathcal{O}[x] \neq \mathcal{O}[x]$ and let $\mathfrak{n}$ be a maximal ideal of $\mathcal{O}[x]$ containing it; then $\mathcal{O}[x]_{\mathfrak{n}}$ is a local ring strictly dominating $\mathcal{O}$ (it contains $x \notin \mathcal{O}$), contradicting maximality. Every valuation ring is integrally closed (proposition 5.1.8), so $\overline{A} \subseteq \bigcap \mathcal{O}$. Conversely, if $x \in k$ is not integral over A, then x^{-1} is a non-unit of $A[x^{-1}]$, hence lies in a maximal ideal $\mathfrak{n}$; the localization $A[x^{-1}]_{\mathfrak{n}}$ is dominated by a valuation ring $\mathcal{O} \supseteq A$ with $x^{-1} \in \mathfrak{m}_{\mathcal{O}}$, so $\nu(x) < 0$ and $x \notin \mathcal{O}$. Thus $\bigcap \mathcal{O} \subseteq \overline{A}$.

Proposition 5.2.6: Extension of Valuations

Let $K \subseteq L$ be a field extension and ν a valuation on K with valuation ring $\mathcal{O}_\nu$. Then ν extends to a valuation on L.

***Proof*:**
The local ring $\mathcal{O}_\nu \subseteq L$ is dominated by a valuation ring $\mathcal{O}$ of L (proposition 5.2.5). Then $\mathcal{O} \cap K$ is a valuation ring of K dominating $\mathcal{O}_\nu$, hence equals $\mathcal{O}_\nu$, so the valuation of $\mathcal{O}$ restricts to ν on K.

Exercise 5.2.1

1. Show that an abelian group admits a translation-invariant total order if and only if it is torsion-free.
2. Let $\mathcal{O}$ be a DVR with unique prime $\mathfrak{p}$. Show that $\mathfrak{p}^n/\mathfrak{p}^{n+1} \cong \mathcal{O}/\mathfrak{p}$ for $n \geq 0$
3. Let $x \in \mathfrak{m}$ be a non-unit of a complete DVR $\mathcal{O}$. Show that $1 - x$ is a unit, its inverse being the convergent series $\sum_{n \geq 0} x^n$.

6

Artinian Rings

Being Noetherian brings a consistent notion of finiteness to rings, letting us define concepts like dimension. It is natural to ask what the dual *descending chain condition* yields. This chapter studies rings and modules satisfying the D.C.C.; it turns out to be far more restrictive than the A.C.C., forcing an Artinian ring to be zero-dimensional and, in fact, a finite product of local Artinian rings.

6.1 Artinian Rings and Modules

Definition 6.1.1: Artinian module and ring

Let M be a R-module. Then if M satisfies the *descending chain condition* on submodules, it is an *artinian module.* More precisely, for any collection of submodules such that

$$M_1 \supseteq M_2 \supseteq \cdots \supseteq M_k \supseteq \cdots$$

then for some n, we get $M_n = M_{n+1} = \cdots$.
If we let $M = R$ be a module over itself, then we say that R is an *Artinian ring.*

Naturally, an Artinian ring has the D.C.C. condition on its ideals.

Proposition 6.1.2: Artinian Module Equivalent

M is Artinian if and only if every non-empty set of submodules of M contains a minimal element under inclusion

***Proof*:**
If M satisfies the D.C.C. and a nonempty set $\mathcal{S}$ of submodules had no minimal element, choose $M_1 \in \mathcal{S}$ and inductively $M_{i+1} \subsetneq M_i$ in $\mathcal{S}$, an infinite strictly descending chain, a contradiction. Conversely, if every nonempty set has a minimal element, a descending chain $M_1 \supseteq M_2 \supseteq \cdots$ has a minimal M_n, whence $M_n = M_{n+1} = \cdots$.

Just like Noetherian modules, a quotient of an Artinian module is Artinian.

Lemma 6.1.3: Artinian Module Closed Under Quotient

Let M be an Artinian module and $N \subseteq M$. Then M/N is also Artinian

***Proof*:**
A descending chain of submodules of M/N pulls back, via the correspondence for $M \twoheadrightarrow M/N$, to a descending chain of submodules of M containing N, which stabilizes; hence the chain in M/N stabilizes.

Example 6.1.4: Artinian, But Not Noetherian Module
Take $\mathbb{Z}[1/2]/\mathbb{Z}$ as a $\mathbb{Z}$-module. Its submodules form the increasing chain

$$0 \subsetneq \tfrac{1}{2}\mathbb{Z}/\mathbb{Z} \subsetneq \tfrac{1}{4}\mathbb{Z}/\mathbb{Z} \subsetneq \cdots, \qquad \tfrac{1}{2^n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/2^n\mathbb{Z},$$

and these are all of its submodules. Any descending chain among them is eventually constant, so the module is Artinian, while the increasing chain shows it is not Noetherian. (It is a module, not a ring.)

However, unlike Noetherian rings, Artinian rings need to satisfy stronger properties on their ideals:

Proposition 6.1.5: Artinian, Prime $\Rightarrow$ Maximal

Let R be an Artinian ring and $P \subseteq R$ be a prime ideal. Then P is maximal

Notice how this differs from Noetherian rings: $\mathbb{Z}[x]$ is Noetherian, (x) is prime, but not maximal since $(x) \subsetneq (2, x)$. Thus, being Artinian immediately imposes stronger conditions on its ideals.

***Proof*:**
Consider R/P, which is also artinian by lemma 6.1.3. We'll show $R/P = F$ is a field. COnsider $x \in F \setminus \{0\}$. Then by the descendign chain condition:

$$(x) \supseteq (x^2) \supseteq \cdots$$

must staxalize, that is for some n:

$$(x^n) = (x^{n+1})$$

which implies $x^n = x^{n+1}c$. By the Cancellion law, we get $1 = xc$, showing that x is in fact a unit. Since every element is a unit in F, we have that F is in fact a field, and so P is maximal, as we sought to show.

Theorem 6.1.6: Structure Theorem for Artinian Rings

Let R be an Artinian ring. Then:

1. There are only finitely many maximal ideals
2. The quotient $R/(\text{Jac}(R))$ is the direct product of a finite number of fields, where if there are n maximal ideals, we have a direct product of n fields R/M_i.
3. Every prime ideal of R is maximal. Thus $\text{kdim}(R) = 0$. As a consequence, $\text{Jac}(R) = \text{Nil}(R)$ and is a nilpotent ideal, $(\text{Jac}(R))^n = 0$ for some $n \geq 1$
4. R is isomorphic to a direct product of a finite number of local artinian rings
5. Every Artinian ring is Noetherian

Note that the last conditions is not true for Artinian Modules.

***Proof*:**

1. Consider the chain:
$$m_1 \supseteq m_1 \cap m_2 \supseteq \cdots$$
I first claim that each inclusion of this chain is proper. To see this, recall that if m_i and m_j are non-equal maximal ideal, it must be that $m_i + m_j = 1$. Thus, any collection of maximal ideals has this pair-wise condition. By the Chinese Remainder Theorem:
$$R/(m_1 \cap \cdots \cap m_n) \cong R/m_1 \times R/m_2 \times \cdots \times R/m_n$$
Now simply see that the element $(\overline{0}, ..., \overline{0}, \overline{1})$ is non-zero, and hence not contain in $m_1 \cap \cdots \cap m_n$, showing that each inclusion *must* be proper! Thus, by the D.C.C., the above chain must stabalize, meaning there can only be finitely many maximal ideals.
2. The proof follows immediately from the previous observation of the CRT
3. First, $J := \text{Jac}(R)$ is nilpotent: by the D.C.C. the chain $J \supseteq J^2 \supseteq \cdots$ stabilizes, say $J^k = J^{k+1}$. If $J^k \neq 0$, the nonempty set of ideals I with $IJ^k \neq 0$ has a minimal element I_0; pick $a \in I_0$ with $aJ^k \neq 0$, so $(a) \subseteq I_0$ is in the set and $(a) = I_0$ by minimality. Then $((a)J)J^k = (a)J^{k+1} = (a)J^k \neq 0$ and $(a)J \subseteq (a)$, so $(a)J = (a)$ by minimality, and Nakayama (lemma 1.4.27, as $J \subseteq \text{Jac}(R)$) forces $(a) = 0$, a contradiction. Hence $J^k = 0$, so $J \subseteq \sqrt{(0)} = \text{Nil}(R)$; the reverse inclusion always holds, giving $\text{Jac}(R) = \text{Nil}(R)$. Now let P be prime. As $P \supseteq \text{Nil}(R) = J$, its image in $R/J \cong k_1 \times \cdots \times k_n$ is a prime; since ideals of a product are products of ideals, that image is $k_1 \times \cdots \times 0 \times \cdots \times k_n$, which is maximal, so P is maximal by the correspondence theorem. Thus every prime is maximal and $\text{kdim} R = 0$.
4. Let $m_1, ..., m_n$ be the maximal ideals of R and let k be such that $(\text{Jac}(R))^k = 0$. Then
$$\prod_i^n M_i^k \subseteq \left(\prod_i^n M_i\right)^k \subseteq (\text{Jac}(R))^k = 0$$

Thus, by the CRT

$$R \cong (R/M_1^k) \times \cdots \times (R/M_n^k)$$

Notice that each R/M_i^k is an artinian with unique maximal ideal M_i/M_i^k.

5. By part (4) it suffices to show local Artinian rings are Noetherian. Let $\mathfrak{m}$ be the maximal ideal; then $\mathfrak{m} = \mathrm{Jac}(R)$ and $\mathfrak{m}^k = 0$ for some k (part 3), giving the filtration $R \supseteq \mathfrak{m} \supseteq \mathfrak{m}^2 \supseteq \cdots \supseteq \mathfrak{m}^k = 0$. Each quotient $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a vector space over the field $R/\mathfrak{m}$ satisfying the D.C.C., hence finite-dimensional (lemma 6.1.11) and of finite length; concatenating their composition series gives one for R, so R is Noetherian.

Corollary 6.1.7: Uniqueness of the Local Decomposition

The decomposition of an Artinian ring $R \cong \prod_{i=1}^n R_i$ into local Artinian rings is unique up to reordering: the factors are the localizations $R_i \cong R_{\mathfrak{m}_i}$ at the maximal ideals (which here coincide with the completions $\hat{R}_{\mathfrak{m}_i}$), corresponding to the primitive idempotents of R.

***Proof*:**
In a product $\prod R_j$ of local rings the idempotents are the tuples of 0s and 1s, the primitive ones being the standard vectors e_i; hence the factors $R_i = e_i R$ are intrinsic. Localizing at $\mathfrak{m}_i$ kills every e_j with $j \neq i$ (each $1 - e_j \notin \mathfrak{m}_i$ becomes a unit and $e_j(1-e_j) = 0$), leaving R_i, so $R_{\mathfrak{m}_i} \cong R_i$; and R_i, being local Artinian, is complete.

Corollary 6.1.8: Reduced Artinian Rings

An Artinian ring is reduced if and only if it is a finite product of fields.

***Proof*:**
A finite product of fields is reduced. Conversely, if R is reduced Artinian then $\mathrm{Nil}(R) = 0$, so by the structure theorem $R \cong \prod_i R/\mathfrak{m}_i^k$ with $\bigcap_i \mathfrak{m}_i^k = \mathrm{Nil}(R) = 0$; reducedness forces each local factor to have zero nilradical, i.e. $\mathfrak{m}_i^k = 0$ collapses to $\mathfrak{m}_i = 0$ there, so each factor is the field $R/\mathfrak{m}_i$.

Corollary 6.1.9: Artinian Ring Equivalent Condition I

Let R be a ring. Then R is Artinian if and only if R is Noetherian and $\dim(R) = 0$

***Proof*:**
R Artinian implies R Noetherian (theorem 6.1.6) with all primes maximal, i.e. $\dim R = 0$. For the converse, let R be Noetherian with $\dim R = 0$. Its minimal primes $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ are finite in number (Noetherian) and all maximal ($\dim R = 0$), and $\mathrm{Nil}(R) = \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$ is nilpotent (finitely generated), so $(\mathfrak{m}_1 \cdots \mathfrak{m}_n)^k \subseteq \mathrm{Nil}(R)^k = 0$ for some k. By the Chinese Remainder Theorem $R \cong \prod_i R/\mathfrak{m}_i^k$, each factor Noetherian local with nilpotent maximal ideal, hence of finite length and Artinian; so R is Artinian.

Corollary 6.1.10: Artinian Ring Equivalent Condition II

Let k be a field and R a finitely generated k-algebra. Then R is artinian if and only if R is finitely generated as a k-module.

***Proof*:**
If R is a finite-dimensional k-vector space, its ideals are k-subspaces, so the D.C.C. holds and R is Artinian. Conversely, if R is Artinian then $\dim R = 0$ (theorem 6.1.6), and a finitely generated k-algebra of dimension 0 is a finite k-module (chapter 7).

For artinian modules over a field $R = k$, there is a clean characterization:

Lemma 6.1.11: Artin+Noetherian Module

Let M be a vector space over k. Then the following are equivalent:

1. M satisfies the A.C.C.
2. M satisfies the D.C.C.
3. M is finite-dimensional

***Proof*:**
Naturally, (3) implies both (1) and (2). Conversely, assume $\dim_k(M) = \infty$, so pick some infinite collection of linearly independent vectors. Then we have an infinite strict chain:

$$\langle e_1 \rangle \subsetneq \langle e_1, e_2 \rangle \cdots$$

and an infinite descending chain:

$$\langle e_1, e_2, ... \rangle \supsetneq \langle e_2, e_3, ... \rangle \supsetneq \cdots$$

completing the proof

Example 6.1.12: Artinian Rings

1. Let $N > 1$ be an integer and $R = \mathbb{Z}/N\mathbb{Z}$. Since R is finite it is Artinian. If $N = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ for distinct primes p_i and $\alpha_i \geq 1$, then by the Chinese Remainder Theorem
$$\mathbb{Z}/N\mathbb{Z} \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_r^{\alpha_r}\mathbb{Z})$$
where each $\mathbb{Z}/p_i^{\alpha_i}\mathbb{Z}$ is a local Artinian ring with maximal ideal $(p_i)/(p_i^{\alpha_i})$. The Jacobson radical of R is $(p_1 p_2 \cdots p_r)$.
2. Let R be a finite dimensional k-algebra over some field k (it need not be commutative!). Then R is an Artinian ring, since any descending chain of R is a descending chain of k-subspces of R, which is bounded by $\dim_k(R)$. Note that if R is not commutative, R is not the finite direct product of local artinian rings, but the finite product of indecomposable artinian rings (i.e. artinian rings which cannot be split into direct sum of submodules).

3. Let $k[x]$ be the polynomial ring over a field k and $f \in k[x]$ be a nonzero polynomial. Then if $f(x) = f_1^{n_1}(x) \cdots f_k^{n_k}(x)$, by the CRT we have:

$$k[x]/(f(x)) \cong k[x]/(f_1^{n_1}(x)) \times \cdots \times k[x]/(f_k^{n_k}(x))$$

which is artinian, since $k[x]/(f(x))$ is a finite dimensional k-module (each $k[x]/(f_i^{n_i}(x))$ is a local artinian ring, with maximla ideal $(f_i(x))/(f_i^{n_i}(x))$). If $n_i = 1$ for all i, $k[x]/(f(x))$ is a product of fields, and if furtheremore there was only 1 n_i (i.e. $i \in \{1\}$), then $k[x]/(f(x))$ would be a field extension

We shall give names to modules that satisfy both the Artinian and Noetherian condition:

Definition 6.1.13: Modules of Finite Length

Let M be an R-module that is Noetherian and Artinian. Then M is an R-module of *finite length.*

By proposition 1.1.8 each finite-length module has a well-defined length $\ell(M) \in \mathbb{N}$. In fact the composition factors themselves are determined:

Theorem 6.1.14: Jordan-Hölder

Let M be a module of finite length. Then any two composition series of M have the same length and the same multiset of simple quotients, up to isomorphism and reordering.

***Proof*:**
Length invariance is proposition 1.1.8. Induct on $\ell(M)$ for the factors: given two composition series with maximal submodules M_1, M_1', either $M_1 = M_1'$ (apply the inductive hypothesis to M_1), or $M_1 + M_1' = M$, whence $M/M_1 \cong M_1'/(M_1 \cap M_1')$ and $M/M_1' \cong M_1/(M_1 \cap M_1')$; inserting a composition series of $M_1 \cap M_1'$ into each shows the two factor multisets agree.

Example 6.1.15: Module of Finite Length
The simplest example would be when $R = k$ is a field, which gives vector-spaces. In this case, for any V

$$\ell(V) = \dim_k(V)$$

Proposition 6.1.16: Finite Length Additive Under Extensions

Let

$$0 \to M' \to M \to M'' \to 0$$

Be a short exact sequence of finite length modules. Then:

$$\ell(M) = \ell(M') + \ell(M'')$$

***Proof*:**
Concatenating a composition series $0 = M'_0 \subsetneq \cdots \subsetneq M'_{\ell(M')} = M'$ of M' with the preimage in M of a composition series of $M'' = M/M'$ yields a composition series of M of length $\ell(M') + \ell(M'')$; by invariance of length (proposition 1.1.8) this is $\ell(M)$.

For a finite-length module the length is computed locally, and the module splits as a direct sum over its (finitely many, maximal) associated points.

Proposition 6.1.17: Length, Support, and Local Decomposition

Let M be a finite-length R-module. Then $\operatorname{Supp}(M)$ is a finite set of maximal ideals, the natural map $M \to \bigoplus_{\mathfrak{m} \in \operatorname{Supp}(M)} M_{\mathfrak{m}}$ is an isomorphism, and

$$\ell_R(M) = \sum_{\mathfrak{m} \in \operatorname{Supp}(M)} \ell_{R_{\mathfrak{m}}}(M_{\mathfrak{m}})$$

***Proof*:**
A composition series of M has simple factors $R/\mathfrak{m}_i$ with the $\mathfrak{m}_i$ maximal, so $\operatorname{Supp}(M) = \{\mathfrak{m}_i\}$ is a finite set of maximal ideals and some product $\mathfrak{m}_1^{a_1} \cdots \mathfrak{m}_r^{a_r}$ annihilates M. Thus M is a module over the Artinian ring $R/\operatorname{Ann}(M) \cong \prod_i R/\mathfrak{m}_i^{a_i}$, which gives the direct-sum decomposition $M \cong \bigoplus_i M_{\mathfrak{m}_i}$. Localizing a composition series at $\mathfrak{m}$ deletes the factors $R/\mathfrak{m}'$ with $\mathfrak{m}' \neq \mathfrak{m}$ and preserves those with $\mathfrak{m}' = \mathfrak{m}$, so $\ell_{R_{\mathfrak{m}}}(M_{\mathfrak{m}})$ counts the factors isomorphic to $R/\mathfrak{m}$; summing over $\mathfrak{m}$ gives $\ell_R(M)$.

Proposition 6.1.18: Length Along a Flat Local Homomorphism

Let $\varphi\colon (A, \mathfrak{m}) \to (B, \mathfrak{n})$ be a flat local homomorphism of Noetherian local rings whose closed fibre $B/\mathfrak{m}B$ has finite length over B, and let M be an A-module of finite length. Then $M \otimes_A B$ has finite length over B, and

$$\ell_B(M \otimes_A B) \;=\; \ell_A(M) \cdot \ell_B(B/\mathfrak{m}B)$$

In particular, if A is Artinian then $\ell_B(B) = \ell_A(A) \cdot \ell_B(B/\mathfrak{m}B)$.

***Proof*:**
Induct on $n = \ell_A(M)$. If $n = 0$ then $M = 0$ and both sides vanish. If $n = 1$ then M is simple, so $M \cong A/\mathfrak{m}$ because A is local, and $M \otimes_A B \cong B/\mathfrak{m}B$, whose length is $\ell_B(B/\mathfrak{m}B)$ by hypothesis; this is the claimed product since $\ell_A(M) = 1$.

For $n > 1$, choose a submodule $M' \subseteq M$ with $\ell_A(M') = n - 1$ and $M/M' \cong A/\mathfrak{m}$, which exists by taking a composition series. Tensoring

$$0 \to M' \to M \to A/\mathfrak{m} \to 0$$

with B over A leaves it exact, since B is flat over A. Length is additive on short exact sequences (proposition 6.1.16), so

$$\ell_B(M \otimes_A B) = \ell_B(M' \otimes_A B) + \ell_B(B/\mathfrak{m}B) = (n-1)\ell_B(B/\mathfrak{m}B) + \ell_B(B/\mathfrak{m}B)$$

using the inductive hypothesis, and the right side is $n \cdot \ell_B(B/\mathfrak{m}B)$, as we sought to show.

The multiplicativity is the algebraic content behind the geometric statement that a flat morphism pulls the fundamental cycle of the base back to the fundamental cycle of the total space: the multiplicity with which a component occurs is multiplied by the length of the fibre, uniformly.

Proposition 6.1.19: Finiteness over an Artinian Ring

Let R be an Artinian ring and M an R-module. Then M is finitely generated iff it has finite length iff it is Noetherian iff it is Artinian. In particular R has finite length over itself.

***Proof*:**
Finite length trivially implies the other three. If M is finitely generated it is a quotient of some R^n; since R has finite length over itself (its composition series comes from the filtration by the $\mathfrak{m}_i^j$), so do R^n and its quotient M. A Noetherian module is finitely generated, hence of finite length by the above. An Artinian module M over the Artinian R also has finite length: with $J = \mathrm{Jac}(R)$ and $J^k = 0$, the filtration $M \supseteq JM \supseteq \cdots \supseteq J^k M = 0$ has each piece $J^i M / J^{i+1} M$ a module over the product of fields R/J satisfying the D.C.C., hence a finite direct sum of simple modules.

Exercise 6.1.1

1. Let M be a Noetherian and Artinian A-module, $\varphi : M \to M$ an A-module homomorphism. Show that there exists an $n \geq 0$ such that $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$.

2. Let M be an R-module. Then M is of finite length iff it has a composition series.

7

Dimension Theory

How large is a ring? For the coordinate ring of a variety the answer should be the variety's dimension, and the algebra offers three ways to measure it, each capturing a different intuition. One counts how many independent parameters the ring carries: the length of the longest chain of prime ideals, since primes correspond to irreducible closed subsets and a chain of them is a tower of nested subvarieties. A second counts how many equations it takes to cut the ring down to something finite: a zero-dimensional, Artinian quotient. A third reads the dimension off a growth rate, the rate at which $R/\mathfrak{m}^n$ grows with n. The central result of the chapter, the *dimension theorem*, is that these three numbers coincide. It is the payoff of the finiteness and locality threads: the Noetherian hypothesis makes the growth rate polynomial, localization concentrates the question at a single prime, and Noether normalization ties the whole theory back to the transparent case of a polynomial ring.

The geometric cash-out (that the Krull dimension of a coordinate ring is the topological dimension of its variety, that a hypersurface drops dimension by one) is developed in [9], which cites this chapter for the ring theory.

7.1 Noether Normalization

The theory rests on the fact that a finitely generated algebra over a field is only finitely far from a polynomial ring.

Theorem 7.1.1: Noether Normalization

Let k be a field and $A = k[r_1, \ldots, r_n]$ a finitely generated k-algebra. Then there are elements $y_1, \ldots, y_q \in A$, algebraically independent over k, such that A is a finite (module-finite) extension of $k[y_1, \ldots, y_q]$.

***Proof*:**
Induct on n, the number of generators. If $r_1, \ldots, r_n$ are algebraically independent, take $y_i = r_i$ and $A = k[y_1, \ldots, y_n]$. Otherwise a nonzero $f \in k[x_1, \ldots, x_n]$ satisfies $f(r_1, \ldots, r_n) = 0$; let d exceed every exponent appearing in f, and set

$$x_i' = x_i - x_n^{d^i} \quad (1 \le i \le n-1)$$

Rewriting f in the variables $x_1', \ldots, x_{n-1}', x_n$, each monomial $x_1^{a_1} \cdots x_n^{a_n}$ of f contributes a top x_n-power $x_n^{a_1 d + a_2 d^2 + \cdots + a_{n-1} d^{n-1} + a_n}$; base-$d$ uniqueness makes these exponents distinct across monomials, so f becomes, up to a nonzero scalar from k, monic in x_n of some degree N with coefficients in $k[x_1', \ldots, x_{n-1}']$. Putting $s_i = r_i - r_n^{d^i}$, the relation $f(r_1, \ldots, r_n) = 0$ exhibits r_n as a root of a monic polynomial over $B = k[s_1, \ldots, s_{n-1}]$, so r_n is integral over B; each $r_i = s_i + r_n^{d^i}$ is then integral over $B[r_n]$, and by transitivity A is integral, hence (being finitely generated) module-finite, over B. As B is generated by $n-1$ elements, induction gives a polynomial subalgebra $k[y_1, \ldots, y_q] \subseteq B$ over which B, and therefore A, is module-finite.

The number q is an invariant: it equals the transcendence degree of A over k when A is a domain, as theorem 7.6.2 will record once dimension is in hand.

7.2 Krull Dimension

Definition 7.2.1: Krull Dimension and Height

The *height* of a prime $\mathfrak{p}$, written $\operatorname{ht}\mathfrak{p}$, is the supremum of lengths r of chains $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r = \mathfrak{p}$ of primes descending from $\mathfrak{p}$. The *Krull dimension* $\dim R$ is the supremum of $\operatorname{ht}\mathfrak{p}$ over all primes $\mathfrak{p}$, equivalently the supremum of lengths of chains of primes in R.

A field has dimension 0; a principal ideal domain that is not a field has dimension 1, its chains being $0 \subsetneq (\pi)$ for irreducible π. For a local ring $(R, \mathfrak{m})$ the dimension is $\operatorname{ht}\mathfrak{m}$, since $\mathfrak{m}$ contains every prime. The chain $0 \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \ldots, x_n)$ in $k[x_1, \ldots, x_n]$ shows $\dim k[x_1, \ldots, x_n] \ge n$; equality is theorem 7.6.2.

Dimension is controlled locally: $\dim R = \sup_{\mathfrak{m}} \dim R_{\mathfrak{m}}$ over maximal ideals, because a chain of primes below $\mathfrak{p}$ survives in $R_{\mathfrak{p}}$. The substance of the theory is therefore the local case, to which the next two sections are devoted.

7.3 The Hilbert-Samuel Function

Throughout this section $(R, \mathfrak{m})$ is a Noetherian local ring and $\mathfrak{q}$ an $\mathfrak{m}$-*primary* ideal, so that $\mathfrak{m}^r \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ for some r and $R/\mathfrak{q}$ is Artinian. We measure R through the lengths $\ell(R/\mathfrak{q}^n)$. The tool is the associated graded ring $\operatorname{gr}_{\mathfrak{q}} R = \bigoplus_{n \ge 0} \mathfrak{q}^n/\mathfrak{q}^{n+1}$, a Noetherian graded ring, finitely generated over the Artinian ring $R/\mathfrak{q}$ by the finitely many degree-one elements $\mathfrak{q}/\mathfrak{q}^2$.

Lemma 7.3.1: Hilbert Polynomial of a Graded Module

Let $S = S_0[t_1, \ldots, t_s]$ be a graded ring, finitely generated over an Artinian ring S_0 by degree-one elements t_i, and $N = \bigoplus_n N_n$ a finitely generated graded S-module. Then $\ell(N_n)$ is finite for each n and agrees, for $n \gg 0$, with a polynomial in n of degree $\leq s-1$ (rational coefficients).

***Proof*:**
Each N_n is a finitely generated S_0-module, hence of finite length. Induct on s. If $s = 0$ then $S = S_0$ and N, being finitely generated, is concentrated in finitely many degrees, so $\ell(N_n) = 0$ for $n \gg 0$ (a polynomial of degree ≤ -1). For the step, multiplication by t_s gives an exact sequence of $S_0[t_1, \ldots, t_{s-1}]$-modules in each degree,

$$0 \to K_n \to N_n \xrightarrow{t_s} N_{n+1} \to (N/t_sN)_{n+1} \to 0,$$

where $K = (0 :_N t_s)$ and N/t_sN are finitely generated and killed by t_s, hence graded modules over $S_0[t_1, \ldots, t_{s-1}]$. Additivity of length gives $\ell(N_{n+1}) - \ell(N_n) = \ell\big((N/t_sN)_{n+1}\big) - \ell(K_n)$, which by induction agrees with a polynomial of degree $\leq s-2$ for $n \gg 0$. A function whose first difference is eventually a polynomial of degree $\leq s-2$ is itself eventually a polynomial of degree $\leq s-1$.

Proposition 7.3.2: The Hilbert-Samuel Polynomial

For a finitely generated R-module M, the *characteristic function* $\chi_{\mathfrak{q}}^M(n) = \ell(M/\mathfrak{q}^nM)$ agrees for $n \gg 0$ with a polynomial in n of degree $\leq s$, where s is the least number of generators of $\mathfrak{q}$. Moreover $\deg \chi_{\mathfrak{q}}^M = \deg \chi_{\mathfrak{m}}^M$, so the degree is independent of the $\mathfrak{m}$-primary ideal $\mathfrak{q}$.

***Proof*:**
The associated graded module $\mathrm{gr}(M) = \bigoplus_n \mathfrak{q}^nM/\mathfrak{q}^{n+1}M$ is a finitely generated graded module over $\mathrm{gr}_{\mathfrak{q}} R$, itself generated over the Artinian ring $R/\mathfrak{q}$ by the s degree-one generators of $\mathfrak{q}/\mathfrak{q}^2$. By lemma 7.3.1, $\ell(\mathfrak{q}^nM/\mathfrak{q}^{n+1}M)$ is eventually a polynomial of degree $\leq s-1$. Since $\chi_{\mathfrak{q}}^M(n) = \ell(M/\mathfrak{q}^nM) = \sum_{r=0}^{n-1} \ell(\mathfrak{q}^rM/\mathfrak{q}^{r+1}M)$, summing raises the degree by one: $\chi_{\mathfrak{q}}^M$ is eventually a polynomial of degree $\leq s$. For the last claim, $\mathfrak{m}^r \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ gives $\mathfrak{m}^{rn} \subseteq \mathfrak{q}^n \subseteq \mathfrak{m}^n$, hence $\chi_{\mathfrak{m}}^M(n) \leq \chi_{\mathfrak{q}}^M(n) \leq \chi_{\mathfrak{m}}^M(rn)$ for all n; the outer terms are polynomials of equal degree, so the middle one shares it.

Write $d(M) = \deg \chi_{\mathfrak{m}}^M(n)$ and $d(R) = \deg \chi_{\mathfrak{m}}(n)$ where $\chi_{\mathfrak{m}}(n) = \ell(R/\mathfrak{m}^n)$. The next proposition is the inductive engine.

Proposition 7.3.3: Dropping the Degree by a Non-Zero-Divisor

If $x \in \mathfrak{m}$ is not a zero-divisor on M, then $d(M/xM) \leq d(M) - 1$.

***Proof*:**
Set $M' = M/xM$. Additivity of length in $0 \to (xM + \mathfrak{q}^n M)/\mathfrak{q}^n M \to M/\mathfrak{q}^n M \to M'/\mathfrak{q}^n M' \to 0$ gives $\chi_{\mathfrak{q}}^{M'}(n) = \chi_{\mathfrak{q}}^{M}(n) - g(n)$, where $g(n) = \ell\big(xM/(xM \cap \mathfrak{q}^n M)\big)$. As x is a non-zero-divisor, $M \xrightarrow{x} xM$ is an isomorphism carrying $\mathfrak{q}^n M$ onto $\mathfrak{q}^n(xM)$, so $\ell(xM/\mathfrak{q}^n(xM)) = \chi_{\mathfrak{q}}^M(n)$. Now $\mathfrak{q}^n(xM) \subseteq xM \cap \mathfrak{q}^n M$, and by the Artin-Rees lemma (theorem 3.4.1) the filtration $\{xM \cap \mathfrak{q}^n M\}$ is $\mathfrak{q}$-stable, so $xM \cap \mathfrak{q}^n M \subseteq \mathfrak{q}^{n-c}(xM)$ for some c and all $n \geq c$. Taking lengths,

$$\chi_{\mathfrak{q}}^M(n-c) \leq g(n) \leq \chi_{\mathfrak{q}}^M(n)$$

Both bounds are $\chi_{\mathfrak{q}}^M$ evaluated at $n-c$ and n, so g agrees for large n with a polynomial of degree $d(M)$ and the same leading coefficient as $\chi_{\mathfrak{q}}^M$. Hence $\chi_{\mathfrak{q}}^{M'} = \chi_{\mathfrak{q}}^M - g$ has degree $\leq d(M) - 1$.

7.4 The Dimension Theorem

Theorem 7.4.1: Dimension Theorem

For a Noetherian local ring $(R, \mathfrak{m})$ the following three numbers coincide:

$$d(R) = \delta(R) = \dim R,$$

where $d(R) = \deg \ell(R/\mathfrak{m}^n)$, and $\delta(R)$ is the least number of generators of an $\mathfrak{m}$-primary ideal.

***Proof*:**
We show $d(R) \leq \delta(R) \leq \dim R \leq d(R)$.

$d(R) \leq \delta(R)$. If $\mathfrak{q}$ is $\mathfrak{m}$-primary and generated by $s = \delta(R)$ elements, then $\deg \chi_{\mathfrak{q}} \leq s$ by proposition 7.3.2, and $\deg \chi_{\mathfrak{q}} = \deg \chi_{\mathfrak{m}} = d(R)$; so $d(R) \leq \delta(R)$.

$\dim R \leq d(R)$. Induct on $d(R)$. If $d(R) = 0$ then $\ell(R/\mathfrak{m}^n)$ is eventually constant, so $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for some n; Nakayama's lemma (lemma 1.4.27) gives $\mathfrak{m}^n = 0$, so R is Artinian and $\dim R = 0$. Suppose $d(R) = d > 0$ and let $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ be a chain of primes; we bound r. We may take $\mathfrak{p}_0$ minimal, and choosing $x \in \mathfrak{p}_1 \setminus \mathfrak{p}_0$, the ring $\bar{R} = R/\mathfrak{p}_0$ is a domain in which x is a nonzero, hence non-zero-divisor, element. By proposition 7.3.3, $d(\bar{R}/x\bar{R}) \leq d(\bar{R}) - 1 \leq d(R) - 1$ (the last as $\bar{R}$ is a quotient of R, whose characteristic polynomial dominates). The chain $\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ descends to a chain of length $r - 1$ in $\bar{R}/x\bar{R}$, so by induction $r - 1 \leq d(R) - 1$, i.e. $r \leq d(R)$. Hence $\dim R \leq d(R)$.

$\delta(R) \leq \dim R$. Let $d = \dim R$; we build an $\mathfrak{m}$-primary ideal from d elements. Choose $x_1, \ldots, x_d \in \mathfrak{m}$ so that x_i lies in no minimal prime of $(x_1, \ldots, x_{i-1})$; this is possible by prime avoidance, as there are only finitely many such minimal primes (corollary 2.4.3), unless one of them is $\mathfrak{m}$ itself, in which case $(x_1, \ldots, x_{i-1})$ is already $\mathfrak{m}$-primary and fewer elements suffice. An induction then shows every prime $\mathfrak{p} \supseteq (x_1, \ldots, x_i)$ has $\operatorname{ht} \mathfrak{p} \geq i$: such $\mathfrak{p}$ contains a minimal prime $\mathfrak{p}'$ of $(x_1, \ldots, x_{i-1})$, which has $\operatorname{ht} \mathfrak{p}' \geq i - 1$ by induction, and $\mathfrak{p} \neq \mathfrak{p}'$ because $x_i \in \mathfrak{p} \setminus \mathfrak{p}'$, so $\operatorname{ht} \mathfrak{p} \geq \operatorname{ht} \mathfrak{p}' + 1 \geq i$. At $i = d$ every prime over $(x_1, \ldots, x_d)$ has height $\geq d = \operatorname{ht} \mathfrak{m}$; since $\mathfrak{m}$ is the only prime of that height in the local ring R, the ideal $(x_1, \ldots, x_d)$ is $\mathfrak{m}$-primary. Thus $\delta(R) \leq d$.

Krull's height theorem, proved next, is now a corollary of the dimension theorem, with no circularity:

its proof invokes only the equalities just established.

7.5 Systems of Parameters and Krull's Principal Ideal Theorem

Definition 7.5.1: System of Parameters

A *system of parameters* for a Noetherian local ring $(R, \mathfrak{m})$ of dimension d is a set of d elements generating an $\mathfrak{m}$-primary ideal. The dimension theorem guarantees one exists, and d is the least possible number of such generators.

Theorem 7.5.2: Krull's Height Theorem

Let R be Noetherian and $\mathfrak{a} = (x_1, \dots, x_c)$ generated by c elements. Then every prime $\mathfrak{p}$ minimal over $\mathfrak{a}$ has $\operatorname{ht}\mathfrak{p} \le c$. In particular (the *Hauptidealsatz*, $c = 1$) a prime minimal over a single nonunit x has height at most 1.

***Proof*:**
Localizing at $\mathfrak{p}$, we may assume $(R, \mathfrak{p})$ is local and $\mathfrak{p}$ is minimal over $\mathfrak{a}$, so that $\mathfrak{a}$ is $\mathfrak{p}$-primary and $\operatorname{ht}\mathfrak{p} = \dim R$. Then $\mathfrak{a}$ is an $\mathfrak{p}$-primary ideal with c generators, so $\dim R = \delta(R) \le c$ by the inequality $\dim R \le d(R) \le \delta(R)$ established in theorem 7.4.1; here only $\dim R \le d(R)$ and $d(R) \le \delta(R)$ are used, and $\delta(R) \le c$ because $\mathfrak{a}$ itself is $\mathfrak{p}$-primary with c generators.

A converse rounds out the picture: in a Noetherian ring a prime of height c is minimal over some ideal generated by c elements, so height is exactly the least number of generators needed to cut out a prime as a component. The Hauptidealsatz is the algebraic form of the geometric principle that one equation cuts dimension by one: passing to $R/(x)$ for x a non-unit non-zero-divisor lowers dimension by exactly one,

$$\dim R/(x) = \dim R - 1,$$

the inequality $\ge$ from theorem 7.5.2 and $\le$ from proposition 7.3.3.

7.6 Dimension and Transcendence Degree

Noether normalization now identifies the dimension of a geometric ring with a transcendence degree, and pins down the dimension of affine space. The engine is that a proper prime quotient of a finitely generated domain loses transcendence degree.

Lemma 7.6.1: Transcendence Degree Drops on a Prime Quotient

Let A be a finitely generated domain over a field k and $\mathfrak{p} \ne 0$ a prime of A. Then $\operatorname{trdeg}_k \operatorname{Frac}(A/\mathfrak{p}) < \operatorname{trdeg}_k \operatorname{Frac}(A)$.

***Proof*:**
Throughout, transcendence bases and the transcendence degree of a field extension are those of *Core Algebra* [8], where in particular the degree is shown to be well-defined.

Put $d = \operatorname{trdeg}_k A$ and $e = \operatorname{trdeg}_k A/\mathfrak{p} \le d$, and suppose $e = d$. Choose $x_1, \dots, x_d \in A$ whose images in $A/\mathfrak{p}$ form a transcendence basis; they are algebraically independent in A, hence (as $d = \operatorname{trdeg} A$) a transcendence basis of A as well, so every element of A is algebraic over $B = k[x_1, \dots, x_d]$. Pick $0 \neq a \in \mathfrak{p}$. An algebraic relation for a over $\operatorname{Frac}(B)$, cleared of denominators and divided by the largest power of a that factors out (valid as A is a domain), becomes $c_0 + c_1 a + \cdots + c_m a^m = 0$ with $c_i \in B$ and $c_0 \neq 0$; then $c_0 = -a(c_1 + \cdots + c_m a^{m-1}) \in \mathfrak{p}$. But $\mathfrak{p} \cap B = 0$, since a nonzero element of $\mathfrak{p} \cap B$ would be a nontrivial algebraic relation among the algebraically independent images $\bar{x}_i$ in $A/\mathfrak{p}$. This contradicts $c_0 \neq 0$, so $e < d$.

Theorem 7.6.2: Dimension of Finitely Generated Algebras

Let A be a finitely generated algebra over a field k that is a domain. Then

$$\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A),$$

and in particular $\dim k[x_1, \dots, x_n] = n$. More generally the number q in Noether normalization (theorem 7.1.1) equals $\dim A$.

***Proof*:**
By theorem 7.1.1, A is module-finite, hence integral, over a polynomial subalgebra $k[y_1, \dots, y_q]$ with the y_i algebraically independent, so $q = \operatorname{trdeg}_k \operatorname{Frac}(A)$; integral extensions preserve Krull dimension (theorem 4.3.2 supplies lying-over, going-up, and incomparability), so $\dim A = \dim k[y_1, \dots, y_q]$. We prove the general claim $\dim B \le \operatorname{trdeg}_k \operatorname{Frac}(B)$ for every finitely generated domain B over k, by induction on transcendence degree. Given a chain $0 = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ in B, the tail descends to a chain of length $r - 1$ in the finitely generated domain $B/\mathfrak{p}_1$, whose transcendence degree is at most $\operatorname{trdeg} B - 1$ by lemma 7.6.1; induction gives $r - 1 \le \operatorname{trdeg} B - 1$, so $r \le \operatorname{trdeg} B$. Applied to $B = k[y_1, \dots, y_q]$ this yields $\dim k[y_1, \dots, y_q] \le q$, while the chain of definition 7.2.1 gives $\ge q$. Hence $\dim A = \dim k[y_1, \dots, y_q] = q = \operatorname{trdeg}_k \operatorname{Frac}(A)$.

The equality $\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A)$ is the algebraic form of the statement that dimension counts independent coordinates; in [9] it becomes the assertion that the dimension of an irreducible affine variety is the transcendence degree of its function field.

7.7 Regular Local Rings

The dimension theorem produced three numbers and showed them equal. A fourth is always available for a local ring and is never smaller: the least number of generators of the maximal ideal, which by Nakayama is the dimension of $\mathfrak{m}/\mathfrak{m}^2$ as a vector space over the residue field. Rings for which the fourth number agrees with the other three are the regular ones, and they are the subject of this section.

The inequality has a geometric reading that fixes the terminology. For a variety X and a point p, the vector space $(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$ is the tangent space at p, so the inequality says a variety has at least as many tangent directions at a point as it has dimensions, and regularity is the case where it has exactly as many. That reading, and the identification of regular local rings with the smooth points of a variety, is [9]; what follows is the ring theory it cites.

Corollary 7.7.1: Dimension and Embedding Dimension

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then

$$\dim R \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$$

***Proof*:**
Put $s = \dim_k \mathfrak{m}/\mathfrak{m}^2$ and choose $x_1, \ldots, x_s \in \mathfrak{m}$ whose images form a k-basis of $\mathfrak{m}/\mathfrak{m}^2$. By Nakayama's lemma (lemma 1.4.27) they generate $\mathfrak{m}$, which is $\mathfrak{m}$-primary, so $\delta(R) \leq s$. The dimension theorem (theorem 7.4.1) gives $\dim R = \delta(R) \leq s$, as we sought to show.

The same argument identifies $\dim_k \mathfrak{m}/\mathfrak{m}^2$ with the least number of generators of $\mathfrak{m}$: Nakayama makes any lift of a basis a generating set, and no smaller set can generate, since a generating set spans $\mathfrak{m}/\mathfrak{m}^2$. This is the number called the *embedding dimension*, and corollary 7.7.1 says it bounds the dimension from above.

Definition 7.7.2: Regular Local Ring

A Noetherian local ring $(R, \mathfrak{m})$ with residue field k is *regular* if

$$\dim R = \dim_k \mathfrak{m}/\mathfrak{m}^2,$$

equivalently if $\mathfrak{m}$ can be generated by $\dim R$ elements. Such a generating set is a *regular system of parameters*: it is a system of parameters (definition 7.5.1) that generates $\mathfrak{m}$ itself rather than merely an $\mathfrak{m}$-primary ideal.

Regularity is therefore the statement that the maximal ideal is as small as the dimension permits. The next theorem replaces that numerical condition by a structural one, and the structural form is what every later argument uses: the associated graded ring, which for a general local ring records how the powers of $\mathfrak{m}$ shrink, is for a regular local ring a polynomial ring on the regular system of parameters. Following this chapter's convention it is written $\operatorname{gr}_{\mathfrak{m}} R$; it is the ring called $G_{\mathfrak{m}}(R)$ in definition 3.5.1.

Theorem 7.7.3: Characterizations of a Regular Local Ring

Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension d with residue field k. The following are equivalent.

1. R is regular.
2. $\mathfrak{m}$ is generated by d elements.
3. $\operatorname{gr}_{\mathfrak{m}} R \cong k[t_1, \ldots, t_d]$ as graded k-algebras.

***Proof*:**

(1) $\iff$ (2) The least number of generators of $\mathfrak{m}$ is $\dim_k \mathfrak{m}/\mathfrak{m}^2$, as recorded after corollary 7.7.1, and that number is at least d by the same corollary. So $\mathfrak{m}$ is generated by d elements exactly when $\dim_k \mathfrak{m}/\mathfrak{m}^2 = d$, which is definition 7.7.2.

(2) $\Rightarrow$ (3) Let $x_1, \dots, x_d$ generate $\mathfrak{m}$ and let

$$\varphi\colon k[t_1, \dots, t_d] \to \operatorname{gr}_{\mathfrak{m}} R, \qquad t_i \mapsto x_i + \mathfrak{m}^2 \in \mathfrak{m}/\mathfrak{m}^2$$

be the graded k-algebra map they determine. It is surjective, because $\mathfrak{m}^n$ is generated by the degree-n monomials in the x_i, so $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is spanned over k by their classes. If $d = 0$ then $\mathfrak{m} = \mathfrak{m}^2$, Nakayama gives $\mathfrak{m} = 0$, and both sides are k; assume $d \geq 1$.

Suppose $\ker\varphi \neq 0$. The kernel of a graded map is graded, so it contains a nonzero homogeneous F of some degree e. For $n \geq e$ multiplication by F is an injection $k[t]_{n-e} \hookrightarrow k[t]_n$, the polynomial ring being a domain, and $(\operatorname{gr}_{\mathfrak{m}} R)_n$ is a quotient of $k[t]_n/F\,k[t]_{n-e}$. Hence

$$\dim_k \mathfrak{m}^n/\mathfrak{m}^{n+1} \leq \binom{n+d-1}{d-1} - \binom{n-e+d-1}{d-1}$$

Both binomials are polynomials in n of degree $d-1$ with the same leading coefficient $1/(d-1)!$, so their difference has degree at most $d-2$.

Since $R/\mathfrak{m} = k$, the length $\ell(\mathfrak{m}^r/\mathfrak{m}^{r+1})$ is its k-dimension, and

$$\ell(R/\mathfrak{m}^n) = \sum_{r=0}^{n-1} \ell(\mathfrak{m}^r/\mathfrak{m}^{r+1})$$

Summing a function that is eventually bounded by a polynomial of degree at most $d-2$ produces one eventually bounded by a polynomial of degree at most $d-1$; the finitely many terms with $r < e$ contribute a constant and do not affect the degree. So $d(R) \leq d-1$. But $d(R) = \dim R = d$ by theorem 7.4.1, a contradiction. Therefore $\ker\varphi = 0$ and φ is an isomorphism.

(3) $\Rightarrow$ (1) Comparing degree-one pieces, $\mathfrak{m}/\mathfrak{m}^2 \cong k^d$, so $\dim_k \mathfrak{m}/\mathfrak{m}^2 = d = \dim R$.

This closes the cycle, completing the proof.

Criterion (3) is the one that travels. Everything below is extracted from it by the same mechanism: a property of the polynomial ring $\operatorname{gr}_{\mathfrak{m}} R$ is transferred back to R along the filtration by powers of $\mathfrak{m}$, and the transfer works because Krull's theorem makes the filtration separated. The first instance is that a regular local ring has no zero-divisors, which is not visible from the numerical definition at all.

Lemma 7.7.4: A Graded Domain Lifts

Let $(R, \mathfrak{m})$ be a local ring with $\bigcap_{n\geq 0} \mathfrak{m}^n = 0$. If $\operatorname{gr}_{\mathfrak{m}} R$ is an integral domain, then so is R.

***Proof*:**
Let $a, b \in R$ be nonzero. Because $\bigcap_n \mathfrak{m}^n = 0$, neither lies in every power of $\mathfrak{m}$, so there are largest integers $r, s \geq 0$ with $a \in \mathfrak{m}^r$ and $b \in \mathfrak{m}^s$ (reading $\mathfrak{m}^0 = R$); thus $a \notin \mathfrak{m}^{r+1}$ and $b \notin \mathfrak{m}^{s+1}$. Their classes $a^* = a + \mathfrak{m}^{r+1}$ and $b^* = b + \mathfrak{m}^{s+1}$ are then nonzero in the graded pieces $(\operatorname{gr}_{\mathfrak{m}} R)_r$ and $(\operatorname{gr}_{\mathfrak{m}} R)_s$.

Since $\operatorname{gr}_{\mathfrak{m}} R$ is a domain, $a^* b^* \neq 0$ in $(\operatorname{gr}_{\mathfrak{m}} R)_{r+s}$, which says exactly that $ab \notin \mathfrak{m}^{r+s+1}$. In particular $ab \neq 0$, completing the proof.

Corollary 7.7.5: A Regular Local Ring Is a Domain

Every regular local ring is an integral domain.

***Proof*:**
Let $(R, \mathfrak{m})$ be regular of dimension d. In a local ring $\mathfrak{m}$ is the Jacobson radical, so corollary 3.4.12 applies and gives $\bigcap_n \mathfrak{m}^n = 0$. By theorem 7.7.3(3) the ring $\operatorname{gr}_{\mathfrak{m}} R$ is a polynomial ring over a field, hence a domain, and lemma 7.7.4 transfers this to R, as we sought to show.

Regularity is also insensitive to completion, for the same reason: passing to $\hat{R}$ changes neither the associated graded ring nor the dimension, and criterion (3) sees nothing else. This is what makes it legitimate to test regularity after completing, where the Cohen structure theorem (theorem 3.6.1) puts formal coordinates on the ring.

Proposition 7.7.6: Regularity and Completion

Let $(R, \mathfrak{m})$ be a Noetherian local ring and $\hat{R}$ its $\mathfrak{m}$-adic completion. Then R is regular if and only if $\hat{R}$ is regular.

***Proof*:**
The ring $\hat{R}$ is Noetherian by theorem 3.5.7, and it is local with maximal ideal $\hat{\mathfrak{m}}$ and the same residue field, since $R/\mathfrak{m}^n \cong \hat{R}/\hat{\mathfrak{m}}^n$ for every n. By proposition 3.5.2(2) the two associated graded rings agree, $\operatorname{gr}_{\mathfrak{m}} R \cong \operatorname{gr}_{\hat{\mathfrak{m}}} \hat{R}$, and by corollary 3.5.9 the two dimensions agree. Criterion (3) of theorem 7.7.3 is a statement about those two data alone, so it holds for R exactly when it holds for $\hat{R}$, completing the proof.

Example 7.7.7: Regular and Singular Local Rings

1. *Formal power series.* Let $R = k[[x_1, \dots, x_n]]$, the completion of $A = k[x_1, \dots, x_n]_{(x_1,\dots,x_n)}$ (example 3.1.2). Then $\mathfrak{m} = (x_1, \dots, x_n)$ and $\mathfrak{m}/\mathfrak{m}^2$ has the n classes $\bar{x}_i$ as a basis, so the embedding dimension is n. The dimension is also n: $\dim A = n$ because $\dim k[x_1, \dots, x_n] = n$ by theorem 7.6.2 and localizing at a maximal ideal of height n retains it, and $\dim R = \dim A$ by corollary 3.5.9. So R is regular, with $x_1, \dots, x_n$ a regular system of parameters, and $\operatorname{gr}_{\mathfrak{m}} R \cong k[t_1, \dots, t_n]$ recovers the fact that a power series has a well-defined lowest-degree homogeneous part.

2. *Discrete valuation rings.* A Noetherian local domain of dimension 1 is regular exactly when

$\mathfrak{m}$ is principal, which by theorem 5.2.2 is exactly when it is a discrete valuation ring. So in dimension one, regular, normal and discrete-valuation all coincide; this is the case [9] uses to identify the smooth points of a curve.

3. *The cusp.* Let $A = k[x,y]/(y^2 - x^3)$ and $R = A_{\mathfrak{m}}$ with $\mathfrak{m} = (\bar{x}, \bar{y})$. Then $\dim R = 1$, since A is a one-dimensional domain by theorem 7.6.2 ($\mathrm{Frac}(A) = k(t)$ under $\bar{x} \mapsto t^2$, $\bar{y} \mapsto t^3$) and $\mathfrak{m}$ is maximal. But the defining relation $y^2 - x^3$ lies in $(x,y)^2$, so it imposes no linear condition: writing $\mathfrak{n} = (x,y) \subseteq k[x,y]$, one has $\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{n}/(\mathfrak{n}^2 + (y^2 - x^3)) = \mathfrak{n}/\mathfrak{n}^2$, which is two-dimensional. So the embedding dimension is $2 > 1 = \dim R$ and R is not regular. This is the algebraic signature of the cusp of the curve $y^2 = x^3$ at the origin.

4. *A regular ring of dimension zero.* If $\dim R = 0$ then regularity forces $\mathfrak{m} = \mathfrak{m}^2$, hence $\mathfrak{m} = 0$ by Nakayama, so the regular local rings of dimension zero are exactly the fields. Contrast the Artinian local ring $k[x]/(x^2)$, which has dimension 0 and embedding dimension 1, so it is not regular; by corollary 7.7.5 it could not be, having the zero-divisor $\bar{x}$.

7.8 Regular Local Rings Are Factorial

Section 7.7 characterized regularity three ways and drew one consequence from the graded criterion, that a regular local ring is a domain. The theorem of this section is deeper and is not suggested by any of the three criteria: a regular local ring has unique factorization. It is due to Auslander and Buchsbaum, and the proof is an induction on dimension that leaves the ring at every step, passing to a localization in which the dimension has dropped.

Two criteria carry the induction, and both are worth having on their own. The first converts factoriality into a statement about height-one primes, which is what makes it accessible to dimension theory at all. The second says factoriality may be checked after inverting a single prime element. Read geometrically the theorem says that on a smooth variety every codimension-one subvariety is cut out near each point by one equation, which is how [9] uses it.

Lemma 7.8.1: Factoriality by Height-One Primes

Let R be a Noetherian domain. Then R is a unique factorization domain if and only if every prime of height one is principal.

***Proof*:**
($\Rightarrow$) Let $\mathfrak{p}$ have height one and pick $a \in \mathfrak{p}$ nonzero; a is a nonunit because $\mathfrak{p}$ is proper. Factor $a = p_1 \cdots p_r$ into irreducibles. As $\mathfrak{p}$ is prime, some $p_i \in \mathfrak{p}$, and in a unique factorization domain an irreducible element is prime ([8, Prime If And Only If Irreducible In UFD]), so (p_i) is a nonzero prime with $0 \subsetneq (p_i) \subseteq \mathfrak{p}$. A strict inclusion $(p_i) \subsetneq \mathfrak{p}$ would exhibit a chain of length two descending from $\mathfrak{p}$, contradicting $\operatorname{ht}\mathfrak{p} = 1$. Hence $\mathfrak{p} = (p_i)$.

($\Leftarrow$) A Noetherian ring satisfies the ascending chain condition on ideals, in particular on principal ideals, so by [8, Unique Factorization Domain Equivalence] it suffices to show every irreducible element is prime. Let a be irreducible. Then (a) is a proper nonzero ideal, so it has a minimal prime $\mathfrak{p}$. Krull's height theorem (theorem 7.5.2) with $c = 1$ gives $\operatorname{ht}\mathfrak{p} \leq 1$, and $\mathfrak{p} \neq 0$ since $0 \neq a \in \mathfrak{p}$, so $\operatorname{ht}\mathfrak{p} = 1$. By hypothesis $\mathfrak{p} = (p)$ for some p, and $a \in (p)$ gives $a = pb$. Here p is a nonunit, so irreducibility of a forces b to be a unit; therefore $(a) = (p) = \mathfrak{p}$ is prime and a is a prime element, completing the proof.

The criterion is what ties factoriality to the rest of this chapter: height is a dimension-theoretic invariant, and Krull's height theorem is exactly what produced the height-one prime in the second half. The next lemma is the descent step. Inverting one prime element destroys only the primes containing it, and those form a single height-one class that can be checked by hand.

Lemma 7.8.2: Nagata's Criterion

Let R be a Noetherian domain and $x \in R$ a prime element. If the localization $R_x = R[1/x]$ is a unique factorization domain, then so is R.

***Proof*:**
Both R and R_x are Noetherian domains, the latter by proposition 1.4.7(1),(2), so lemma 7.8.1 applies to each. Let $\mathfrak{p} \subseteq R$ be a prime of height one; it suffices to show $\mathfrak{p}$ is principal.

Case $x \in \mathfrak{p}$. Then $0 \subsetneq (x) \subseteq \mathfrak{p}$ with (x) prime, and $\operatorname{ht}\mathfrak{p} = 1$ forces $\mathfrak{p} = (x)$.

Case $x \notin \mathfrak{p}$. The primes of R_x are the extensions of the primes of R not meeting $\{x^n \mid n \geq 0\}$, and the correspondence preserves inclusions (proposition 1.4.5), so $\mathfrak{p}R_x$ is a prime of height one. As R_x is a unique factorization domain, lemma 7.8.1 makes $\mathfrak{p}R_x$ principal. Multiplying a generator by a power of the unit x we may take the generator to be some $a \in R$, and then $a \in \mathfrak{p}R_x \cap R = \mathfrak{p}$, the contraction of the extension of $\mathfrak{p}$ being $\mathfrak{p}$ again since $\mathfrak{p}$ misses the powers of x.

We may further assume $x \nmid a$. If $a = xa'$ then $a' \in \mathfrak{p}$, because $a \in \mathfrak{p}$, $x \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime; and a' still generates $\mathfrak{p}R_x$, since x is a unit there. This replacement cannot repeat indefinitely: it would place a in $\bigcap_n (x^n)$, which is 0 by corollary 3.4.11, while $a \neq 0$.

Now $\mathfrak{p} = (a)$. One inclusion is $a \in \mathfrak{p}$. For the other, let $b \in \mathfrak{p}$. Then $b \in \mathfrak{p}R_x = aR_x$, so $x^n b = ac$ for some $c \in R$ and $n \geq 0$. If $n \geq 1$ then x divides ac; as x is prime and $x \nmid a$, it divides c, and cancelling x from both sides (legitimate in a domain) lowers n by one. Iterating gives $b \in (a)$. So every height-one prime of R is principal, and lemma 7.8.1 makes R a unique factorization domain, completing the proof.

What remains is to produce a generator for a height-one prime after inverting x, and this is where the induction's homological input enters. The prime will be visibly principal at every localization, and the problem is to pass from that local information to a single global generator. The next two lemmas do so; the first is elementary, and the second is the one place this section imports a fact it does not prove.

Lemma 7.8.3: A Finite Free Resolution Makes a Projective Module Stably Free

Let S be a ring and P a finitely generated projective S-module admitting a resolution $0 \to F_n \to \cdots \to F_0 \to P \to 0$ by finitely generated free modules. Then P is *stably free*: $P \oplus S^a \cong S^b$ for some $a, b \geq 0$.

***Proof*:**
Induct on n. If $n = 0$ then $P \cong F_0$ is free. For $n \geq 1$ put $K = \ker(F_0 \to P)$. Since P is projective the sequence $0 \to K \to F_0 \to P \to 0$ splits, so $F_0 \cong P \oplus K$; in particular K is a direct summand of a finitely generated free module, hence finitely generated projective, and $0 \to F_n \to \cdots \to F_1 \to K \to 0$ is a free resolution of length $n - 1$. By induction $K \oplus S^a \cong S^b$ for some a, b. Then

$$P \oplus S^b \cong P \oplus K \oplus S^a \cong F_0 \oplus S^a$$

which is free, as we sought to show.

Lemma 7.8.4: A Locally Principal Ideal of Finite Global Dimension Is Principal

Let S be a Noetherian domain of finite global dimension and $I \subseteq S$ a nonzero ideal such that $I_{\mathfrak{q}}$ is a principal ideal of $S_{\mathfrak{q}}$ for every prime $\mathfrak{q}$ of S. Then I is principal.

***Proof*:**
Step 1: I is invertible, hence projective. Work inside $K = \mathrm{Frac}(S)$ and put $I^{-1} = \{z \in K \mid zI \subseteq S\}$, an S-submodule of K containing S. The product II^{-1} is an ideal of S; it equals S, because for each prime $\mathfrak{q}$ the localization $I_{\mathfrak{q}}$ is generated by one element $t_{\mathfrak{q}} \neq 0$, so $t_{\mathfrak{q}}^{-1} \in (I^{-1})_{\mathfrak{q}}$ and $(II^{-1})_{\mathfrak{q}} \ni t_{\mathfrak{q}} t_{\mathfrak{q}}^{-1} = 1$, and an ideal that is the unit ideal at every prime is the unit ideal (proposition 1.4.13 applied to S/II^{-1}). Choose $a_1, \ldots, a_r \in I$ and $b_1, \ldots, b_r \in I^{-1}$ with $\sum_i a_i b_i = 1$. Then $x \mapsto (b_1 x, \ldots, b_r x)$ is an S-module map $I \to S^r$, and composing with $(s_1, \ldots, s_r) \mapsto \sum_i s_i a_i$ returns x. So I is a direct summand of S^r, hence finitely generated projective, and $I_{\mathfrak{q}} \cong S_{\mathfrak{q}}$ at every prime.

Step 2: I is stably free. Finite global dimension gives I a finite resolution by finitely generated free modules, so lemma 7.8.3 applies: $I \oplus S^a \cong S^b$ for some a, b. Localizing at any prime and comparing ranks gives $1 + a = b$, so the relation reads $I \oplus S^a \cong S^{a+1}$.

Step 3: rank one and stably free give free. This is [8, A Stably Free Module of Rank One Is Free], applied to I: it is finitely generated projective with $I_{\mathfrak{q}} \cong S_{\mathfrak{q}}$ at every prime, and Step 2 supplies the required relation. So $I \cong S$ as an S-module, and the image of 1 under such an isomorphism generates I, completing the proof.

Theorem 7.8.5: Regular Local Rings Are Factorial

Every regular local ring is a unique factorization domain.

***Proof*:**
Let $(R, \mathfrak{m})$ be regular of dimension d; induct on d.

Base cases. If $d = 0$ then R is a field (example 7.7.7), which has no nonzero nonunits and is trivially factorial. If $d = 1$ then R is a Noetherian local domain whose maximal ideal is principal, hence a discrete valuation ring (theorem 5.2.2); its only height-one prime is $\mathfrak{m}$, which is principal, so lemma 7.8.1 applies.

Producing a prime element. Let $d \geq 2$. By corollary 7.7.5 R is a domain. Choose $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Its class is nonzero in $\mathfrak{m}/\mathfrak{m}^2$, so it extends to a basis, and lifting gives a minimal generating set $x, x_2, \ldots, x_d$ of $\mathfrak{m}$. Then $\mathfrak{m}/(x)$ is generated by the $d - 1$ classes of $x_2, \ldots, x_d$, while

$\dim R/(x) = d - 1$ because x is a nonzero nonunit in a domain (the consequence of theorem 7.5.2 recorded after it). So $R/(x)$ is regular by theorem 7.7.3(2), hence a domain by corollary 7.7.5, and therefore x is a prime element of R.

Reduction to a localization. By lemma 7.8.2 it is enough that $S = R_x$ is a unique factorization domain, and by lemma 7.8.1 it is enough that every height-one prime of S is principal. Note S is a Noetherian domain by proposition 1.4.7.

Global dimension is finite along the way. The Auslander-Buchsbaum-Serre theorem ([5, Auslander-Buchsbaum-Serre Theorem]) makes $\operatorname{gldim} R$ finite, R being regular. Global dimension does not rise under localization: an S'-module over a localization S' of R is the localization of itself viewed as an R-module, and localizing an R-projective resolution gives an S'-projective resolution of the same length, localization being exact (proposition 1.4.9) and carrying projectives to projectives. Hence $\operatorname{gldim} S \leq \operatorname{gldim} R < \infty$, and likewise for every $R_{\mathfrak{p}}$.

The localizations are smaller regular rings. Let $\mathfrak{q}$ be a prime of S and $\mathfrak{p}$ the corresponding prime of R, so $S_{\mathfrak{q}} \cong R_{\mathfrak{p}}$ and $x \notin \mathfrak{p}$. Then $\mathfrak{p} \subsetneq \mathfrak{m}$, so $\dim R_{\mathfrak{p}} = \operatorname{ht} \mathfrak{p} < d$. Since $\operatorname{gldim} R_{\mathfrak{p}}$ is finite, the Auslander-Buchsbaum-Serre theorem in the other direction makes $R_{\mathfrak{p}}$ regular, and the induction hypothesis makes it a unique factorization domain.

Conclusion. Let $\mathfrak{P}$ be a height-one prime of S. For a prime $\mathfrak{q}$ of S, either $\mathfrak{P} \not\subseteq \mathfrak{q}$, in which case $\mathfrak{P} S_{\mathfrak{q}} = S_{\mathfrak{q}}$ is principal, or $\mathfrak{P} \subseteq \mathfrak{q}$, in which case $\mathfrak{P} S_{\mathfrak{q}}$ is a height-one prime of the unique factorization domain $S_{\mathfrak{q}}$ and is principal by lemma 7.8.1. So $\mathfrak{P}$ is a nonzero locally principal ideal of a Noetherian domain of finite global dimension, and lemma 7.8.4 makes it principal. Therefore S is a unique factorization domain, and lemma 7.8.2 carries this back to R, completing the proof.

Factoriality is the strongest of the standard finiteness conditions a local ring can carry, and the two weaker ones follow from it at once. The first is integral closure.

Corollary 7.8.6: A Unique Factorization Domain Is Integrally Closed

Every unique factorization domain is integrally closed in its fraction field.

***Proof*:**
Let R be a unique factorization domain with fraction field K and let $z \in K$ be integral over R. Write $z = a/b$ with $a, b \in R$, $b \neq 0$, sharing no irreducible factor, which unique factorization permits. From $z^n + c_{n-1} z^{n-1} + \cdots + c_0 = 0$ with $c_i \in R$, multiplying by b^n gives

$$a^n = -b \left(c_{n-1} a^{n-1} + c_{n-2} a^{n-2} b + \cdots + c_0 b^{n-1} \right)$$

So every irreducible factor p of b divides a^n, and p is prime ([8, Prime If And Only If Irreducible In UFD]), so p divides a, contrary to the choice of a and b. Hence b has no irreducible factor at all, so it is a unit and $z = ab^{-1} \in R$, as we sought to show.

Corollary 7.8.7: A Regular Local Ring Is Normal

Every regular local ring is a normal domain, and in particular is regular in codimension one.

***Proof*:**
Combine theorem 7.8.5 with corollary 7.8.6; the ring is a domain by corollary 7.7.5. For the last clause, a localization of a normal domain is normal by proposition 1.4.7(3), completing the proof.

Example 7.8.8: Normal but Not Factorial
Neither implication in

$$\text{regular} \Rightarrow \text{factorial} \Rightarrow \text{normal}$$

reverses. Let k have characteristic $\neq 2$ and let $B = k[u^2, uv, v^2] \subseteq k[u, v]$, the subring of polynomials of even total degree. Sending $x \mapsto u^2$, $y \mapsto v^2$, $z \mapsto uv$ identifies B with $k[x, y, z]/(xy - z^2)$, so B is the coordinate ring of the quadric cone.

B is normal. Let σ be the involution $u \mapsto -u$, $v \mapsto -v$ of $k[u, v]$. Its fixed ring is the even part, which is B, and its fixed field inside $k(u, v)$ is $\mathrm{Frac}(B) = k(u^2, u/v)$. Let $w \in \mathrm{Frac}(B)$ be integral over B. Then w is integral over $k[u, v]$, which is a unique factorization domain ([8, Polynomial Rings in Several Variables over a UFD are UFDs]) and hence integrally closed by corollary 7.8.6; so $w \in k[u, v]$. Being in $\mathrm{Frac}(B)$, w is fixed by σ, so $w \in k[u, v]^\sigma = B$.

B is not factorial. In B the identity $u^2 \cdot v^2 = (uv)^2$ gives two factorizations of one element into irreducibles: each of u^2, v^2, uv has degree two, which is the least positive degree occurring in B, so none of them factors further in B, and no two of them are associates ($B^\times = k^\times$).

B is not regular. Localize at $\mathfrak{m} = (u^2, uv, v^2)$. The fraction field $k(u^2, u/v)$ has transcendence degree two over k, so $\dim B_{\mathfrak{m}} = 2$ by theorem 7.6.2, while $\mathfrak{m}/\mathfrak{m}^2$ has the three classes of u^2, uv, v^2 as a basis, $\mathfrak{m}^2$ being spanned by the monomials of degree four. So the embedding dimension is $3 > 2$.

The three computations are consistent with the displayed implications and show both are strict: $B_{\mathfrak{m}}$ is normal and not factorial, and by theorem 7.8.5 a non-factorial ring cannot be regular, which the third computation confirms directly.

7.9 Normal Domains in Codimension One

A normal domain is determined by its behaviour in codimension one. That is the content of this section, and it is what makes normality a condition one can check by looking only at the height-one primes: the ring is the intersection of the discrete valuation rings sitting at them. The result is used by the homological characterizations of normality in *Homological Algebra* [5] and by the divisor theory of [9] and [1].

Everything rests on one fact about principal ideals, which is where normality enters.

Lemma 7.9.1: Associated Primes of a Principal Ideal Have Height One

Let R be a Noetherian normal domain and $b \in R$ a nonzero nonunit. Then every associated prime of (b) has height one.

***Proof*:**
Let $\mathfrak{p} \in \mathrm{Ass}(R/(b))$. Associated primes localize (corollary 2.6.2), so $\mathfrak{p}R_\mathfrak{p} \in \mathrm{Ass}\big(R_\mathfrak{p}/bR_\mathfrak{p}\big)$, and $R_\mathfrak{p}$ is again a Noetherian normal domain by proposition 1.4.7. Since $\mathrm{ht}\,\mathfrak{p} = \mathrm{kdim} R_\mathfrak{p}$, it suffices to treat the case where $(R, \mathfrak{m})$ is local with $\mathfrak{m} \in \mathrm{Ass}(R/(b))$ and to show $\mathrm{kdim} R = 1$.

By definition 2.3.1 there is $a \in R$ with $\mathfrak{m} = \mathrm{Ann}(\overline{a}) = (b : a)$, where $\overline{a}$ is the class of a in $R/(b)$; in particular $a \notin (b)$, since otherwise the annihilator would be all of R. Put $x = a/b \in \mathrm{Frac}(R)$, so $x \notin R$. For every $r \in \mathfrak{m}$ one has $ra \in (b)$, hence $rx = ra/b \in R$: that is, $x\mathfrak{m} \subseteq R$, and $x\mathfrak{m}$ is an R-submodule of R, so an ideal.

Case 1: $x\mathfrak{m} \subseteq \mathfrak{m}$. Then $x^n\mathfrak{m} \subseteq \mathfrak{m}$ for every n, so $R[x]\mathfrak{m} \subseteq \mathfrak{m} \subseteq R$. Choose $0 \neq c \in \mathfrak{m}$, which exists as $b \in \mathfrak{m}$ is nonzero. Then $cR[x] \subseteq R$, so $R[x] \subseteq c^{-1}R$, a cyclic R-module; as R is Noetherian, the submodule $R[x]$ is a finitely generated R-module, and proposition 4.1.4(2) makes x integral over R. Normality gives $x \in R$, contradicting $x \notin R$.

Case 2: $x\mathfrak{m} \not\subseteq \mathfrak{m}$. Then the ideal $x\mathfrak{m}$ is not contained in the unique maximal ideal, so $x\mathfrak{m} = R$. Pick $t \in \mathfrak{m}$ with $xt = 1$. For any $p \in \mathfrak{m}$ we have $xp \in R$ and $p = t\,(xp)$, so $\mathfrak{m} = tR$ is principal, and it is nonzero. A Noetherian local domain whose maximal ideal is principal and nonzero is a discrete valuation ring by theorem 5.2.2, hence of dimension one.

Case 1 is impossible, so Case 2 holds and $\mathrm{kdim} R = 1$, as we sought to show.

Theorem 7.9.2: A Normal Domain Is the Intersection of Its Height-One Localizations

Let R be a Noetherian normal domain with fraction field K. Then $R_\mathfrak{p}$ is a discrete valuation ring for every prime $\mathfrak{p}$ of height one, and

$$R = \bigcap_{\mathrm{ht}\,\mathfrak{p}=1} R_\mathfrak{p}$$

the intersection taken inside K.

***Proof*:**
For $\mathrm{ht}\,\mathfrak{p} = 1$ the ring $R_\mathfrak{p}$ is a Noetherian local domain of dimension one, normal by proposition 1.4.7, hence a discrete valuation ring by theorem 5.2.2.

The inclusion $R \subseteq \bigcap R_\mathfrak{p}$ is immediate. For the reverse, let x lie in every $R_\mathfrak{p}$ and write $x = a/b$ with $a, b \in R$ and $b \neq 0$. If b is a unit then $x \in R$, so assume otherwise and suppose $a \notin (b)$, aiming at a contradiction.

The class $\overline{a} \in R/(b)$ is then nonzero, and $I = \mathrm{Ann}(\overline{a}) = (b : a)$ is a proper ideal every element of which kills $\overline{a}$, hence is a zero-divisor on $R/(b)$. By corollary 2.5.2 the zero-divisors on $R/(b)$ are the union of the associated primes of (b), of which there are finitely many (corollary 2.4.3), so prime avoidance (proposition 2.5.1) places $I \subseteq \mathfrak{q}$ for a single $\mathfrak{q} \in \mathrm{Ass}(R/(b))$. By lemma 7.9.1, $\mathrm{ht}\,\mathfrak{q} = 1$.

Now $x \in R_\mathfrak{q}$ by hypothesis, so $a/b = c/s$ for some $c \in R$ and $s \notin \mathfrak{q}$, whence $sa \in bR$ and $s \in (b : a) = I \subseteq \mathfrak{q}$. That contradicts $s \notin \mathfrak{q}$. Therefore $a \in (b)$ and $x \in R$, completing the proof.

The theorem is the reason divisor theory works on a normal variety: a rational function regular at every codimension-one point is regular, so a function is controlled by its orders of vanishing along the height-one primes alone. That reading, and the class group it supports, is [9].

8

Flatness and Faithfully-Flat Descent

Flatness is the algebraic form of continuous variation. A flat module is one against which tensoring preserves injections, so that submodule relations survive base change; for a ring map $R \to S$ it says the fibers of $\operatorname{Spec} S \to \operatorname{Spec} R$ vary without jumping. When in addition the base change loses no information (when it is *faithfully flat*), one can argue in reverse: a property that holds after tensoring up already held below. This is *descent*, and it is the reason the local models of a space determine the space. The chapter records the ring-theoretic core that the geometry books cash out as faithfully-flat and fppf descent, closing with the local criterion that detects flatness one fiber at a time, the tool deformation theory runs on.

Basic flatness is assumed from [8]; the homological criterion and its local refinement are proved in [5] and cited here.

8.1 Flat Modules

Recall from [8] that an R-module M is *flat* if $- \otimes_R M$ is exact, equivalently if it preserves injections; a ring map $R \to S$ is flat if S is flat as an R-module. Flatness is preserved under base change and composition, and is detected locally: M is flat over R iff $M_{\mathfrak{p}}$ is flat over $R_{\mathfrak{p}}$ for every prime $\mathfrak{p}$ (proposition 1.4.16). Localizations are flat, and for a Noetherian local ring the completion is flat (corollary 3.4.6). The homological measure of the failure of flatness is Tor: M is flat iff $\operatorname{Tor}_1^R(M, N) = 0$ for all N, and it suffices to test $N = R/\mathfrak{a}$ on finitely generated ideals; this is proved in [5] and used freely below.

8.2 Faithfully Flat Modules and Algebras

Definition 8.2.1: Faithfully Flat

An R-module M is *faithfully flat* if it is flat and, for every R-module N, the vanishing $N \otimes_R M = 0$ forces $N = 0$. A ring map $R \to S$ is faithfully flat if S is faithfully flat as an R-module.

Proposition 8.2.2: Characterizations of Faithful Flatness

For a flat R-module M the following are equivalent:

1. M is faithfully flat;
2. a sequence of R-modules is exact iff it stays exact after $- \otimes_R M$;
3. $\mathfrak{m}M \neq M$ for every maximal ideal $\mathfrak{m}$, i.e. $M \otimes_R \kappa(\mathfrak{m}) \neq 0$ for all $\mathfrak{m}$.

For a flat ring map $R \to S$ these are equivalent to $\operatorname{Spec} S \to \operatorname{Spec} R$ being surjective.

***Proof*:**
(1) $\Rightarrow$ (2): flatness already preserves exactness; conversely if $C_\bullet \otimes M$ is exact then $H(C_\bullet) \otimes M = H(C_\bullet \otimes M) = 0$ (a flat functor commutes with homology), so $H(C_\bullet) = 0$ by (1). (2) $\Rightarrow$ (3): the sequence $0 \to R/\mathfrak{m} \to 0$ is not exact, so by (2) it remains non-exact after $- \otimes M$, forcing $M/\mathfrak{m}M = (R/\mathfrak{m}) \otimes M \neq 0$. (3) $\Rightarrow$ (1): let $N \neq 0$ and $0 \neq x \in N$, so $Rx \cong R/\mathfrak{a}$ with $\mathfrak{a} = \operatorname{Ann}(x) \subseteq \mathfrak{m}$ for some maximal $\mathfrak{m}$. Tensoring with the flat M: the inclusion $Rx \hookrightarrow N$ gives an injection $Rx \otimes M \hookrightarrow N \otimes M$, while the surjection $R/\mathfrak{a} \to R/\mathfrak{m}$ gives a surjection $Rx \otimes M = M/\mathfrak{a}M \twoheadrightarrow M/\mathfrak{m}M \neq 0$; hence $Rx \otimes M \neq 0$, so $N \otimes M \neq 0$. For a flat ring map these are further equivalent to surjectivity of $\operatorname{Spec} S \to \operatorname{Spec} R$: (1) gives $S \otimes_R \kappa(\mathfrak{p}) \neq 0$ for every prime $\mathfrak{p}$, so every fibre is nonempty and the map is surjective; conversely a surjective map hits every maximal ideal, which is (3).

Corollary 8.2.3: Faithfully Flat Maps are Injective

A faithfully flat ring map $R \to S$ is injective, and $\mathfrak{a} = \mathfrak{a}S \cap R$ for every ideal $\mathfrak{a}$ of R.

***Proof*:**
Apply faithful flatness to the kernel K of $R \to S$: tensoring $0 \to K \to R \to S$ with the flat S gives $K \otimes_R S = \ker(S \to S \otimes_R S)$, the map being $s \mapsto 1 \otimes s$, which is split by the multiplication $S \otimes_R S \to S$; so $K \otimes S = 0$, whence $K = 0$. For the second claim, the map $R/\mathfrak{a} \to (R/\mathfrak{a}) \otimes_R S = S/\mathfrak{a}S$ has kernel $(\mathfrak{a}S \cap R)/\mathfrak{a}$; since $S/\mathfrak{a}S$ is faithfully flat over $R/\mathfrak{a}$ (base change of a faithfully flat map), it is injective by the first part, so $\mathfrak{a}S \cap R = \mathfrak{a}$.

Localization $R \to R_f$ is flat but rarely faithfully flat (its image in $\operatorname{Spec} R$ omits $V(f)$); by contrast a nonzero free module is faithfully flat, a finite free ring extension is faithfully flat, and for a Noetherian local ring the completion $R \to \hat{R}$ is faithfully flat.

8.3 Faithfully-Flat Descent

Faithful flatness lets a property established upstairs be brought back down.

Proposition 8.3.1: Descent of Module Properties

Let $R \to S$ be faithfully flat and M an R-module. Then M is finitely generated (respectively finitely presented, flat, or finitely generated projective) over R iff $M \otimes_R S$ has the corresponding property over S.

***Proof*:**
Necessity is base change. For sufficiency: if $M \otimes S$ is generated by finitely many elements, clear denominators so they are images of $m_i \otimes 1$; the cokernel C of $R^n \xrightarrow{(m_i)} M$ satisfies $C \otimes S = 0$, so $C = 0$ and M is finitely generated. Given that, finite presentation descends by applying the same to the kernel of $R^n \to M$ (finitely generated as $\ker \otimes S = \ker(S^n \to M \otimes S)$ is). Flatness descends because $\mathrm{Tor}_1^R(M, N) \otimes S = \mathrm{Tor}_1^S(M \otimes S, N \otimes S) = 0$ for all N, whence $\mathrm{Tor}_1^R(M, N) = 0$ by faithful flatness. Finitely generated projective descends since it equals finitely presented plus flat.

The deeper statement recovers a module, not merely a property, from data upstairs. Write $S^{\otimes 2} = S \otimes_R S$ and $S^{\otimes 3} = S \otimes_R S \otimes_R S$, with the two coprojections $p_1, p_2 \colon S \to S^{\otimes 2}$.

Theorem 8.3.2: Faithfully-Flat Descent for Modules

Let $R \to S$ be faithfully flat. Then $M \mapsto M \otimes_R S$ is an equivalence from R-modules to the category of *descent data*: pairs (N, φ) with N an S-module and $\varphi \colon p_1^* N \xrightarrow{\sim} p_2^* N$ an isomorphism over $S^{\otimes 2}$ satisfying the cocycle condition over $S^{\otimes 3}$. The inverse sends (N, φ) to the equalizer of φ-twisted coprojections $N \rightrightarrows N \otimes_S S^{\otimes 2}$.

***Proof*:**
The heart is the exactness, for every R-module M, of the *Amitsur complex*

$$0 \to M \to M \otimes_R S \to M \otimes_R S^{\otimes 2},$$

the second map being $m \otimes s \mapsto m \otimes (p_1 - p_2)(s)$. It suffices to check this after the faithfully flat base change $S \otimes_R -$, where the augmented complex acquires the contracting homotopy $s_0 \otimes s_1 \otimes \cdots \mapsto (s_0 s_1) \otimes s_2 \otimes \cdots$ (multiplication of the first two tensor factors), which shows exactness. Faithful flatness then gives exactness over R. Applying this with the canonical datum shows $M \otimes S$ recovers M as the stated equalizer, so the functor is fully faithful; the cocycle condition is exactly what makes the equalizer of a general (N, φ) an R-module whose base change is (N, φ), giving essential surjectivity.

This is the affine case of the descent that lets sheaves and torsors be glued along faithfully flat (fppf) covers, developed in [7].

8.4 The Local Criterion for Flatness

For finitely generated modules over a Noetherian local ring, flatness collapses to a single Tor, and this is what makes flatness checkable in families.

Theorem 8.4.1: Local Criterion for Flatness

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field κ and M a finitely generated R-module. Then

$$M \text{ flat} \iff M \text{ free} \iff \mathrm{Tor}_1^R(\kappa, M) = 0$$

More generally, for a local homomorphism $(R, \mathfrak{m}) \to (S, \mathfrak{n})$ of Noetherian local rings and a finitely generated S-module M, one has M flat over R iff $\mathrm{Tor}_1^R(\kappa, M) = 0$; over an Artinian local base this is the criterion by which flatness of a deformation is tested one infinitesimal step at a time.

The equivalences and the general criterion are proved homologically in [5, Local Criterion for Flatness], which owns this theorem (DECISIONS 2026-08-17); here the point is that they reduce flatness (a condition on all modules) to the vanishing of one Tor against the residue field, and it is in this form that [7] invokes flatness over Artinian bases.

8.5 Fitting Ideals

The Fitting ideals of a finitely presented module refine the notion of rank: they cut out the loci where the number of generators jumps, commute with base change, and are the standard tool behind flattening stratifications.

Definition 8.5.1: Fitting Ideals

Let M be a finitely presented R-module with presentation $R^m \xrightarrow{A} R^n \to M \to 0$, so A is an $n \times m$ matrix. For $i \geq 0$ the *i-th Fitting ideal* $\mathrm{Fitt}_i(M)$ is generated by the $(n-i) \times (n-i)$ minors of A, with the conventions $\mathrm{Fitt}_i(M) = R$ when $n - i \leq 0$ and $\mathrm{Fitt}_i(M) = 0$ when $n - i > m$.

Proposition 8.5.2: Properties of Fitting Ideals

The Fitting ideals are independent of the chosen presentation, and:

1. $\mathrm{Fitt}_0(M) \subseteq \mathrm{Fitt}_1(M) \subseteq \cdots$, with $\mathrm{Fitt}_i(M) = R$ for $i \geq n$;
2. base change: $\mathrm{Fitt}_i(M \otimes_R R') = \mathrm{Fitt}_i(M)R'$ for any $R \to R'$;
3. $\mathrm{Fitt}_0(M) \subseteq \mathrm{Ann}(M)$ and $\mathrm{Ann}(M)^n \subseteq \mathrm{Fitt}_0(M)$;
4. for a prime $\mathfrak{p}$, the minimal number of generators of $M_{\mathfrak{p}}$ is $> d$ iff $\mathrm{Fitt}_d(M) \subseteq \mathfrak{p}$.

***Proof*:**
Any two finite presentations of M are related by adding free summands and invertible row/column operations, under which the ideal of $(n-i)$-minors is unchanged (a standard determinantal computation), giving well-definedness; base change is immediate since the minors of A map to those of $A \otimes R'$. The chain and the values for $i \geq n$ are clear. For (3), Cramer's rule applied to A gives $\mathrm{Fitt}_0(M)\, M = 0$, and expanding determinants shows $\mathrm{Ann}(M)^n \subseteq \mathrm{Fitt}_0(M)$. For (4), by Nakayama the minimal number of generators of $M_{\mathfrak{p}}$ equals $\dim_{\kappa(\mathfrak{p})} M \otimes \kappa(\mathfrak{p}) = n - \mathrm{rank}(A \bmod \mathfrak{p})$, and $\mathrm{Fitt}_d(M) \not\subseteq \mathfrak{p}$ says exactly that some $(n-d)$-minor is a unit mod $\mathfrak{p}$, i.e. $\mathrm{rank}(A \bmod \mathfrak{p}) \geq n - d$.

Thus the rank-jump locus $V(\mathrm{Fitt}_d(M)) = \{\mathfrak{p} \mid M_{\mathfrak{p}} \text{ needs} > d \text{ generators}\}$ is closed and base-change compatible, giving the flattening stratification used to build Quot and Hilbert schemes ([7]); $V(\mathrm{Fitt}_0(M))$ cuts out the support of M (with its Fitting scheme structure; the scheme-theoretic support proper is $V(\mathrm{Ann}\, M)$, with the same underlying closed set). Over a domain, a finitely presented M is locally free of rank r iff $\mathrm{Fitt}_{r-1}(M) = 0$ and $\mathrm{Fitt}_r(M) = R$, i.e. iff $\mathrm{Fitt}_r(M) = R$ and A has constant rank $n - r$.

8.6 Generic Flatness

Away from a proper closed subset of the base, a coherent family becomes flat. This Grothendieck finiteness underlies dimension counts in families and the construction of moduli.

Theorem 8.6.1: Generic Flatness

Let R be a Noetherian integral domain, S a finitely generated R-algebra, and M a finitely generated S-module. Then there is a nonzero $f \in R$ such that M_f is a free R_f-module (in particular flat over R_f).

***Proof*:**
We sketch the standard argument (Eisenbud, *Commutative Algebra*, Theorem 14.4; equivalently the Stacks Project generic-flatness lemma). By dévissage a finite filtration of M with quotients $S/\mathfrak{p}_i$ ($\mathfrak{p}_i \subseteq S$ prime) reduces the claim to $M = S/\mathfrak{p}$, since a module filtered by pieces that are free after localizing is itself free after localizing. If $\mathfrak{q} := \ker(R \to S/\mathfrak{p})$ is nonzero, then $\mathfrak{q}$ annihilates M, so $M_f = 0$ is free for any $0 \neq f \in \mathfrak{q}$. Otherwise $R \hookrightarrow S/\mathfrak{p}$ is a domain of finite type over R; by Noether normalization $(S/\mathfrak{p}) \otimes_R K$ is a *finite* module (not in general free) over a polynomial ring $K[t_1, \ldots, t_d]$, where $K = \mathrm{Frac}(R)$, and spreading out gives $0 \neq f_0$ with S_{f_0} finite over $R_{f_0}[t_1, \ldots, t_d]$. A Noetherian induction on the relative dimension d then trivializes, after inverting finitely many further elements of R, the torsion pieces of this finite module, producing a single f with M_f free over R_f.

What Next?

In the EYNTKA series, commutative algebra is put to work at once. *Classical Algebraic Geometry* [9] reads the prime spectrum as a space and turns the dictionary above into the geometry of varieties; *Algebraic Number Theory* [3] specializes to the rings of integers of number fields, where the one-dimensional regular local rings of this book become the discrete valuations that govern ramification. The scheme theory of *Algebraic Geometry* [1] globalizes the affine picture, and *Algebraic K-Theory* [2] studies the projective modules met here as the objects of a homology theory.

Two neighbours refine the material rather than move past it. *Homological Algebra* [5] supplies the derived-functor account of flatness together with the depth, Auslander-Buchsbaum, and Serre theory that turns dimension into a homological invariant; *Étale Cohomology* [4] builds on the completion theory of this book for its local models, developing the Henselian and strictly Henselian local rings it needs there. For the reader heading toward research, the p-adic Hodge theory and perfectoid geometry at the frontier of the series rest on exactly the completion and flatness of this book.

Index

Artin-Rees Lemma, 62
Artinian Module, 97
Artinian Ring, 97
Ass(M), 43
Associated Graded Ring, 67
Associated Prime, 43

Chain, 11
Characteristic Polynomial, 107
Cohen Structure Theorem, 72
Complete Ring, 58
Completion
    $\mathfrak{a}$-adic of Ring, 52
Composition Series
    R-module, 11

Descent
    Faithfully Flat, 123
Descent Datum, 123

Embedded Prime, 47

Faithfully Flat, 122
Filtration, 59
Fitting Ideal, 124

Gröbner Basis, 17

Hauptidealsatz, 109
Height of a Prime, 106
Hilbert's Basis Theorem, 12
Hilbert-Samuel Polynomial, 107

Ideal
    Contains a Product of Primes, 10
Ideal Contraction, 82
Ideal Extension, 82
Integral Closure
    Finiteness, 81
Integral Element, 76
Isolated Prime, 47

Jacobson Radical, 34

Krull Dimension, 106
Krull's Principal Ideal Theorem, 109
Krull's Theorem, 65

length of Composition Series, 11
Local Property, 29
Local Ring, 32
Localization, 21
Lying Over, 84

Monomial Order, 17

Nagata's Criterion, 115
Noether Normalization, 105
Noetherian
    Ring, 9

Primary Decomposition, 43
Primary Ideal, 41
Prime Avoidance, 46

R-module of Finite Length, 102

Regular Local Ring, 111
Regular Sequence
    associated graded ring of, 68
Regular System of Parameters, 111
Residue Field, 33
Ring Extension, 75

S-polynomial, 18
Saturated Ideal, 25
Separated Filtration, 58
Supp(M), 46
Support of a Module, 46
System of Parameters, 109

Unique Factorization Domain
    height-one criterion, 114
Universal Property
    Localization, 20

Valuation
    Discrete, 90
    Real, 89
Valuation Ring, 91
Valuation Totally Ordered Abelian Group, 89

Weierstrass Division Theorem, 73
Weierstrass Preparation Theorem, 73

Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need to Know about Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic K-Theory*. 1st ed.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Number THeory*. Vol. 1. URL: https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-algebraic-number-theory.pdf&name=Algebraic%20Number%20Theory&source=books.

[4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Étale Cohomology*. Corequisite; in preparation. 2026.

[5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Homological Algebra*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_homological_algebra.pdf.

[6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Local Field Theory*. URL: https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-local-field-theory.pdf&name=Local%20Field%20Theory&source=books.

[7] Nathanael Chwojko-Srawley. *Everything You Need to Know about Moduli, Deformation Theory, and Stacks*. 1 vols. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA-moduli-stacks.pdf.

[8] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.

[9] Nathanael Chwojko-Srawley. *EYNTKA Classical Algebraic Geometry*.

[10] Hideyuki Matsumura. *Commutative Algebra*. 2nd ed. URL: https://aareyanmanzoor.github.io/assets/matsumura-CA.pdf.

[11] The Stacks project authors. *The Stacks Project*. 2026. URL: https://stacks.math.columbia.edu.